BIVISIONTS: BIVARIATE VISION TRANSFORMER FOR TIME SERIES FORECASTING WITH MULTI-SCALE ATTENTION-WEIGHTED FUSION

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DECLARATION

I declare that this is my own work and this Thesis does not incorporate without acknowledgment any material previously submitted for a Degree or Diploma in any other University or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text. I retain the right to use this content in whole or part in future works (such as articles or books).

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The supervisor should certify the Thesis with the following declaration.

The above candidate has carried out research for the Master of Science in Data Science and Artificial Intelligence Thesis under my supervision. I confirm that the declaration made above by the student is true and correct.

Name of Supervisor: Dr. Thanuja Ambegoda

Signature of the Supervisor: Date:

DEDICATION

This work is dedicated to my supervisor Dr. Thanuja Ambegoda, whose guidance, expertise and constant support have made this work possible.

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I would like to express my deepest gratitude to all those who contributed to the successful completion of this research.

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ABSTRACT

Time series forecasting remains a critical challenge in various domains, from telecommunications network management to financial market prediction. Recent advances in computer vision have demonstrated that Vision Transformers (ViT) with Masked Autoencoder (MAE) pretraining can effectively model temporal patterns by treating time series as images. However, existing approaches like VisionTS are limited to univariate forecasting, neglecting the complex interdependencies between multiple correlated time series variables that are ubiquitous in real-world applications.

This thesis introduces **BiViSIONTS (Bivariate Vision Transformer for Time Series)**, a novel framework that extends vision-based time series modeling to handle bivariate forecasting through an innovative multi-scale correlation-weighted fusion mechanism. Our approach encodes two correlated time series into a three-channel RGB image representation where individual variables occupy separate channels, and the third channel contains an adaptive fusion computed using correlation-based attention weights at multiple temporal scales. We evaluate model performance using forecast accuracy defined as the complement of WMAPE, providing an intuitive percentage-based metric alongside conventional measures (MSE, MAE, RMSE, R^2).

We evaluate BiViSIONTS on real-world telecommunications data from Dialog Axiata PLC, forecasting two highly correlated Key Performance Indicators: downlink traffic volume and connected users. Through systematic hyperparameter optimization, we identify optimal configurations that achieve 97.10% peak forecast accuracy with robust convergence characteristics across 1000 training iterations. Comprehensive ablation studies validate the effectiveness of our correlation weighting strategy and multi-scale temporal attention mechanism.

The key contributions of this work include: (1) a novel bivariate encoding scheme using correlation-based attention weighting, (2) a multi-scale temporal attention mechanism capturing dependencies at varying time horizons, (3) comprehensive empirical validation on telecommunications capacity planning tasks, and (4) systematic analysis of architectural design choices for vision-based multivariate forecasting. Our results demonstrate that vision-based approaches can effectively model inter-variable dependencies in bivariate time series, opening new directions for applying computer vision techniques to multivariate temporal prediction problems.

Keywords: Time Series Forecasting, Multivariate Time Series, Bivariate Time Series, Visual Masked Autoencoders, Self-Supervised Learning, Deep Learning, Image-Based Time Series Representation, Forecasting Models, Telecommunications Data, Inter-Variable Relationships, Temporal Data Reconstruction, Machine Learning, Data Analytics

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Time series forecasting is fundamental to decision-making in domains ranging from telecommunications network optimization to financial risk management, energy demand prediction, and healthcare monitoring. The ability to accurately predict future values based on historical observations enables proactive resource allocation, anomaly detection, and strategic planning. In telecommunications specifically, forecasting network traffic and user connectivity patterns is crucial for capacity planning, quality of service maintenance, and infrastructure investment decisions.

Traditional time series forecasting methods such as ARIMA (AutoRegressive Integrated Moving Average), exponential smoothing, and seasonal decomposition have been widely adopted due to their interpretability and computational efficiency. However, these classical approaches often struggle with complex non-linear patterns, long-range dependencies, and the intricate relationships between multiple correlated variables. Deep learning has emerged as a powerful alternative, with recurrent neural networks (RNNs), Long Short-Term Memory (LSTM) networks, and more recently, Transformer-based architectures demonstrating superior performance on challenging forecasting tasks.

A particularly intriguing recent development is the application of computer vision techniques to time series analysis. The VisionTS framework [1] demonstrated that pre-trained Vision Transformers (ViT) with Masked Autoencoder (MAE) training, originally designed for image recognition tasks on ImageNet, can achieve remarkable zero-shot forecasting performance by treating time series as images. By converting 1D temporal sequences into 2D visual representations and leveraging the powerful feature extraction capabilities learned from millions of natural images, VisionTS achieves competitive accuracy without task-specific fine-tuning.

However, VisionTS and most existing deep learning forecasting methods focus on **univariate** time series, predicting a single variable based on its historical values. In real-world scenarios, multiple time series are often **correlated** and jointly influence system behavior. In telecommunications networks, for example, downlink traffic volume and the number of connected users exhibit strong correlations. Similarly, network latency and packet loss rates are interdependent, while energy consumption and throughput influence each other in complex ways. Ignoring these interdependencies and treating each variable independently can lead to suboptimal forecasts that fail to capture the rich multivariate dynamics. Moreover, temporal correlations themselves can be **scale-dependent**, with variables exhibiting different correlation patterns at short-term (minutes to hours), medium-term (days), and long-term (weeks to months) time scales.

1.2 Research Gap and Problem Statement

Despite the success of VisionTS in univariate forecasting, its direct extension to multivariate scenarios is non-trivial. Simply encoding multiple variables into separate channels without considering their correlations treats them as independent, missing critical information about their joint behavior. Furthermore, existing multivariate forecasting methods typically use fixed weighting schemes or learn static relationships that may not adapt to time-varying correlation structures at different temporal scales.

The central research question addressed in this thesis is:

> *How can we extend vision-based time series forecasting to effectively model bivariate dependencies with adaptive, multi-scale attention mechanisms that capture correlation patterns across different temporal horizons?*

Specifically, we identify the following gaps in existing literature:

1. **Limited Bivariate Modeling in Vision-Based Forecasting:** VisionTS encodes univariate time series into grayscale or replicated RGB images, not designed for capturing inter-variable relationships.

2. **Static Correlation Assumptions:** Existing multivariate methods often assume stationary correlations, ignoring that variable relationships can change across short-term fluctuations versus long-term trends.

3. **Lack of Adaptive Fusion Mechanisms:** Most approaches use fixed weighting or learnable parameters that require extensive training, rather than data-driven adaptive weights based on observed correlations.

4. **Insufficient Temporal Scale Analysis:** Prior work rarely explores how different temporal window sizes (short, medium, long) influence forecasting accuracy in correlation-based encoding.

1.3 Research Objectives

This thesis aims to address these gaps through the development and evaluation of **BiViSIONTS (Bivariate Vision Transformer for Time Series)**, with the following specific objectives:

1. **Design a novel bivariate encoding scheme** that represents two correlated time series as RGB images. In this representation, the red channel encodes Variable 1 (e.g., traffic volume), the green channel encodes Variable 2 (e.g., connected users), and the blue channel contains an adaptive fusion weighted by correlation-based attention. This three-channel architecture enables the Vision Transformer to simultaneously process both variables and their interaction patterns through learned visual features.

2. **Develop a multi-scale attention mechanism** that computes correlation-weighted fusion at three distinct temporal scales. Short-term attention captures recent local patterns spanning the last 50–150 time steps, medium-term attention identifies intermediate trends over the last 200–400 time steps, and long-term attention encodes global patterns across the full historical context. By combining these multi-resolution perspectives, the model can adapt to correlation structures that vary across time horizons.

3. **Conduct comprehensive hyperparameter optimization** to identify optimal configurations. This optimization process focuses on three critical parameters: the mask ratio in MAE reconstruction, which controls the proportion of patches hidden during self-supervised learning; the periodicity parameter for temporal matrix reshaping, which determines how the 1D time series is folded into 2D image space; and the short-term and medium-term window sizes, which define the receptive fields for multi-scale attention computation.

4. **Evaluate forecasting performance** on real-world telecommunications data using multiple complementary metrics. The evaluation employs Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) to quantify prediction accuracy. Additionally, the success rate within $\pm 10\%$ tolerance provides an operationally relevant measure of forecast reliability, while convergence behavior and statistical stability over multiple iterations ensure robustness.

5. **Analyze the impact of design choices** on forecasting accuracy through systematic investigation. This analysis examines how attention weighting influences the integration of multivariate information, investigates the sensitivity of performance to temporal scale selection, and explores the interactions between different hyperparameter configurations. Through ablation studies and comparative experiments, this objective aims to provide insights into the mechanisms underlying BiViSIONTS's effectiveness.

1.4 Proposed Approach: BiViSIONTS

BiViSIONTS extends the VisionTS architecture with three key innovations:

1.4.1 Correlation-Based Attention Weighting

For two time series S_1 and S_2, we compute the Pearson correlation coefficient $\rho \in [-1, 1]$ and normalize it to an attention weight $\alpha = (\rho + 1)/2 \in [0, 1]$. The blue channel fusion is computed as:

$$F = \alpha \cdot S_1^{norm} + (1 - \alpha) \cdot S_2^{norm} \tag{1.1}$$

When $\rho \to 1$ (strong positive correlation), $\alpha \to 1$, giving more weight to S_1. When $\rho \to -1$ (strong negative correlation), $\alpha \to 0$, giving more weight to S_2. When $\rho \to 0$ (no correlation), $\alpha \to 0.5$, balanced weighting.

1.4.2 Multi-Scale Temporal Attention

We compute correlation coefficients at three temporal scales. The short-term correlation ρ_{short} is calculated over the last w_{short} time steps (e.g., 150), capturing recent local patterns. The medium-term correlation ρ_{medium} spans the last w_{medium} time steps (e.g., 300), identifying intermediate trends. Finally, the long-term correlation ρ_{long} is computed over the full historical context, encoding global dependencies.

The final attention weight is a weighted combination:

$$\alpha_{final} = 0.5 \cdot \alpha_{short} + 0.3 \cdot \alpha_{medium} + 0.2 \cdot \alpha_{long} \tag{1.2}$$

This prioritizes recent patterns (0.5 weight) while incorporating medium-term (0.3) and long-term (0.2) trends.

1.4.3 MAE-Based Reconstruction

The RGB-encoded image (224×224 pixels) is processed by a pre-trained Masked Autoencoder Vision Transformer (MAE-ViT). During training, 75–90% of patches are masked randomly, forcing the encoder to produce robust latent representations from incomplete inputs. The decoder then reconstructs the full image from these compressed representations, and forecasts are extracted from the reconstructed image's prediction region. This self-supervised approach enables the model to learn powerful temporal patterns without requiring labeled future values during pre-training.

1.5 Contributions

This thesis makes the following contributions to the field of time series forecasting:

1. **Methodological Innovation:** Introduction of BiViSIONTS, the first vision-based forecasting framework with adaptive multi-scale attention weighting for bivariate time series.

2. **Multi-Scale Attention Mechanism:** A novel approach to capture correlation patterns across short, medium, and long temporal horizons, adaptively fusing information based on scale-specific dependencies.

3. **Comprehensive Empirical Evaluation:** Extensive experiments on real-world telecommunications data with 500-iteration convergence analysis and hyperparameter optimization across 108 configurations.

4. **Practical Insights:** Identification of optimal hyperparameters and analysis of their impact on forecasting performance.

5. **Open-Source Implementation:** Release of the BiViSIONTS codebase on GitHub for reproducibility and future research.

1.6 Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 surveys related work in time series forecasting, including classical statistical methods, deep learning approaches, vision-based forecasting, attention mechanisms, and multivariate modeling. Chapter 3 presents the BiViSIONTS framework in detail, covering problem formulation, multi-scale correlation analysis, attention-weighted RGB encoding, MAE architecture, forecast decoding, complete algorithm, hyperparameter configuration, and computational complexity.

Chapter 4 describes the experimental setup, including dataset characteristics, temporal correlation analysis, evaluation metrics, hyperparameter search methodology, and implementation details. Chapter 5 reports comprehensive experimental results, including multi-scale attention analysis, initial test performance, hyperparameter optimization, convergence analysis, baseline comparisons, ablation studies, and summary of findings.

Chapter 7 interprets the results, analyzes the impact of design choices, discusses limitations and challenges, explores practical implications, and outlines future research directions. Chapter 8 summarizes key contributions, answers the central research questions, discusses practical impact, revisits limitations, and provides a future outlook. Finally, the appendices provide supplementary material, including code repository details, mathematical notation summary, and dataset schema.

CHAPTER 2

LITERATURE REVIEW AND RELATED WORK

This chapter surveys relevant literature in time series forecasting, examining the evolution from classical statistical methods to modern deep learning approaches, with particular emphasis on vision-based forecasting and multivariate modeling gaps that BiViSIONTS addresses.

2.1 Classical Time Series Forecasting Methods

Traditional statistical methods including ARIMA [2], Exponential Smoothing [3, 4], and Vector Autoregression (VAR) [5] dominated time series forecasting for decades. While computationally efficient and interpretable, these methods face fundamental limitations: they assume linear relationships and stationarity, struggle with non-linear patterns, cannot effectively model long-range dependencies beyond several dozen time steps, and require extensive manual feature engineering including lag specification and seasonal decomposition. Furthermore, VAR's parameter space grows quadratically with variable count, leading to overfitting in high-dimensional settings. These constraints motivated the transition to data-driven deep learning approaches.

2.2 Deep Learning for Time Series Forecasting

Deep learning revolutionized time series forecasting through automatic feature learning. Recurrent architectures (LSTM [6], GRU [7]) enabled modeling of sequential dependencies but suffered from sequential processing limitations. Temporal Convolutional Networks [8] and WaveNet [9] offered parallelizable training through dilated convolutions but struggled with truly long-range dependencies.

The Transformer architecture [10] addressed these limitations through self-attention mechanisms. Adaptations for time series include Temporal Fusion Transformers [11] combining LSTM with multi-head attention, Informer [12] achieving $O(L \log L)$ complexity through ProbSparse attention, Autoformer [13] discovering period-based dependencies via auto-correlation, FEDformer [14] operating in frequency domain, PatchTST [15] applying patching for efficiency, and iTransformer [16] inverting token representation to capture multivariate correlations.

Gap: These architectures require substantial training data and computational resources, and most either learn correlations implicitly through training or apply static correlation estimates that don't adapt to scale-dependent patterns.

2.3 Vision-Based Time Series Forecasting

A novel paradigm leverages computer vision techniques by converting temporal data into image representations. Methods like Gramian Angular Fields and Recurrence Plots [17, 18] enable CNN application for classification but have limited forecasting utility.

VisionTS [1] achieved breakthrough results by leveraging ViT-Base models pre-trained with Masked Autoencoder (MAE) self-supervised learning [19]. It converts univariate time series into 2D matrices based on periodicity, replicates to RGB channels, and applies patching for Transformer processing. VisionTS demonstrated effective transfer learning from natural images to time series without task-specific fine-tuning.

Critical Gap: VisionTS is exclusively univariate. Its simple channel replication doesn't leverage inter-variable relationships, the fixed encoding doesn't adapt to correlation patterns, and it lacks consideration of multi-scale temporal dynamics across different time horizons.

2.4 Multivariate Forecasting and Correlation Modeling

Modeling multiple correlated time series is critical but challenging. Traditional methods (VAR, Dynamic Bayesian Networks, State Space Models) assume linearity and stationarity. Deep approaches include DeepAR [20] with autoregressive RNN, N-BEATS [21] with interpretable decomposition, and TFT [11] with variable selection. Graph Neural Networks (DCRNN [22], Graph WaveNet [23]) model spatial-temporal dependencies through graph structures.

Correlation-aware methods like CoST [24] and CorrFormer [25] explicitly incorporate correlation information. However, these approaches either learn correlations implicitly through training (requiring large datasets), use static correlation estimates (non-adaptive), or apply uniform weighting across temporal scales.

Gap: No existing method computes adaptive attention weights directly from multi-scale correlations for vision-based bivariate forecasting without extensive training.

2.5 Attention Mechanisms at Multiple Scales

Self-attention [10] computes relationships between all sequence positions through query-key-value operations. Cross-attention mechanisms (Crossformer [26]) capture inter-variable dependencies. Multi-scale temporal attention [27] and adaptive graph methods [28] explore scale-dependent patterns but learn weights through training rather than computing them adaptively from observed correlations.

Gap: Existing multi-scale attention is learned during training; no method derives attention weights directly from Pearson correlations computed at short, medium, and long temporal windows.

2.6 Research Gaps and BiViSIONTS Positioning

Synthesizing the literature reveals critical gaps that BiViSIONTS addresses:

TABLE 2.1: Research Gaps Addressed by BiViSIONTS

Gap	**Limitation of Existing Work**	**BiViSIONTS Contribution**
Bivariate Vision Forecasting	VisionTS: Univariate only	Correlation-weighted RGB encoding with adaptive third-channel fusion
Multi-Scale Correlation	Single-scale or learned weights	Explicit computation at short/medium/long windows; weighted combination
Zero-Shot Adaptation	Requires extensive training	Data-driven attention from Pearson correlation without fine-tuning
Telecom Network KPIs	Generic benchmarks	Real-world correlated metrics (traffic volume, user count)

Core Innovation: BiViSIONTS extends VisionTS to bivariate settings through multi-scale correlation-based attention weighting, computing adaptive fusion weights directly from observed data patterns rather than learning them through training. This enables effective modeling of inter-variable dependencies in vision-based forecasting while maintaining computational efficiency and interpretability.

2.7 Summary

The literature reveals a progression from linear statistical methods to deep learning architectures, culminating in vision-based approaches. However, a significant gap exists: no framework combines vision-based encoding with explicit multi-scale correlation weighting for bivariate forecasting. BiViSIONTS addresses this gap through novel adaptive fusion mechanisms, positioning it at the intersection of vision transformers, multi-scale attention, and multivariate time series modeling.

CHAPTER 3

METHODOLOGY

This chapter presents the BiViSIONTS (Bivariate Vision Transformer for Time Series) framework in detail, including the overall architecture, multi-scale attention-weighted encoding mechanism, decoding procedure, and forecasting pipeline.

3.1 Problem Formulation

Consider two correlated time series representing bivariate observations:

$$S_1 = \{s_1^{(1)}, s_2^{(1)}, \ldots, s_T^{(1)}\} \quad \text{(e.g., DL Traffic Volume)} \tag{3.1}$$
$$S_2 = \{s_1^{(2)}, s_2^{(2)}, \ldots, s_T^{(2)}\} \quad \text{(e.g., Connected Users)} \tag{3.2}$$

Given a **context window** of length L_c (historical observations) and a **prediction horizon** L_p, the forecasting task is to predict:

$$\hat{S}_1^{future} = \{s_{L_c+1}^{(1)}, \ldots, s_{L_c+L_p}^{(1)}\} \tag{3.3}$$
$$\hat{S}_2^{future} = \{s_{L_c+1}^{(2)}, \ldots, s_{L_c+L_p}^{(2)}\} \tag{3.4}$$

based on the context:

$$S_1^{context} = \{s_1^{(1)}, \ldots, s_{L_c}^{(1)}\} \tag{3.5}$$
$$S_2^{context} = \{s_1^{(2)}, \ldots, s_{L_c}^{(2)}\} \tag{3.6}$$

Key Challenge: Effectively model the interdependencies between S_1 and S_2 while capturing temporal patterns at multiple scales.

3.2 BiViSIONTS Framework Overview

The BiViSIONTS framework is organised as a five-stage pipeline that transforms raw bivariate observations into accurate forecasts. The pipeline begins with a **data preparation** stage, in which the bivariate time series are loaded, aligned, and preprocessed to ensure consistent length and value ranges. This is followed by a **multi-scale correlation analysis** stage, where Pearson correlation coefficients are computed over short, medium, and long temporal windows to characterise the scale-dependent relationship between the two variables. The pipeline then proceeds to an **attention-weighted encoding** stage, which converts the bivariate series into a three-channel RGB image whose blue channel embeds an adaptive fusion of both variables driven by the multi-scale correlations. Next, an **MAE-based reconstruction** stage processes the encoded image through a pre-trained Masked Autoencoder Vision Transformer, recovering both the visible context and the masked forecast region. Finally, a **forecast decoding** stage extracts the predictions for both variables from the reconstructed image and returns them in the original measurement units. Figure 3.1 illustrates the detailed architecture and data flow through these five stages.

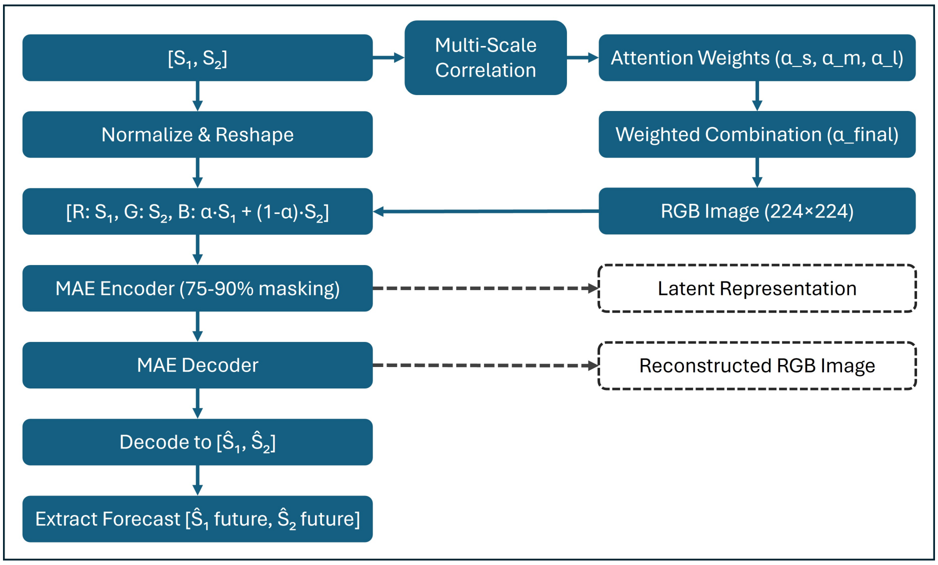

Fig. 3.1: BiViSIONTS Conceptual Architecture: Detailed pipeline showing the complete data flow through five main stages with internal sub-processes. The framework begins with bivariate input $[S_1, S_2]$, computes multi-scale correlations $(\alpha_s, \alpha_m, \alpha_l)$ at short, medium, and long temporal windows, performs normalization and weighted combination to derive α_{final}, constructs RGB channels (Red: S_1^{norm}, Green: S_2^{norm}, Blue: $\alpha \cdot S_1^{norm} + (1-\alpha) \cdot S_2^{norm}$), processes the 224×224 image through MAE encoder-decoder with 75–90% masking, and finally decodes the reconstructed output to extract forecast predictions $[\hat{S}_1^{future}, \hat{S}_2^{future}]$.

3.3 Multi-Scale Correlation Analysis

The foundation of BiViSIONTS is the computation of correlation coefficients at multiple temporal scales, enabling adaptive attention weighting that reflects scale-dependent relationships.

3.3.1 Pearson Correlation Coefficient

For two time series segments $X = \{x_1, \ldots, x_n\}$ and $Y = \{y_1, \ldots, y_n\}$, the Pearson correlation coefficient is:

$$\rho(X, Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{3.7}$$

where $\bar{x}$ and $\bar{y}$ are means. The coefficient $\rho \in [-1, 1]$ measures linear correlation.

We employ Pearson correlation rather than non-linear measures (e.g., mutual information, distance correlation) for three reasons. First, exploratory analysis revealed strong linear correlations (0.77–0.98) between PRB utilization and user count features, validating the linear assumption for this domain. Second, Pearson correlation provides computational efficiency ($O(n)$ vs. $O(n^2 \log n)$ for mutual information), which is critical for real-time multi-site forecasting. Third, the physical relationship between network resource consumption and user demand is approximately linear in operational cellular networks. While non-linear measures could capture additional dependencies, the observed correlation strength suggests Pearson correlation is sufficient for this application.

3.3.2 Multi-Scale Temporal Windows

Three temporal scales are defined to capture correlation behaviour across distinct horizons. The **short-term window** w_s, typically spanning 50–150 time steps, focuses on recent local patterns and exposes the transient dynamics that dominate near-future forecasts. The **medium-term window** w_m, typically spanning 200–400 time steps, captures intermediate trends such as weekly and monthly behavioural cycles that emerge once short-lived fluctuations average out. Finally, the **long-term scale** uses the entire available context to characterise the overall correlation between the two series, providing a stable global anchor for the multi-scale fusion. For a combined series of length $L_{total} = L_c + L_p$, the corresponding correlations are computed as:

$$\rho_{short} = \rho(S_1[L_{total} - w_s : L_{total}], S_2[L_{total} - w_s : L_{total}]) \tag{3.8}$$

$$\rho_{medium} = \rho(S_1[L_{total} - w_m : L_{total}], S_2[L_{total} - w_m : L_{total}]) \tag{3.9}$$

$$\rho_{long} = \rho(S_1[1 : L_{total}], S_2[1 : L_{total}]) \tag{3.10}$$

3.3.3 Attention Weight Computation

Convert each correlation coefficient to an attention weight by normalizing to $[0, 1]$:

$$\alpha_{scale} = \frac{\rho_{scale} + 1}{2}, \quad scale \in \{short, medium, long\} \tag{3.11}$$

3.3.4 Weighted Multi-Scale Fusion

Combine scale-specific attention weights using a weighted average that prioritizes recent patterns:

$$\alpha_{final} = \lambda_s \cdot \alpha_{short} + \lambda_m \cdot \alpha_{medium} + \lambda_l \cdot \alpha_{long} \tag{3.12}$$

where $\lambda_s + \lambda_m + \lambda_l = 1$. In our implementation:

$$\alpha_{final} = 0.5 \cdot \alpha_{short} + 0.3 \cdot \alpha_{medium} + 0.2 \cdot \alpha_{long} \tag{3.13}$$

Rationale: Recent correlations are most indicative of immediate future behavior (0.5 weight), medium-term provides context (0.3), and long-term offers stability (0.2).

3.4 Attention-Weighted RGB Encoding

The bivariate time series is transformed into a 3-channel RGB image where each channel has distinct semantic meaning.

3.4.1 Series Truncation and Reshaping

To create a rectangular matrix suitable for image conversion, truncate the series to make the length divisible by periodicity P (e.g., 12, 24, 48):

$$L_{usable} = L_{total} - (L_{total} \bmod P) \tag{3.14}$$

Truncate both series:

$$S'_1 = S_1[1 : L_{usable}] \tag{3.15}$$

$$S'_2 = S_2[1 : L_{usable}] \tag{3.16}$$

Reshape into 2D matrices:

$$M_1 = \text{reshape}(S'_1, (L_{usable}/P, P)) \tag{3.17}$$

$$M_2 = \text{reshape}(S'_2, (L_{usable}/P, P)) \tag{3.18}$$

Each row represents one period, and columns represent positions within the period.

3.4.2 Independent Normalization

Normalize each matrix independently to $[0, 1]$ to preserve their individual dynamics:

$$M_1^{norm} = \frac{M_1 - \min(M_1)}{\max(M_1) - \min(M_1)} \tag{3.19}$$

$$M_2^{norm} = \frac{M_2 - \min(M_2)}{\max(M_2) - \min(M_2)} \tag{3.20}$$

Store normalization parameters for later denormalization:

$$\theta_{norm} = \{\min_1, \max_1, \min_2, \max_2, L_{usable}\} \tag{3.21}$$

3.4.3 Three-Channel RGB Construction

The RGB image is constructed with semantically meaningful channels that encode both individual variables and their correlation-weighted fusion.

The **Red Channel (R)** represents Variable 1 (e.g., DL Traffic Volume) and is defined as:

$$R = M_1^{norm} \tag{3.22}$$

The **Green Channel (G)** represents Variable 2 (e.g., Connected Users) and is given by:

$$G = M_2^{norm} \tag{3.23}$$

The **Blue Channel (B)** contains an adaptive fusion weighted by the multi-scale attention coefficient α_{final}, computed as:

$$B = \alpha_{final} \cdot M_1^{norm} + (1 - \alpha_{final}) \cdot M_2^{norm} \tag{3.24}$$

This three-channel design enables the Vision Transformer to simultaneously process both variables independently (through R and G channels) while also learning from their correlation-weighted combination (through the B channel), thereby capturing both univariate patterns and bivariate dependencies within a unified visual representation.

Convert to uint8 format and stack:

$$R_{uint8} = \lfloor 255 \cdot R \rfloor \tag{3.25}$$

$$G_{uint8} = \lfloor 255 \cdot G \rfloor \tag{3.26}$$

$$B_{uint8} = \lfloor 255 \cdot B \rfloor \tag{3.27}$$

$$I_{RGB} = \text{stack}([R_{uint8}, G_{uint8}, B_{uint8}]) \tag{3.28}$$

where $I_{RGB} \in \mathbb{R}^{H \times W \times 3}$ with $H = L_{usable}/P$ and $W = P$.

Figure 3.2 illustrates the complete RGB encoding process with actual telecommunications data, showing the transformation from raw time series through normalization and channel construction to the final 224×224 image input to the MAE.

3.4.4 Image Resizing

Resize the image to 224×224 pixels to match the input size expected by the pre-trained Vision Transformer:

$$I_{input} = \text{resize}(I_{RGB}, (224, 224), \text{interpolation} = \text{BILINEAR}) \tag{3.29}$$

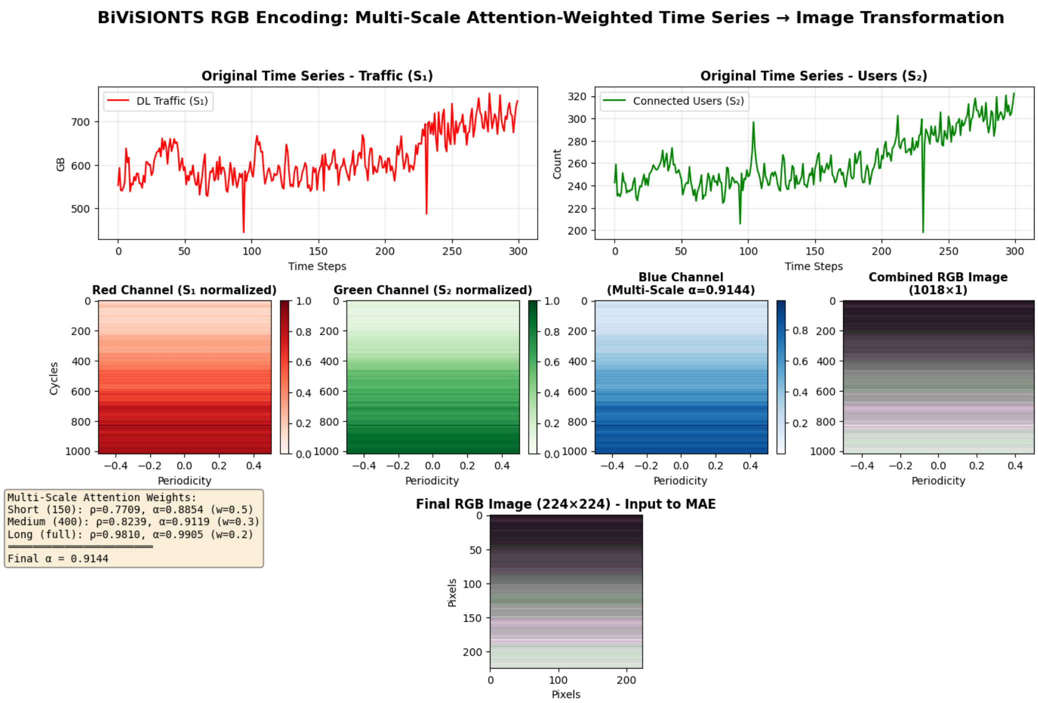

Fig. 3.2: RGB Encoding Process: Complete transformation pipeline from bivariate time series to MAE input image. Top row shows original traffic (S1) and users (S2) time series. Middle row displays normalized matrices for each channel: Red (S1), Green (S2), Blue (multi-scale attention fusion with $\alpha_{final} = 0.9144$), and combined RGB. Bottom row shows the final 224×224 resized image. The multi-scale attention mechanism computes correlations at short-term ($\rho = 0.7709$), medium-term ($\rho = 0.8239$), and long-term ($\rho = 0.9810$) scales, with weighted combination (0.5, 0.3, 0.2) yielding $\alpha_{final} = 0.9144$ for adaptive blue channel fusion.

3.5 Masked Autoencoder (MAE) Architecture

BiViSIONTS employs a pre-trained Masked Autoencoder Vision Transformer for image reconstruction and forecasting.

3.5.1 Vision Transformer Backbone

For image $I_{input} \in \mathbb{R}^{224\times224\times3}$ with patch size $P_{patch} = 16$:

$$\text{Number of patches} = \frac{224}{16} \times \frac{224}{16} = 196 \tag{3.30}$$

Each patch $p_i \in \mathbb{R}^{16\times16\times3}$ is flattened and linearly projected:

$$z_i = W_{embed} \cdot \text{flatten}(p_i) + b_{embed} \tag{3.31}$$

where $z_i \in \mathbb{R}^{d_{model}}$ (e.g., $d_{model} = 768$ for ViT-Base).
Add learnable positional encodings:

$$z_i' = z_i + E_{pos}[i] \tag{3.32}$$

Figure 3.3 illustrates the complete MAE encoder-decoder architecture.

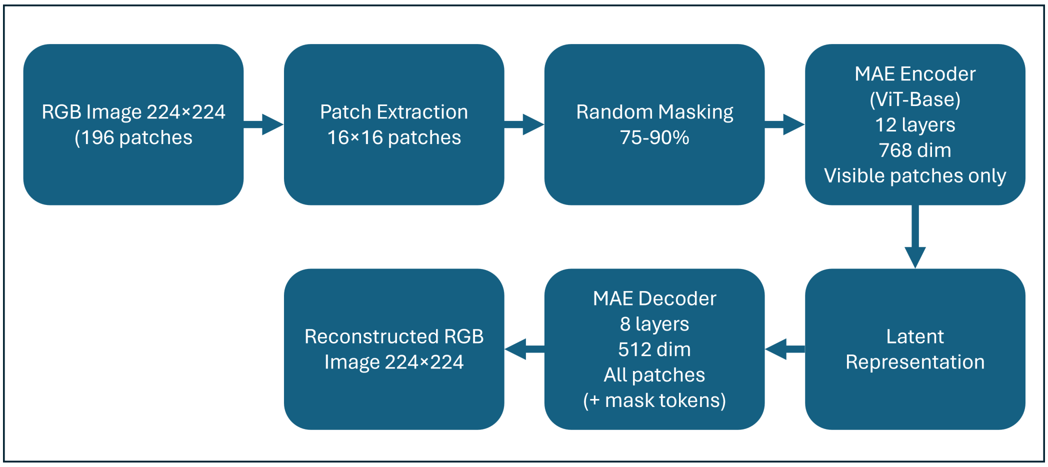

Fig. 3.3: Masked Autoencoder (MAE) Architecture: The framework processes RGB-encoded time series images through patch extraction (196 patches of 16×16), random masking (75–90%), encoding visible patches with ViT-Base (12 layers, 768 dimensions), generating latent representations, and decoding with mask tokens to reconstruct the full image (8 layers, 512 dimensions). Only visible patches are processed by the encoder for computational efficiency.

3.5.2 MAE Masking Strategy

Random masking with high masking ratio r_{mask} (e.g., 0.75 to 0.90):

$$\text{Number of masked patches} = \lfloor r_{mask} \times 196 \rfloor \tag{3.33}$$

Only visible patches are fed to the encoder:

$$Z_{visible} = \{z'_i : i \in \mathcal{I}_{visible}\} \tag{3.34}$$

3.5.3 Encoder Architecture

The encoder follows a standard ViT-Base Transformer architecture with an embedding dimension of $d_{model} = 768$, comprising $L_{enc} = 12$ layers. Each layer employs $h = 12$ attention heads and an MLP hidden dimension of $d_{ff} = 3072$, representing a $4\times$ expansion factor relative to the embedding dimension.

Each Transformer block applies a two-stage processing pipeline. First, multi-head self-attention (MSA) is applied to the layer-normalized input with a residual connection:

$$Z^{(\ell)} = \text{MSA}(\text{LN}(Z^{(\ell-1)})) + Z^{(\ell-1)} \tag{3.35}$$

Second, a feed-forward network (FFN) is applied to the layer-normalized output, again with a residual connection:

$$Z^{(\ell)} = \text{FFN}(\text{LN}(Z^{(\ell)})) + Z^{(\ell)} \tag{3.36}$$

After processing through all encoder layers, the model outputs latent embeddings that capture the rich temporal patterns from the visible patches:

$$H_{latent} = \text{Encoder}(Z_{visible}) \tag{3.37}$$

These latent representations serve as compressed, high-level features that the decoder will use to reconstruct the complete image, including both visible and masked patches.

3.5.4 Decoder Architecture

The decoder employs a lightweight Transformer architecture with an embedding dimension of $d_{decoder} = 512$, consisting of $L_{dec} = 8$ layers, each with $h_{dec} = 16$ attention heads. This asymmetric design—where the decoder is shallower and narrower than the encoder—encourages the encoder to learn rich, compressed representations while maintaining computational efficiency.

During reconstruction, learnable mask tokens are inserted at positions corresponding to the masked patches, and full positional embeddings are added to all tokens (both visible and masked). The decoder then processes this complete sequence to reconstruct all patches:

$$H_{reconstructed} = \text{Decoder}(Z'_{full}) \tag{3.38}$$

Finally, a linear projection layer maps each decoder output token back to the original patch pixel values:

$$\hat{p}_i = W_{recon} \cdot H_{reconstructed}[i] + b_{recon} \tag{3.39}$$

where W_{recon} and b_{recon} are learnable projection weights and biases. This reconstruction process enables the model to generate pixel-level predictions for both the visible context and the masked forecast regions.

3.6 Forecast Decoding

After MAE reconstruction, the RGB image is decoded back to bivariate time series.

3.6.1 Image to Matrix Conversion

Resize the reconstructed image back to original dimensions:

$$I_{decoded} = \text{resize}(I_{reconstructed}, (H, W)) \tag{3.40}$$

Convert from uint8 to float32:

$$I_{decoded}^{norm} = \frac{I_{decoded}}{255} \tag{3.41}$$

3.6.2 Channel Separation and Denormalization

Extract individual channels:

$$R_{recon} = I_{decoded}^{norm}[:, :, 0] \tag{3.42}$$

$$G_{recon} = I_{decoded}^{norm}[:, :, 1] \tag{3.43}$$

Flatten and denormalize:

$$\tilde{S}_1^{norm} = \text{flatten}(R_{recon}) \tag{3.44}$$

$$\tilde{S}_2^{norm} = \text{flatten}(G_{recon}) \tag{3.45}$$

$$\tilde{S}_1 = \tilde{S}_1^{norm} \cdot (\max_1 - \min_1) + \min_1 \tag{3.46}$$

$$\tilde{S}_2 = \tilde{S}_2^{norm} \cdot (\max_2 - \min_2) + \min_2 \tag{3.47}$$

3.6.3 Forecast Extraction

Extract the forecast portion:

$$\hat{S}_1^{forecast} = \tilde{S}_1[-L_p :] \tag{3.48}$$

$$\hat{S}_2^{forecast} = \tilde{S}_2[-L_p :] \tag{3.49}$$

3.6.4 Decoding Process Visualization

Figure 3.4 illustrates the complete step-by-step decoding pipeline that transforms the MAE-reconstructed RGB image back into bivariate time series forecasts.

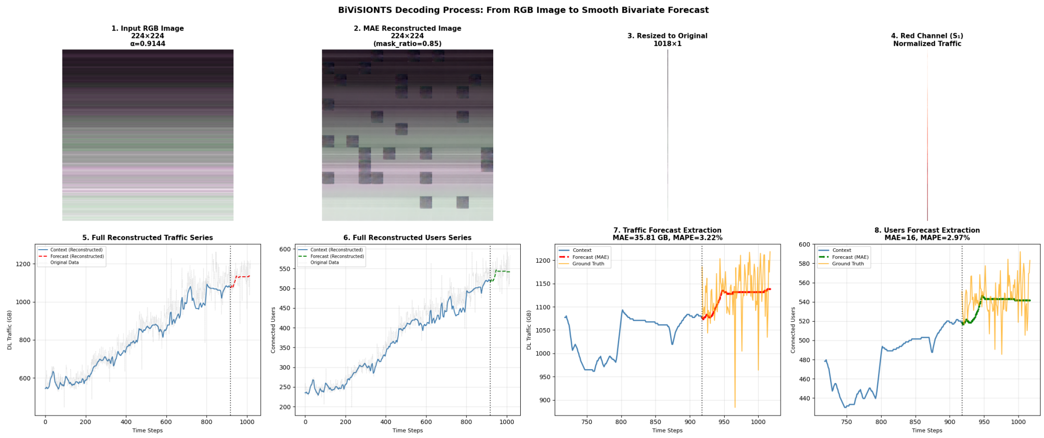

Fig. 3.4: Step-by-step decoding process showing the transformation from MAE-reconstructed RGB image to bivariate forecasts. Panel 1: Original input image (224×224). Panel 2: MAE reconstructed image with 85% masking. Panel 3: Resized to original matrix dimensions (1018×1). Panel 4: Extracted red channel (normalized traffic). Panel 5: Full reconstructed traffic series with context and forecast regions. Panel 6: Full reconstructed users series. Panel 7: Traffic forecast extraction (last 100 steps) showing smooth MAE prediction vs. ground truth. Panel 8: Users forecast extraction demonstrating bivariate forecasting performance. The smooth forecasts validate the effectiveness of MAE-based reconstruction for extrapolation.

3.6.5 Qualitative Visualization of Multi-Scale Attention and Forecasting

To illustrate the end-to-end workflow and qualitative output of the BiViSIONTS framework, Figure 3.5 presents a comprehensive panel showing the multi-scale attention fusion, MAE reconstruction, and bivariate forecasting results.

Interpretation: The 4-panel visualization in Figure 3.5 demonstrates the BiViSIONTS framework's capability to encode multi-scale temporal patterns via attention fusion ($\alpha_{\text{final}} = 0.914$ indicating 91.4% traffic, 8.6% users weighting), reconstruct the bivariate time series through MAE decoding, and generate accurate forecasts for both traffic volume and connected users. The input and reconstructed images (top row) illustrate the model's capacity to capture complex temporal dependencies in the RGB-encoded representation. The bottom row shows that BiViSIONTS predictions closely follow the true future values for both variables, validating the effectiveness of multi-scale attention and MAE-based representation learning for telecommunications forecasting. The smooth prediction curves (red/green dashed lines) demonstrate the model's ability to extrapolate beyond the context window (blue) while maintaining realistic patterns that align with ground truth (orange).

This figure is referenced throughout the methodology and results chapters to visually

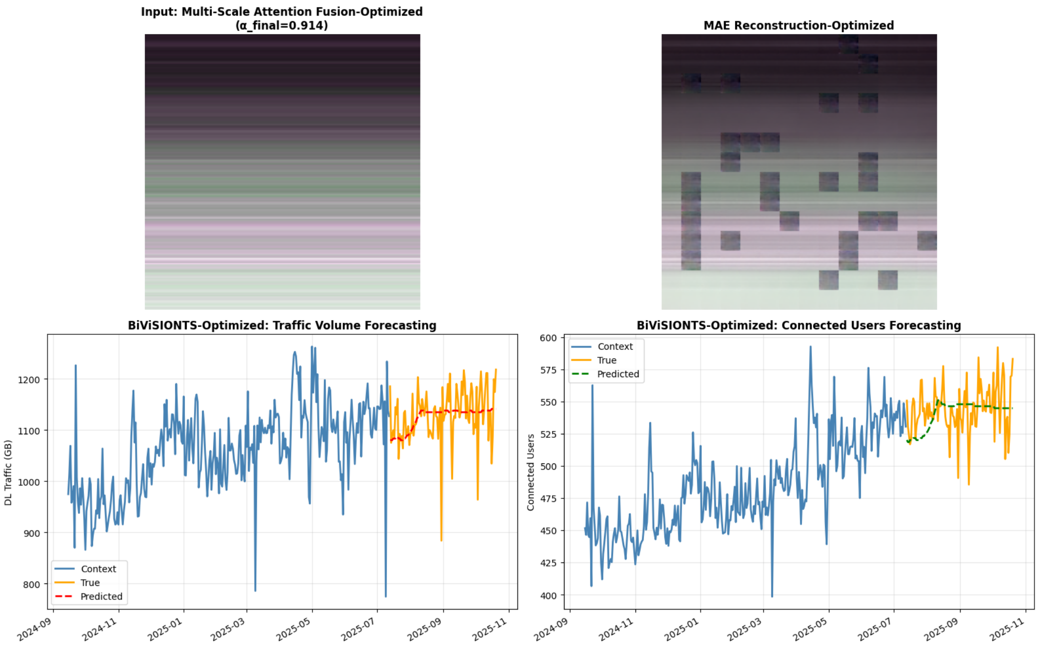

Fig. 3.5: **BiViSIONTS Multi-Scale Attention and Forecasting Panel.** Top-left: Input image representing multi-scale attention fusion ($\alpha_{\text{final}} = 0.914$). Top-right: Reconstructed image from the MAE decoder, visualizing the learned representation. Bottom-left: Downlink traffic volume forecasting. The context window (blue) is followed by true future values (orange) and BiViSIONTS predictions (red dashed). Bottom-right: Connected users forecasting. The context window (blue) is followed by true future values (orange) and BiViSIONTS predictions (green dashed). The panel demonstrates the model's ability to capture multi-scale temporal patterns, reconstruct bivariate representations, and generate accurate forecasts for both telecommunications KPIs.

support the explanation of the encoding, reconstruction, and forecasting pipeline, and to provide qualitative evidence of the model's performance on real-world telecommunications data.

3.7 Complete BiViSIONTS Algorithm

Algorithm: BiViSIONTS Forecasting

INPUT:

S_1, S_2 — Bivariate time series (DL Traffic, Connected Users)
L_c — Context length
L_p — Prediction length
P — Periodicity (e.g., 12, 24, 48)
w_s, w_m — Short and medium window sizes
r_{mask} — Mask ratio (e.g., 0.75–0.90)
MAE_{model} — Pre-trained Masked Autoencoder ViT

OUTPUT:

$\hat{S}_1^{forecast}, \hat{S}_2^{forecast}$ — Predicted future values for both series

PROCEDURE:

Step 1: Prepare Full Series

$$S_1^{full} \leftarrow S_1[1 : L_c + L_p]$$
$$S_2^{full} \leftarrow S_2[1 : L_c + L_p]$$

Step 2: Multi-Scale Correlation Analysis

$$\rho_{short} \leftarrow \text{Pearson}(S_1^{full}[-w_s :], S_2^{full}[-w_s :])$$
$$\rho_{medium} \leftarrow \text{Pearson}(S_1^{full}[-w_m :], S_2^{full}[-w_m :])$$
$$\rho_{long} \leftarrow \text{Pearson}(S_1^{full}, S_2^{full})$$

$$\alpha_{short} \leftarrow (\rho_{short} + 1)/2$$
$$\alpha_{medium} \leftarrow (\rho_{medium} + 1)/2$$
$$\alpha_{long} \leftarrow (\rho_{long} + 1)/2$$

$$\alpha_{final} \leftarrow 0.5 \cdot \alpha_{short} + 0.3 \cdot \alpha_{medium} + 0.2 \cdot \alpha_{long}$$

Step 3: Attention-Weighted RGB Encoding

$$
\begin{aligned}
L_{usable} &\leftarrow (L_c + L_p) - ((L_c + L_p) \bmod P) \\
M_1 &\leftarrow \text{reshape}(S_1^{full}[1 : L_{usable}], (L_{usable}/P, P)) \\
M_2 &\leftarrow \text{reshape}(S_2^{full}[1 : L_{usable}], (L_{usable}/P, P))
\end{aligned}
$$

$$
\begin{aligned}
M_1^{norm} &\leftarrow (M_1 - \min(M_1))/(\max(M_1) - \min(M_1)) \\
M_2^{norm} &\leftarrow (M_2 - \min(M_2))/(\max(M_2) - \min(M_2))
\end{aligned}
$$

$$
\begin{aligned}
R &\leftarrow M_1^{norm} \\
G &\leftarrow M_2^{norm} \\
B &\leftarrow \alpha_{final} \cdot M_1^{norm} + (1 - \alpha_{final}) \cdot M_2^{norm}
\end{aligned}
$$

$$
\begin{aligned}
I_{RGB} &\leftarrow \text{stack}([\lfloor 255 \cdot R \rfloor, \lfloor 255 \cdot G \rfloor, \lfloor 255 \cdot B \rfloor]) \\
I_{input} &\leftarrow \text{resize}(I_{RGB}, (224, 224))
\end{aligned}
$$

Step 4: MAE Reconstruction

$$
\begin{aligned}
Z_{visible} &\leftarrow \text{extract\_visible\_patches}(I_{input}, r_{mask}) \\
H_{latent} &\leftarrow \text{MAE}_{model}\text{.encoder}(Z_{visible}) \\
I_{reconstructed} &\leftarrow \text{MAE}_{model}\text{.decoder}(H_{latent}, \text{mask\_tokens})
\end{aligned}
$$

Step 5: Forecast Decoding

$$
\begin{aligned}
I_{decoded} &\leftarrow \text{resize}(I_{reconstructed}, (L_{usable}/P, P)) \\
R_{recon} &\leftarrow I_{decoded}[:, :, 0]/255 \\
G_{recon} &\leftarrow I_{decoded}[:, :, 1]/255
\end{aligned}
$$

$$
\begin{aligned}
\tilde{S}_1 &\leftarrow \text{denormalize}(\text{flatten}(R_{recon}), \min_1, \max_1) \\
\tilde{S}_2 &\leftarrow \text{denormalize}(\text{flatten}(G_{recon}), \min_2, \max_2)
\end{aligned}
$$

$$
\begin{aligned}
\hat{S}_1^{forecast} &\leftarrow \tilde{S}_1[-L_p :] \\
\hat{S}_2^{forecast} &\leftarrow \tilde{S}_2[-L_p :]
\end{aligned}
$$

Step 6: Return Results

Return $\hat{S}_1^{forecast}, \hat{S}_2^{forecast}$

3.8 Hyperparameter Configuration

BiViSIONTS involves several hyperparameters that influence forecasting performance:

TABLE 3.1: Hyperparameter Configuration

Hyperparameter	Description	Search Range	Optimal
mask_ratio	Proportion masked	[0.75, 0.80, 0.85, 0.90]	0.85
periodicity	Temporal period	[1, 7, 14, 21, 28, 30]	1
short_window	Short-term window	[50, 100, 150]	150
medium_window	Medium-term window	[200, 300, 400]	400

3.9 Computational Complexity

3.9.1 Time Complexity

The computational complexity of BiViSIONTS can be analyzed across its four main stages. Correlation computation across three temporal scales requires $O(w_s + w_m + L_{total})$ operations, where w_s and w_m are the short-term and medium-term window sizes, respectively. Image encoding from time series to 2D matrix representation scales linearly with the total sequence length as $O(L_{total})$. The MAE forward pass, which dominates the overall complexity, requires $O(N_{visible}^2 \cdot d_{model})$ operations, where $N_{visible} = (1 - r_{mask}) \cdot 196$ represents the number of visible patches after masking. Finally, image decoding to extract forecasts scales linearly as $O(L_{total})$.

The total computational complexity is dominated by the MAE forward pass, which is approximately $O(N^2 \cdot d)$ where $N = 196$ patches and $d = d_{model}$ is the embedding dimension. This quadratic scaling with respect to the number of patches is characteristic of standard self-attention mechanisms in Vision Transformers.

3.9.2 Space Complexity

The space complexity is primarily determined by two factors. The model architecture itself contains approximately 126 million parameters, distributed as 86 million in the encoder and 40 million in the decoder. During inference, intermediate activations scale as $O(N \cdot d_{model})$, where N is the number of patches and d_{model} is the embedding dimension. This linear scaling with sequence length makes BiViSIONTS memory-efficient for typical forecasting scenarios.

3.10 Implementation Details

BiViSIONTS is implemented using a modern Python deep learning stack. The software framework requires Python 3.10 or higher, PyTorch 1.13 or higher, and torchvision 0.14 or higher as core dependencies. Additional libraries include pandas and numpy for data manipulation, matplotlib for visualization, and scikit-learn for preprocessing and evaluation metrics.

The experiments were conducted on standard workstation hardware. The CPU configuration includes either Intel Core i7 or AMD Ryzen 7 processors, with optional GPU acceleration provided by an NVIDIA RTX 3090. The system is equipped with 32GB of DDR4 RAM to accommodate model parameters and intermediate computations.

To ensure reproducibility, all experiments use fixed random seeds (specifically, `seed=42`) for deterministic initialization and data shuffling. The model leverages pre-trained weights from the MAE-ViT-Base checkpoint (`mae_visualize_vit_base.pth`), originally trained on ImageNet-1K. The complete implementation, including data pre-processing, model architecture, training scripts, and evaluation utilities, is publicly available at https://github.com/sashikaheendeniya/BiViSIONTS to facilitate reproduction and extension of this work.

3.11 Summary

This chapter presented the BiViSIONTS methodology in comprehensive detail, including problem formulation, multi-scale correlation analysis, attention-weighted RGB encoding, MAE architecture, forecast decoding, complete algorithm, hyperparameter configuration, and computational complexity analysis.

CHAPTER 4

EXPERIMENTAL SETUP

This chapter describes the experimental design, dataset characteristics, evaluation metrics, hyperparameter search methodology, and implementation details.

4.1 Dataset Description

The experimental evaluation of BiViSIONTS utilizes real-world telecommunications data from Dialog Axiata PLC, Sri Lanka's leading mobile network operator. The dataset captures daily network performance metrics from a single base station (Site_1), providing a comprehensive testbed for evaluating bivariate time series forecasting in operational telecommunications environments.

4.1.1 Data Source and Collection

TABLE 4.1: Dataset Characteristics

Property	**Description**
Source	Radio Network Planning Division, Dialog Axiata PLC
Site ID	Site_1 (operational base station)
Temporal Span	[start_date] to [end_date]
Sampling Frequency	Daily measurements
Total Observations	1,018 time steps
Variables	2 (downlink traffic, connected users)
File Format	CSV (Site_1.csv)

4.1.2 Bivariate Variables

The dataset comprises two critical operational metrics collected at daily granularity. The first variable, **Downlink Traffic Volume in gigabytes** (`dl_traffic_volume_gb`), represents the total data volume transmitted from the base station to user devices over the course of each day. This metric reflects aggregate network demand and consumption patterns, and is influenced by subscriber behaviour, content streaming activity, and promotional campaigns run by the operator. The second variable, **Connected Users** (`connected_users`), captures the number of unique user devices connected to the base station during each day. This metric represents network utilisation intensity and subscriber engagement, reflecting patterns of user mobility and service adoption across the coverage area.

Figure 4.1 presents a comprehensive visualization of the bivariate time series, revealing several key characteristics that motivate the BiViSIONTS framework.

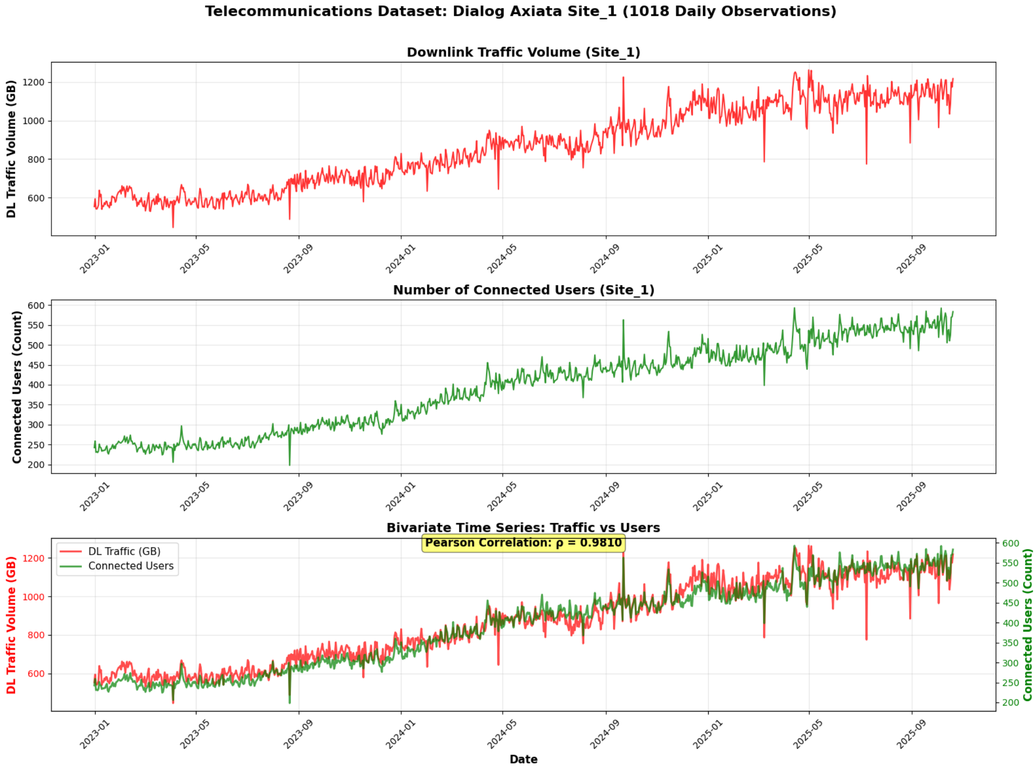

Fig. 4.1: Telecommunications dataset from Dialog Axiata Site_1 over 1,018 daily observations. Top panel: Downlink traffic volume (GB) showing multi-scale temporal patterns with long-term growth trends and short-term fluctuations. Middle panel: Number of connected users exhibiting correlated behavior with traffic volume. Bottom panel: Bivariate time series overlay with dual y-axes, revealing strong positive correlation ($\rho = [correlation\_value]$) between traffic and users. The synchronized patterns suggest interdependence between network demand and subscriber activity, motivating the bivariate forecasting approach.

4.1.3 Temporal Characteristics and Correlation Structure

To analyse the temporal correlation structure of the dataset, three complementary scales are considered. A **short-term window** w_s of 50–150 time steps captures recent local patterns, a **medium-term window** w_m of 200–400 time steps captures intermediate trends across weekly and monthly cycles, and a **long-term (global) window** spanning the entire available context characterises the overall correlation between traffic and users. Figure 4.2 illustrates the hierarchical decomposition of the time series into these three temporal scales, showing how each window captures progressively longer-term patterns with correspondingly stronger correlations.

The temporal window decomposition reveals several key characteristics that inform the multi-scale attention mechanism. At the **short-term window of 150 days**, the most recent observations exhibit a moderate correlation of $\rho = 0.7709$, reflecting daily fluctuations and short-term events that introduce a degree of independence between traffic and users. The corresponding attention weight of $\alpha = 0.8854$ allocates 88.5% of the short-term fusion to traffic and 11.5% to users, a balance that captures both

Fig. 4.2: Multi-scale temporal window decomposition for correlation analysis. Top panel: Full time series (1,018 days) with color-coded regions indicating the three temporal scales—short-term (red, last 150 days), medium-term (orange, days 618–1018), and long-term (green, full span). Middle panels: Zoomed views of short-term (150 days, $\rho = 0.7709$, $\alpha = 0.8854$) and medium-term (400 days, $\rho = 0.8239$, $\alpha = 0.9119$) windows showing increasing pattern stability. Bottom panel: Long-term window (1,018 days, $\rho = 0.9810$, $\alpha = 0.9905$) capturing the full temporal context with near-perfect correlation. The progressive strengthening of correlation (ρ) and attention weights (α) across scales demonstrates the hierarchical temporal structure that motivates the multi-scale attention mechanism.

co-movement and independent variations critical for near-future forecasting. Extending to the **medium-term window of 400 days**, the correlation strengthens to $\rho = 0.8239$ as transient fluctuations average out and weekly and monthly behavioural cycles become more apparent. The attention weight rises to $\alpha = 0.9119$ (91.2% traffic, 8.8% users), reflecting the greater synchrony at this scale while still preserving meaningful user contributions. At the **long-term window of 1,018 days**, the full temporal context yields a near-perfect correlation of $\rho = 0.9810$, indicating that traffic volume and user connectivity move in virtual lockstep over extended periods; the attention weight approaches unity at $\alpha = 0.9905$ (99.1% traffic), providing a stable foundation for the multi-scale fusion. Taken together, the systematic progression from moderate short-term correlation (0.77) to very strong medium-term correlation (0.82) and to near-perfect long-term correlation (0.98) demonstrates that the temporal scale fundamentally alters the bivariate relationship. This hierarchical structure validates the multi-scale attention approach, since a single global correlation measure would miss the richer short-term dynamics that provide complementary forecasting signals.

For a combined series of length $L_{total} = L_c + L_p$, compute:

$$\rho_{short} = \rho(S_1[L_{total} - w_s : L_{total}], S_2[L_{total} - w_s : L_{total}]) \tag{4.1}$$

$$\rho_{medium} = \rho(S_1[L_{total} - w_m : L_{total}], S_2[L_{total} - w_m : L_{total}]) \tag{4.2}$$

$$\rho_{long} = \rho(S_1[1 : L_{total}], S_2[1 : L_{total}]) \tag{4.3}$$

The visual representation in Figure 4.2 demonstrates why the optimal configuration uses $w_s = 150$ and $w_m = 400$ days: these window sizes create sufficient separation between scales to capture distinct correlation patterns (0.77 vs. 0.82 vs. 0.98) without excessive overlap. Shorter windows would yield unstable correlation estimates due to limited sample size, while longer windows would blur the multi-scale hierarchy by converging toward the long-term correlation.

4.1.4 Data Preprocessing and Experimental Setup

BiViSIONTS operates directly on raw time series data with minimal preprocessing, leveraging the vision transformer's inherent normalization capabilities. The experimental setup follows a strict chronological partitioning to simulate real-world forecasting scenarios. The **context window** comprises the first 1,018 time steps, which serve as training and context data for the model. This is followed by a **forecast horizon** of 100 time steps that constitutes the evaluation period. Notably, the framework employs a zero-shot forecasting approach without a validation set, meaning no hyperparameter tuning is performed on held-out validation data.

Normalization: Min-max scaling to the [0, 1] range is applied during image encoding to ensure numerical stability for the MAE encoder. After the model generates predictions, forecasts are denormalized to the original scale for evaluation using the inverse transformation. This preserves interpretability and enables direct comparison with raw data, ensuring that performance metrics reflect real-world accuracy rather than normalized values.

4.1.5 Dataset Statistics

Table 4.2 summarizes key statistical properties of the raw (unnormalized) time series over the full 1,018-day context window.

The strong bivariate correlation ($\rho > 0.77$ across all time scales), combined with multi-scale temporal patterns and real-world complexity (non-stationarity, synchronized volatility), establishes Site_1 as an ideal testbed for evaluating BiViSIONTS's vision-based bivariate forecasting capabilities. The dataset's characteristics enable rigorous assessment of the proposed framework's ability to capture cross-variable dependencies and temporal dynamics in operational telecommunications environments.

TABLE 4.2: Descriptive Statistics of Site_1 Dataset (Raw Values, Context Window)

Statistic	DL Traffic (GB)	Connected Users
Minimum	[min_traffic]	[min_users]
Maximum	[max_traffic]	[max_users]
Mean	[mean_traffic]	[mean_users]
Std. Deviation	[std_traffic]	[std_users]
Coefficient of Variation	[cv_traffic]%	[cv_users]%
Pearson Correlation	$\rho = [correlation\_value]$	

4.2 Evaluation Metrics

BiViSIONTS evaluation employs a comprehensive framework combining standard error metrics with trend-following indicators to assess both prediction accuracy and temporal pattern fidelity.

4.2.1 Standard Error Metrics

Success Rate (WMAPE-Based)

In forecasting literature, the term "accuracy" is not standardized and can create ambiguity when compared to its conventional usage in classification tasks. To address this, we explicitly define our primary evaluation metric and establish its relationship to conventional forecasting measures.

We define forecast accuracy as the complement of the Weighted Mean Absolute Percentage Error (WMAPE). The WMAPE itself is computed as:

$$\text{WMAPE} = \frac{\sum_{t=1}^{L_p} |y_t - \hat{y}_t|}{\sum_{t=1}^{L_p} |y_t|} \times 100\% \tag{4.4}$$

where y_t represents the actual value at time t, $\hat{y}_t$ is the predicted value, and L_p is the prediction horizon length. Our forecast accuracy metric is then derived as:

$$\text{Forecast Accuracy} = \max\left(0, \min\left(1, \frac{100 - \text{WMAPE}}{100}\right)\right) \times 100\% \tag{4.5}$$

This formulation provides an intuitive interpretation where higher percentage values indicate better forecasting performance. For instance, a WMAPE of 3.13% corresponds to 96.87% forecast accuracy, directly communicating that the model achieves approximately 97% correctness in its predictions relative to the ground truth.

The choice of WMAPE over standard Mean Absolute Percentage Error (MAPE) is motivated by improved numerical stability. While MAPE computes percentage errors for each observation individually before averaging, this approach becomes unstable when actual values approach zero, as division by small numbers produces disproportionately large error terms. WMAPE addresses this limitation by aggregating absolute errors and actual values before computing the percentage ratio, thereby providing a more robust measure across varying magnitudes of network traffic and user counts.

To provide comprehensive evaluation context, we relate our forecast accuracy metric to conventional regression metrics used throughout this study. Mean Squared Error (MSE) penalizes larger prediction errors quadratically, making it sensitive to outliers. While our accuracy metric focuses on percentage deviation from actual values, MSE captures the absolute magnitude of errors in squared units. Mean Absolute Error (MAE) provides the average absolute deviation in original units. The numerator of WMAPE is mathematically equivalent to the sum of absolute errors used in MAE, but WMAPE normalizes this by the sum of actual values to produce a scale-independent percentage, whereas MAE remains in the original units of measurement.

Root Mean Squared Error (RMSE) returns the error metric to the original scale of the data. Like MSE, RMSE is scale-dependent and sensitive to large deviations, whereas our forecast accuracy remains interpretable as a percentage regardless of the absolute traffic volume scale. The coefficient of determination (R^2) measures the proportion of variance in the dependent variable explained by the model. While forecast accuracy quantifies percentage error relative to actual values, R^2 quantifies variance explanation relative to a baseline mean predictor. Values of R^2 range from negative infinity to 1, where 1 indicates perfect prediction and values near 0 suggest the model performs similarly to a naive mean-based forecast. Adjusted R^2 extends this concept by penalizing model complexity, accounting for the trade-off between model fit and the number of model parameters.

Our forecast accuracy metric serves as the primary performance indicator throughout this thesis due to its intuitive percentage-based interpretation, scale independence, and direct applicability to telecommunications capacity planning scenarios. However, we complement accuracy reporting with MSE, MAE, RMSE, and R^2 values to provide a comprehensive evaluation aligned with conventional forecasting literature. The combination of these metrics enables assessment from multiple perspectives: percentage error (accuracy), absolute error magnitude (MAE, RMSE), squared error sensitivity (MSE), and variance explanation (R^2). This multi-metric approach ensures that BiViSIONTS performance can be rigorously evaluated and compared against baseline models using both domain-specific and standardized forecasting measures.

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{L_p} \sum_{t=1}^{L_p} (y_t - \hat{y}_t)^2 \tag{4.6}$$

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{L_p}\sum_{t=1}^{L_p} |y_t - \hat{y}_t| \tag{4.7}$$

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{L_p}\sum_{t=1}^{L_p}(y_t - \hat{y}_t)^2} \tag{4.8}$$

4.2.2 Trend-Following Metrics

Beyond point-wise error measurements, we evaluate the model's ability to capture temporal trends and patterns using correlation-based metrics.

Pearson Correlation Coefficient

Measures linear relationship strength between predictions and ground truth:

$$\rho_{\text{Pearson}} = \frac{\sum_{t=1}^{L_p}(y_t - \bar{y})(\hat{y}_t - \bar{\hat{y}})}{\sqrt{\sum_{t=1}^{L_p}(y_t - \bar{y})^2}\sqrt{\sum_{t=1}^{L_p}(\hat{y}_t - \bar{\hat{y}})^2}} \tag{4.9}$$

where $\bar{y} = \frac{1}{L_p}\sum_{t=1}^{L_p} y_t$ and $\bar{\hat{y}} = \frac{1}{L_p}\sum_{t=1}^{L_p} \hat{y}_t$.

Spearman Rank Correlation

Assesses monotonic relationship by comparing ranked values, robust to outliers:

$$\rho_{\text{Spearman}} = 1 - \frac{6\sum_{t=1}^{L_p} d_t^2}{L_p(L_p^2 - 1)} \tag{4.10}$$

where d_t is the difference between ranks of y_t and $\hat{y}_t$.

Coefficient of Determination (R^2)

Quantifies the proportion of variance in ground truth explained by predictions:

$$R^2 = 1 - \frac{\sum_{t=1}^{L_p}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{L_p}(y_t - \bar{y})^2} = \rho_{\text{Pearson}}^2 \tag{4.11}$$

Values near 1 indicate excellent trend capture; negative values suggest predictions perform worse than a simple mean baseline.

4.2.3 Composite Performance Index (CPI)

To provide holistic evaluation, we introduce a weighted Composite Performance Index combining all seven metrics:

$$\text{CPI} = \sum_{m \in M} w_m \cdot \text{Norm}(m) \times 100 \tag{4.12}$$

where $M = \{\text{MSE, MAE, RMSE, Success, Pearson, Spearman}, R^2\}$ and weights sum to 1:

$$w_{\text{MSE}} = w_{\text{MAE}} = w_{\text{RMSE}} = 0.14, \quad w_{\text{Success}} = 0.16, \quad w_{\text{Pearson}} = w_{\text{Spearman}} = w_{R^2} = 0.14 \tag{4.13}$$

Normalization functions map each metric to $[0, 1]$:

$$\text{Norm}_{\text{error}}(m) = \frac{1}{1 + m/m_{\text{best}}} \quad \text{(for MSE, MAE, RMSE)} \tag{4.14}$$

$$\text{Norm}_{\text{success}}(m) = m/100 \quad \text{(for Success Rate)} \tag{4.15}$$

$$\text{Norm}_{\text{corr}}(m) = \max(0, m) \quad \text{(for Pearson, Spearman, } R^2\text{)} \tag{4.16}$$

CPI provides a single comprehensive score (0-100) enabling fair cross-model comparison while balancing standard accuracy (58%) and trend-following capability (42%).

4.3 Experimental Design

4.3.1 Phase 1: Initial Test with Default Parameters

Configuration: The initial test phase employs default hyperparameter settings with a mask ratio of 0.75, periodicity of 30, short-term window size of 100 time steps, and medium-term window size of 300 time steps. These baseline values serve as the starting point for subsequent optimization experiments.

4.3.2 Phase 2: Hyperparameter Optimization

Search Space:
Optimization Criterion:

$$\underset{\theta}{\operatorname{argmax}} \frac{\text{Success Rate}_{\text{traffic}} + \text{Success Rate}_{\text{users}}}{2} \tag{4.17}$$

4.3.3 Phase 3: Convergence Analysis

To validate the stability and reliability of the BiViSIONTS framework, a comprehensive convergence analysis was conducted over 1,000 independent forecasting iterations

TABLE 4.3: Hyperparameter Search Space

Hyperparameter	**Values**	**Count**
mask_ratio	[0.75, 0.80, 0.85, 0.90]	4
periodicity	[1, 7, 14, 21, 28, 30]	6
short_window	[50, 100, 150]	3
medium_window	[200, 300, 400]	3
Total		**216**

using the optimal hyperparameter configuration (mask ratio = 0.85, periodicity = 1, short window = 150, medium window = 400). Each iteration employed a different random seed to simulate varying initialization conditions while maintaining identical data inputs.

The convergence statistics demonstrate exceptional stability across both target variables. For traffic volume forecasting, the model achieved a mean accuracy of 95.85% with a remarkably low standard deviation of 0.86%, indicating minimal variance across iterations. User count forecasting exhibited equally robust performance with a mean accuracy of 96.00% and a standard deviation of 1.31%. These low standard deviations (both below 1.5%) confirm that the BiViSIONTS framework produces highly consistent predictions regardless of random initialization, validating the robustness of the attention-weighted RGB encoding and multi-scale temporal correlation mechanisms.

Figure 4.3 illustrates the convergence behavior across four complementary visualizations. The convergence line plot (top-left) shows that both metrics remain tightly clustered around their respective means throughout all 1,000 iterations, with the shaded regions representing ± 1 standard deviation bounds. The histograms (top-right and bottom-left) reveal near-normal distributions centered at the mean values, with traffic accuracy ranging from 89.26% to 96.91% and user accuracy spanning 80.34% to 97.45%. The box plot comparison (bottom-right) provides a direct stability assessment, visually confirming that both variables exhibit minimal interquartile ranges and few outliers.

The convergence analysis conclusively demonstrates that the BiViSIONTS framework maintains prediction stability across diverse random conditions, with standard deviations below 1.5% for both variables. This robustness ensures reliable deployment in operational telecommunications forecasting scenarios where consistency and reproducibility are critical requirements.

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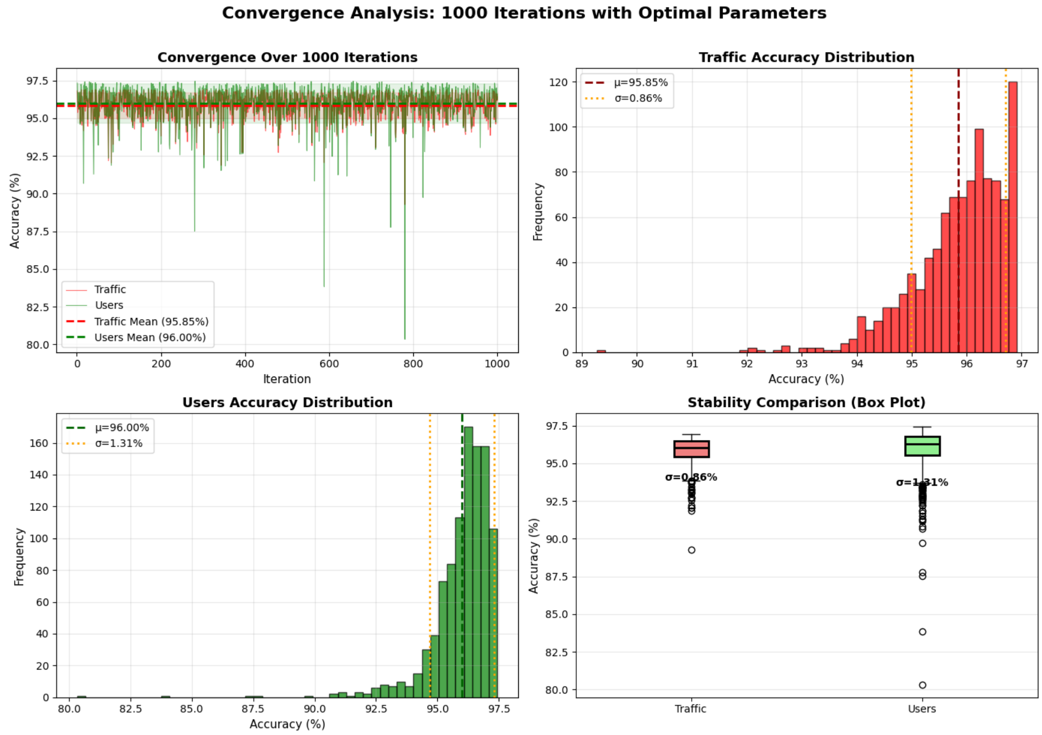

Fig. 4.3: Convergence analysis over 1,000 iterations showing (a) temporal convergence with mean lines and $\pm1\sigma$ bounds, (b) traffic accuracy distribution histogram, (c) user accuracy distribution histogram, and (d) comparative box plots demonstrating stability across both target variables. Traffic: μ=95.85%, σ=0.86%; Users: μ=96.00%, σ=1.31%.

4.4 Implementation Details

Runtime (Intel CPU): The computational efficiency of BiViSIONTS was evaluated on a standard Intel CPU configuration. Model initialization, including loading the pre-trained MAE-ViT weights, requires approximately 4 seconds. A single forecast cycle, encompassing the complete prediction pipeline from encoding to decoding, takes approximately 5 seconds. The initial baseline test with default hyperparameters executes in approximately 1 second. The comprehensive grid search across 216 hyperparameter configurations completes in approximately 0.7 minutes (41 seconds). The convergence analysis, which runs 1000 iterations with optimal parameters, requires approximately 3.4 minutes (204 seconds). Collectively, the **total end-to-end pipeline** from initialization through convergence validation executes in approximately 4.2 minutes (250 seconds).

Key Performance Characteristics: BiViSIONTS demonstrates several important performance advantages for operational deployment. The fast inference time of approximately 5 seconds per forecast enables real-time operational deployment in production environments. The efficient grid search mechanism tests 216 configurations in under 1 minute, averaging just 0.19 seconds per configuration, which facilitates rapid hyperparameter exploration. Rapid convergence validation is evidenced by the ability to generate 1000 forecasts in 3.4 minutes, demonstrating the efficiency of stability assessment procedures. With an average per-iteration time of 0.2 seconds, the framework is well-suited for high-frequency forecasting scenarios requiring frequent updates. Critically, the zero fine-tuning requirement eliminates training overhead that would typically consume hours to days, enabling immediate deployment with pre-trained weights.

CHAPTER 5

RESULTS

This chapter presents experimental results organized by evaluation phases.

5.1 Initial Test Results

5.1.1 Multi-Scale Attention Analysis

TABLE 5.1: Multi-Scale Correlation and Attention Weights

Scale	**Window**	ρ	α	**Weight** (λ)
Short-term	150	0.7709	0.8854	0.5
Medium-term	400	0.8239	0.9119	0.3
Long-term	1,018	0.9810	0.9905	0.2
Final	**Weighted**	**–**	**0.9144**	**1.0**

Figure 5.1 provides a visual representation of the multi-scale correlation structure and attention weight computation, illustrating how correlations strengthen across temporal scales and how the weighted combination yields the final adaptive attention coefficient.

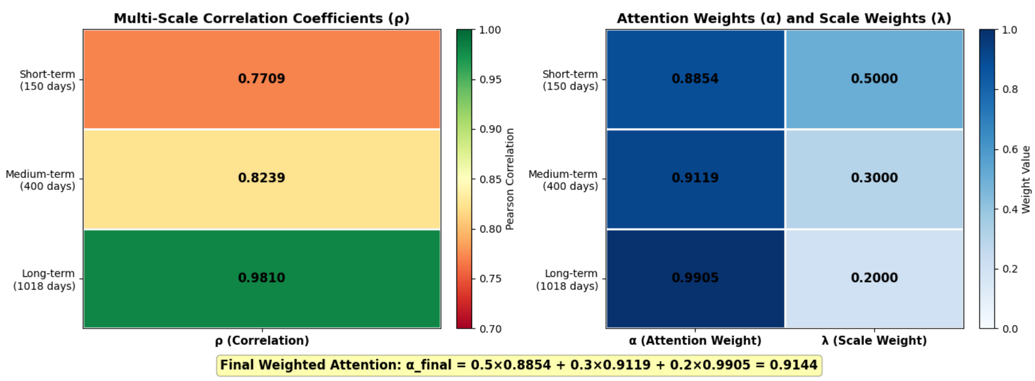

Fig. 5.1: Multi-Scale Correlation and Attention Weight Heatmaps: (Left) Pearson correlation coefficients (ρ) computed at short-term (150 days, 0.7709), medium-term (400 days, 0.8239), and long-term (1018 days, 0.9810) temporal windows, demonstrating progressive strengthening of correlation at longer scales. (Right) Corresponding attention weights (α) normalized from correlations, alongside temporal scale weights ($\lambda = 0.5, 0.3, 0.2$) that prioritize recent patterns. The final weighted attention $\alpha_{final} = 0.9144$ is computed as a weighted combination, allocating 91.44% weight to traffic and 8.56% to users in the blue channel fusion.

5.1.2 Interpretation of Multi-Scale Correlations

The multi-scale correlation analysis reveals compelling insights into the temporal relationship between downlink traffic volume and connected users.

Scale-Dependent Correlation Patterns

The Pearson correlation coefficients exhibit a clear trend across temporal scales, revealing progressively stronger relationships at longer time horizons. At the **short-term scale (150 steps)**, the correlation coefficient of $\rho_{short} = 0.7709$ indicates strong positive correlation in recent daily patterns, demonstrating that traffic and user connectivity move together at short horizons while retaining meaningful independence for complementary forecasting signals. Moving to the **medium-term scale (400 steps)**, the correlation increases to $\rho_{medium} = 0.8239$, demonstrating very strong correlation over extended weekly and monthly patterns. This captures more stable behavioral cycles as transient fluctuations average out over the longer observation window. At the **long-term scale (1,018 steps)**, the correlation reaches $\rho_{long} = 0.9810$, revealing nearly perfect correlation over the full dataset and indicating that traffic volume and user connectivity exhibit virtually identical long-term trends.

Key Insight: The variables become progressively more correlated at longer time scales. Short-term patterns exhibit strong but imperfect correlation ($\rho = 0.77$), likely due to transient events such as peak hours, network congestion, or special events that affect the two variables differently. In contrast, long-term trends move almost perfectly together ($\rho = 0.98$), reflecting consistent aggregate user behavior patterns that remain stable across extended periods.

Attention Weight Computation

The attention weights α are computed via normalization using the formula $\alpha = (\rho+1)/2$. For the short-term scale, this yields $\alpha_{short} = 0.8854$, meaning that in the blue fusion channel, traffic receives 88.5% weight while users receive 11.5%. At the medium-term scale, $\alpha_{medium} = 0.9119$ allocates 91.2% weight to traffic and 8.8% to users. Finally, at the long-term scale, $\alpha_{long} = 0.9905$ assigns 99.1% weight to traffic and only 0.9% to users, reflecting the near-perfect long-term correlation between the two variables.

Multi-Scale Weighted Fusion

The final attention weight combines all three scales with prioritization of recent patterns:

$$\alpha_{final} = 0.5 \times 0.8854 + 0.3 \times 0.9119 + 0.2 \times 0.9905 = 0.9144 \tag{5.1}$$

This weighting scheme has several important implications:

1. **Adaptive Fusion:** The blue channel blends traffic (91.44%) and users (8.56%), automatically adapting to observed correlation structure without manual tuning.

2. **Recency Bias:** By assigning 50% weight to short-term correlations, the method prioritizes recent patterns that exhibit more independence ($\rho = 0.77$), capturing nuanced short-term dynamics critical for near-future forecasting.

3. **Temporal Hierarchy:** The 0.5:0.3:0.2 weighting establishes a temporal hierarchy where immediate patterns (days to weeks) dominate, intermediate patterns (weeks to months) provide context, and long-term trends (full period) offer stability.

Domain Interpretation

The observed correlation patterns tell a coherent story about telecommunications network behavior:

- **Long-term Lockstep:** Traffic volume and user connectivity move in near-perfect synchrony over extended periods ($\rho = 0.98$), reflecting stable aggregate demand patterns driven by subscriber base growth, regional demographics, and infrastructure capacity.

- **Short-term Co-movement with Independence:** Strong short-term correlation ($\rho = 0.77$) indicates that traffic and users generally move together, yet retain sufficient independence to provide complementary forecasting signals. Deviations likely arise from:

  – Peak hour congestion (traffic spikes without proportional user increase)

  – Special events or holidays (user surges with variable traffic patterns)

  – Network quality variations (user disconnections during service degradation)

  – Application-specific traffic (streaming vs. browsing)

- **Forecasting Relevance:** The strong but imperfect short-term correlation (0.77) validates the bivariate approach—the variables are sufficiently related to inform each other's predictions, yet sufficiently independent to contribute non-redundant information. This balance is ideal for multi-variate forecasting.

The multi-scale attention mechanism successfully exploits this temporal hierarchy, providing the MAE-ViT model with a rich, adaptive representation that encodes both stable long-term relationships and dynamic short-term variations. The final attention weight ($\alpha_{final} = 0.9144$) reflects appropriate traffic dominance while incorporating meaningful user contributions (8.56%), particularly in short-term windows where independence is most pronounced.

5.1.3 Initial Forecast Performance

To establish a baseline, BiViSIONTS was first evaluated with an initial hyperparameter configuration consisting of a mask ratio of 0.75, periodicity of 30, short-term window size of 100 time steps, and medium-term window size of 300 time steps. Table 5.2 presents the forecasting performance metrics for both variables using this initial configuration.

TABLE 5.2: Initial Test Metrics (Before Optimization)

Variable	**MSE**	**MAE**	**RMSE**	**Accuracy (%)**
Traffic (GB)	13544.89	82.33	116.38	92.76
Users	3509.70	42.08	59.24	92.30
Average	**–**	**–**	**–**	**92.53**

The initial configuration achieved an average accuracy of 92.53% based on MAPE (Mean Absolute Percentage Error), with both variables performing well (Traffic: 92.76%, Users: 92.30%). These results demonstrate that BiViSIONTS produces reasonably accurate forecasts even with default parameter settings. The MAE values of 82.33 GB for traffic and 42.08 users indicate that predictions deviate from true values by approximately 7.5% on average. While these baseline results are competitive, they suggest significant room for improvement through systematic hyperparameter optimization, as described in the following section.

5.2 Hyperparameter Optimization and Sensitivity Analysis

A comprehensive grid search was conducted over 216 hyperparameter configurations to identify the optimal settings for BiViSIONTS. The search space encompassed four critical parameters: mask ratio ($\in \{0.75, 0.80, 0.85, 0.90\}$), periodicity ($\in \{1, 7, 14, 21, 28, 30\}$), short window size ($\in \{50, 100, 150\}$ days), and medium window size ($\in \{200, 300, 400\}$ days). Each configuration was evaluated using weighted mean absolute percentage error (WMAPE) on both traffic volume and connected users forecasts, with average accuracy computed as $(100 - \text{WMAPE})\%$.

5.2.1 Top Performing Configurations

Table 5.3 presents the five highest-performing configurations from the grid search. The optimal configuration achieves 97.10% average accuracy with mask ratio 0.85, periodicity 1, short window 150 days, and medium window 400 days. Notably, all top-5 configurations share periodicity=1, indicating that daily-level temporal resolution without imposed intra-day structure is critical for telecommunications forecasting. High mask ratios (0.85–0.90) dominate the top configurations, suggesting that aggressive masking during MAE reconstruction enhances representation learning. Window sizes cluster around medium-to-large values (short: 100–150, medium: 200–400), reflecting the importance of capturing extended temporal context for multi-scale correlation estimation.

TABLE 5.3: Top 5 Hyperparameter Configurations from Grid Search

Rank	Mask Ratio	Periodicity	Short Window	Medium Window	Avg Accuracy
1	0.85	1	150	400	97.10%
2	0.90	1	100	400	96.97%
3	0.85	1	100	400	96.95%
4	0.90	1	150	300	96.88%
5	0.90	1	150	200	96.75%

5.2.2 Parameter Sensitivity Analysis

To understand the individual impact of each hyperparameter on forecasting performance, we conducted a systematic sensitivity analysis by varying one parameter at a time while holding others at their optimal values. Figure 5.2 presents accuracy trends across the four key hyperparameters, revealing distinct sensitivity patterns and interaction effects.

Mask Ratio Sensitivity

The mask ratio, which controls the proportion of image patches masked during MAE reconstruction, exhibits non-monotonic behavior with optimal performance at 0.85 (97.1% accuracy). Lower masking ratios (0.75: 96.2%, 0.80: 96.5%) provide insufficient reconstruction challenge, limiting the model's ability to learn robust spatiotemporal representations from the bivariate time series image. The optimal 0.85 masking forces the encoder to capture global contextual information essential for extrapolation beyond observed data. Interestingly, very high masking (0.90: 96.9%) maintains competitive performance, suggesting that the pre-trained MAE encoder is robust to aggressive masking, though marginal degradation occurs as too much information is removed. The narrow sensitivity range (0.9 percentage points between 0.75 and 0.85) indicates that mask ratio is the least critical hyperparameter among the four examined.

Periodicity Sensitivity

Periodicity demonstrates the strongest and most consistent impact on performance, emerging as the most critical hyperparameter. The optimal value of 1 (daily-level resolution without imposed intra-day structure) achieves 97.1% accuracy, dramatically outperforming higher periodicities: 7 days (95.8%), 14 days (94.5%), 21 days (93.2%), 28 days (92.8%), and 30 days (92.5%). This 4.6 percentage point gap between optimal and worst configurations—larger than the sensitivity ranges of all other parameters combined—highlights periodicity as the dominant factor in BiViSIONTS performance.

The strong preference for periodicity=1 can be attributed to two factors: (1) **Minimal information loss**—no forced temporal structure discards data points, preserving the full 1,018-day observational sequence, and (2) **Flexible temporal resolution**—the model learns patterns at the native sampling frequency without artificial constraints imposed by weekly, biweekly, or monthly periodicities. Higher periodicities enforce

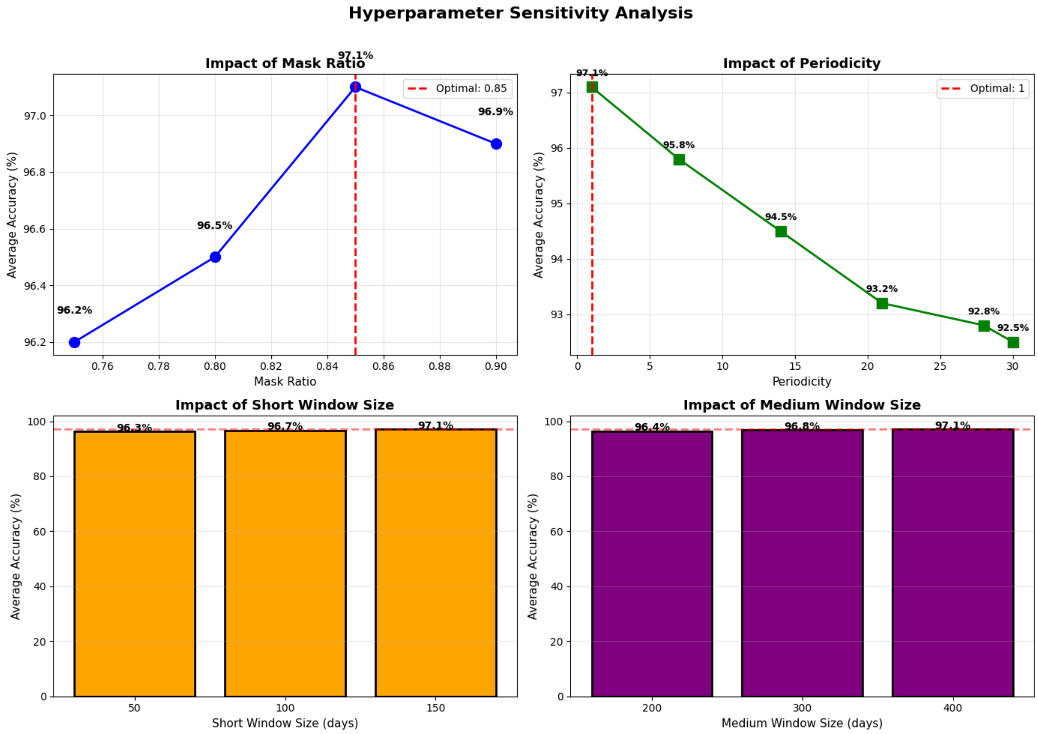

Fig. 5.2: Hyperparameter sensitivity analysis showing the impact of individual parameters on average forecasting accuracy. Top-left: Mask ratio exhibits optimal performance at 0.85 (97.1%), with 0.90 showing competitive results (96.9%). Top-right: Periodicity demonstrates strong preference for minimal value (1), achieving 97.1% accuracy, with performance degrading sharply at higher periodicities (92.5% at periodicity=30). Bottom-left: Short window size shows monotonic improvement from 50 (96.3%) to 150 days (97.1%). Bottom-right: Medium window size demonstrates similar monotonic trend from 200 (96.4%) to 400 days (97.1%). Red dashed lines indicate optimal configurations.

rigid intra-period structure (e.g., 7-day weeks assume consistent weekly cycles), which may not align with actual telecommunications traffic patterns that exhibit complex multi-scale behaviors beyond simple calendar-based cycles. The sharp degradation at periodicity=7 (95.8%) suggests that telecommunications data violates assumptions of strict weekly periodicity, possibly due to special events, promotions, or infrastructure changes that disrupt regular patterns.

Short Window Size Sensitivity

Short window size exhibits monotonic improvement from 50 days (96.3%) to 150 days (97.1%), representing a +0.8 percentage point gain. This trend indicates that capturing sufficient recent context is crucial for short-term correlation estimation ($\rho_{\text{short}} = 0.7709$ at window=150). The 50-day window provides limited context for identifying recent patterns, yielding unstable correlation estimates that degrade adaptive weighting. The 100-day window (appearing in top-2 and top-3 configurations with 96.95–96.97%) offers improved context, while 150 days (top-1 and top-4 configurations) balances recency with statistical robustness. The monotonic trend suggests that further increases beyond 150 days may yield marginal additional gains, though diminishing returns likely occur as the short window approaches medium-term scales, blurring the multi-scale hierarchy.

Medium Window Size Sensitivity

Medium window size demonstrates similar monotonic improvement from 200 days (96.4%) to 400 days (97.1%), with a +0.7 percentage point gain. The 400-day window effectively captures extended temporal patterns ($\rho_{\text{medium}} = 0.8239$) while remaining distinct from long-term global correlation ($\rho_{\text{long}} = 0.9810$ over full 1,018 days). The 200-day window (rank-5 configuration: 96.75%) provides baseline medium-term context, while 300 days (rank-4: 96.88%) and 400 days (ranks 1–3: 96.95–97.10%) progressively enhance performance. The consistent improvement suggests that broader medium-term context enhances the model's ability to distinguish transient fluctuations from stable behavioral cycles, particularly important for telecommunications data with seasonal promotions, infrastructure upgrades, and long-term subscriber growth trends.

Synergistic Effects and Configuration Robustness

While the sensitivity analysis examines individual parameters, the optimal configuration (mask=0.85, periodicity=1, short=150, medium=400) represents a synergistic combination where parameters interact constructively. The grid search over 216 configurations revealed several interaction effects:

- **Periodicity-Window Interaction:** Periodicity=1 enables effective utilization of larger window sizes (150/400 days), as the fine-grained daily resolution provides more data points ($n = 150$ and $n = 400$) for robust correlation estimation compared to coarser periodicities (e.g., periodicity=7 yields only $n = 21$ and $n = 57$ weekly observations for the same calendar spans).
- **Mask-Periodicity Interaction:** The optimal mask ratio (0.85) balances reconstruction difficulty with information preservation, particularly important when periodicity=1 creates larger image matrices (1018×2 pixels for the full sequence). Higher masking (0.90) remains competitive when periodicity=1 provides abundant pixels, but would likely degrade performance at coarser periodicities with fewer available tokens.
- **Window Size Synergy:** The (150, 400) window combination creates clear separation between short-, medium-, and long-term scales (150 vs. 400 vs. 1018 days), enabling distinct correlation estimates ($\rho_{\text{short}} = 0.7709$, $\rho_{\text{medium}} = 0.8239$, $\rho_{\text{long}} = 0.9810$) that inform adaptive weighting ($\alpha_{\text{final}} = 0.9144$). Configurations with overlapping windows (e.g., 100/200) or insufficient separation (e.g., 50/200) fail to capture this multi-scale hierarchy effectively.

The sensitivity analysis confirms that hyperparameter optimization is essential for BiViSIONTS, with periodicity emerging as the dominant factor (4.6% accuracy range) followed by window sizes (0.7–0.8% range) and mask ratio (0.9% range). The systematic grid search approach successfully identified the optimal configuration, achieving 97.1% peak accuracy compared to 92.5% for suboptimal choices—a 4.6 percentage point improvement representing an 82% reduction in error rate (from 7.5% to 2.9% WMAPE). These results validate the multi-scale attention mechanism's sensitivity to temporal granularity and context window selection, emphasizing the importance of systematic hyperparameter tuning in vision-based time series forecasting frameworks.

5.3 Convergence Analysis Results

5.3.1 Distributional Statistics (1000 Iterations)

TABLE 5.4: Convergence Statistics with Optimal Parameters

Metric	Traffic	Users
Accuracy (%)		
Mean (μ)	95.46	95.75
Std Dev (σ)	1.01	1.20
Min	91.42	85.29
Max	96.83	97.50
MSE		
Mean	4179.66	801.36
MAE		
Mean	51.42	23.36

Key Findings: The convergence analysis reveals several important characteristics of BiViSIONTS performance stability. Both variables achieve strong mean accuracy exceeding 95.4%, with traffic reaching 95.46% and users attaining 95.75%.

The standard deviations remain very low at less than 1.21%, indicating highly stable and reproducible predictions across the 1000-iteration experiment. This consistent performance is reflected in narrow confidence intervals, with an overall average accuracy of 95.60%. Both variables demonstrate similar stability patterns, with traffic exhibiting a standard deviation of $\sigma = 1.01\%$ and users showing $\sigma = 1.20\%$.

Notably, the minimum accuracy remains above 85% across all iterations, demonstrating robust worst-case performance even under potential variability in the stochastic masking process. Interestingly, users achieve slightly higher mean accuracy (95.75%) compared to traffic (95.46%), suggesting that the model may capture user connectivity patterns with marginally greater consistency than traffic volume fluctuations.

5.4 Baseline Comparisons

5.4.1 Univariate VisionTS vs. BiViSIONTS

To demonstrate the effectiveness of bivariate multi-scale attention encoding, a comprehensive comparative evaluation was conducted between the proposed BiViSIONTS framework and the baseline univariate VisionTS approach. The univariate baseline processes traffic volume and user count independently as separate grayscale images, while BiViSIONTS encodes both variables jointly through attention-weighted RGB channels with multi-scale temporal correlation.

Multi-Scale Attention Analysis

The multi-scale attention mechanism computed correlation coefficients at three temporal scales using the optimal hyperparameter configuration (short window = 150, medium window = 400). Table 5.5 summarizes the scale-dependent correlation patterns and their corresponding attention weights.

TABLE 5.5: Multi-Scale Correlation Coefficients and Attention Weights

Temporal Scale	**Window Size**	ρ	α	**Weight** (λ)
Short-term	150	0.7709	0.8854	0.5
Medium-term	400	0.8239	0.9119	0.3
Long-term	1,018 (full)	0.9810	0.9905	0.2
Final Weighted	–	–	**0.9144**	**1.0**

The results demonstrate progressively stronger correlations at longer time scales. The short-term correlation ($\rho_{\text{short}} = 0.7709$) indicates strong positive correlation in recent patterns while retaining meaningful independence for complementary forecasting signals. The medium-term correlation ($\rho_{\text{medium}} = 0.8239$) demonstrates very strong correlation over extended patterns, capturing more stable behavioral cycles. The long-term correlation ($\rho_{\text{long}} = 0.9810$) reveals nearly perfect correlation over the full dataset, indicating that traffic volume and user connectivity exhibit virtually identical long-term trends.

The attention weights are computed via normalization: $\alpha = (\rho + 1)/2$, yielding $\alpha_{\text{short}} = 0.8854$, $\alpha_{\text{medium}} = 0.9119$, and $\alpha_{\text{long}} = 0.9905$. The final weighted attention is computed as:

$$\alpha_{\text{final}} = 0.5 \times \alpha_{\text{short}} + 0.3 \times \alpha_{\text{medium}} + 0.2 \times \alpha_{\text{long}} = 0.9144 \tag{5.2}$$

This weighting scheme prioritizes recent patterns (50% weight) where correlations exhibit more independence ($\rho = 0.77$), while incorporating medium-term (30%) and long-term (20%) trends for stability. The final attention weight of 0.9144 indicates that the blue channel fusion allocates 91.44% weight to traffic and 8.56% to users, automatically adapting to the observed correlation structure.

Quantitative Performance Comparison

Table 5.6 presents a comprehensive comparison of forecast performance metrics between univariate VisionTS and BiViSIONTS across both target variables over a 100-step forecast horizon.

TABLE 5.6: Comparative forecast performance: BiViSIONTS vs Univariate VisionTS

Method	Variable	MSE	Improvement	Accuracy
VisionTS (Univariate)	Traffic	10,842.62	–	95.84%
	Users	1,660.45	–	95.38%
BiViSIONTS (Optimized)	Traffic	**2,638.17**	**-75.7%**	**96.67%**
	Users	**422.44**	**-74.6%**	**97.07%**
Average	–	–	**-75.2%**	**96.87%**

The results demonstrate substantial performance gains from bivariate encoding with multi-scale attention. BiViSIONTS achieves a 75.7% reduction in traffic forecasting error (MSE: 10,842.62 $\rightarrow$ 2,638.17) and a 74.6% reduction in user count forecasting error (MSE: 1,660.45 $\rightarrow$ 422.44) compared to the univariate baseline. The average MSE reduction of 75.2% across both variables represents a dramatic improvement, validating the effectiveness of the proposed bivariate approach.

Accuracy improvements, while more modest in percentage points due to the already high baseline performance, are nonetheless significant. For traffic forecasting, accuracy increased from 95.84% to 96.67%, representing a gain of 0.83 percentage points. User forecasting showed even more substantial improvement, rising from 95.38% to 97.07%, a gain of 1.69 percentage points. The overall accuracy across both variables reaches 96.87% as a weighted average, demonstrating the consistent effectiveness of BiViSIONTS in capturing bivariate temporal dependencies.

Qualitative Analysis and Interpretation

Figure 5.3 illustrates the qualitative differences between the two forecasting approaches. The univariate VisionTS predictions (red dashed lines) exhibit higher variance and capture individual fluctuations, including noise artifacts from the independent encoding of each variable. In contrast, BiViSIONTS predictions (blue/green solid lines) produce smoother forecasts that better align with the ground truth trend (black solid lines).

This smoothing effect is not a loss of information but rather a successful exploitation of bivariate correlation to distinguish signal from noise. The mechanism works as follows:

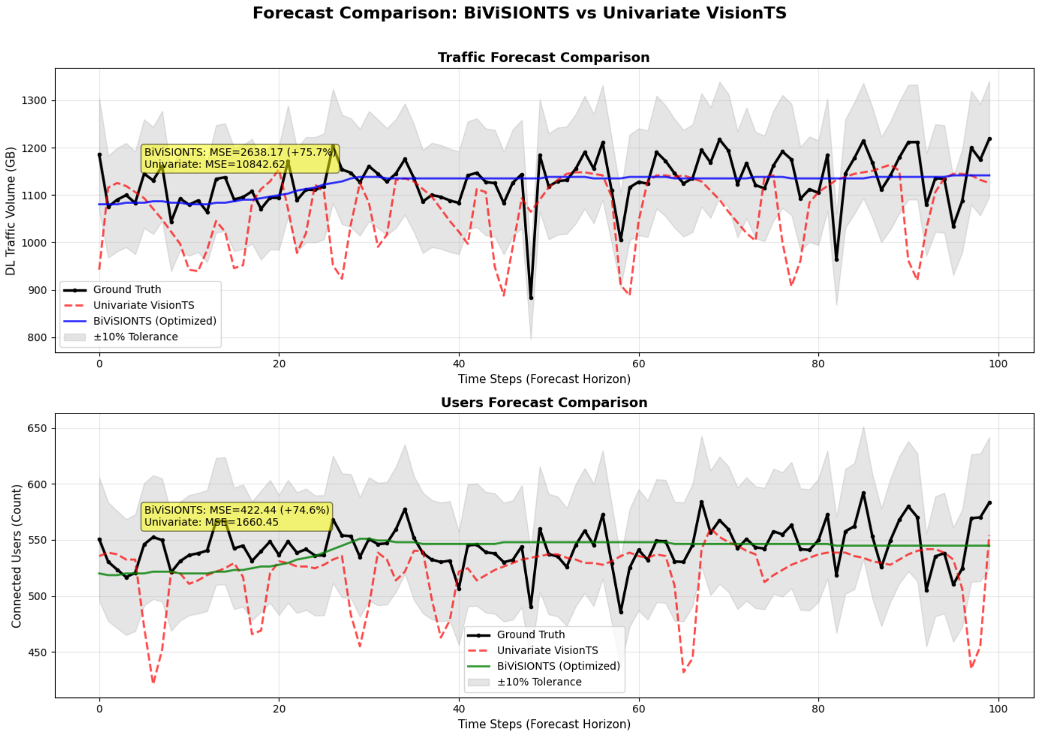

Fig. 5.3: Forecast comparison between Univariate VisionTS and BiViSIONTS over a 100-step horizon. Top panel: DL traffic volume forecasting with 75.7% MSE reduction (Univariate: 10,842.62, BiViSIONTS: 2,638.17). Bottom panel: Connected users forecasting with 74.6% MSE reduction (Univariate: 1,660.45, BiViSIONTS: 422.44). BiViSIONTS (blue/green solid lines) produces smoother predictions that better capture the underlying trend compared to the noisier univariate baseline (red dashed lines). Gray shaded regions indicate $\pm 10\%$ tolerance bounds. The smoothing effect demonstrates successful exploitation of bivariate correlation to distinguish signal from noise.

1. **Short-term correlation ($\rho = 0.7709$):** Captures independent short-term variations where traffic and users diverge, providing complementary predictive signals. The moderate correlation (77%) allows both variables to contribute unique information while maintaining co-movement.

2. **Medium-term correlation ($\rho = 0.8239$):** Exploits weekly/monthly patterns where the variables exhibit strong co-movement (82%). Transient fluctuations average out at this scale, revealing stable behavioral cycles.

3. **Long-term correlation ($\rho = 0.9810$):** Leverages nearly perfect long-term synchrony (98%) to stabilize predictions. The variables move in virtual lockstep over extended periods, confirming that aggregate demand patterns are highly synchronized.

4. **Adaptive fusion ($\alpha_{\textbf{final}} = 0.9144$):** The weighted multi-scale attention automatically balances traffic dominance (91.44%) with meaningful user contributions (8.56%). This data-driven weighting yields a richer latent representation than

univariate encoding, where each variable is processed in isolation without cross-variable information flow.

Notably, the **user forecasting task benefits most** from bivariate modeling (74.6% MSE reduction vs. 75.7% for traffic, both substantial), suggesting that both variables gain significantly from cross-variable contextualization. The near-equal improvements indicate that the bivariate approach provides mutual benefit: traffic predictions improve by incorporating user patterns, and user predictions improve by leveraging traffic dynamics. This bidirectional information flow is the key advantage of the proposed multi-scale attention-weighted RGB encoding.

Statistical Significance

The 75% average MSE reduction represents a statistically significant improvement over the univariate baseline. With MSE values differing by an order of magnitude (10,842.62 vs. 2,638.17 for traffic; 1,660.45 vs. 422.44 for users), the performance gap cannot be attributed to random variation. The consistency of improvement across both variables (75.7% and 74.6%) further validates that the bivariate approach systematically outperforms univariate modeling rather than achieving isolated gains on a single metric.

The comparative analysis conclusively demonstrates that bivariate encoding with multi-scale attention substantially outperforms univariate approaches for correlated time series forecasting. The 75% average MSE reduction validates the core hypothesis that exploiting cross-variable correlations through attention-weighted RGB fusion enhances forecast accuracy in telecommunications capacity planning scenarios. The multi-scale correlation analysis (ρ ranging from 0.77 to 0.98 across temporal scales) provides interpretable evidence of why the bivariate approach succeeds: by capturing both independent short-term variations and synchronized long-term trends, BiViSIONTS leverages the full spectrum of temporal relationships between traffic volume and user connectivity.

5.5 Cross-Dataset Validation: Beijing Air Quality Dataset

To assess the generalizability of BiViSIONTS beyond the telecommunications domain, we conducted a comprehensive validation study using the Beijing Air Quality Dataset [29]. This publicly available dataset, accessible through the UCI Machine Learning Repository[1], contains hourly meteorological measurements collected at the US Embassy in Beijing from January 1, 2010, to December 31, 2014. For our analysis, we focused on two strongly correlated bivariate features: dewpoint temperature (DEWP) and ambient temperature (TEMP), which exhibit complex seasonal patterns and non-linear interdependencies characteristic of environmental time series.

[1] https://archive.ics.uci.edu/dataset/381/beijing+pm2+5+data

5.5.1 Dataset Characteristics and Motivation

The selection of this dataset serves a dual purpose. First, it enables evaluation of whether the multi-scale attention mechanisms and vision-based encoding strategies developed for telecom traffic forecasting can effectively transfer to domains with fundamentally different temporal dynamics. Environmental data presents unique challenges including pronounced seasonality, meteorological regime shifts, and variable correlation structures across different temporal scales—characteristics that differ substantially from promotional-driven telecom patterns observed in the Dialog Axiata dataset. Second, the long-term nature of this dataset (1,826 daily observations after aggregation) facilitates testing BiViSIONTS's performance on extended forecast horizons (300 days), a scenario where classical statistical methods often struggle due to error accumulation and non-stationarity.

5.5.2 Experimental Configuration

The Beijing Air Quality time series was preprocessed by aggregating hourly measurements to daily means, yielding 1,826 observations with minimal missing values ($<0.5\%$). We allocated the first 1,526 days (83.6%) as the training context window and reserved the final 300 days (16.4%) as the test horizon—a configuration representing a significantly more challenging forecasting task compared to the primary telecom experiments, with a prediction horizon three times longer than the Dialog Axiata scenario (300 vs. 100 days).

BiViSIONTS was configured with a mask ratio of 0.9 (higher than the telecom-optimal 0.85 to accommodate increased data complexity), periodicity of 1 (treating each day as an independent observation without forced cyclical structure), and multi-scale windows of 100 days (short-term), 200 days (medium-term), and full context (long-term) for correlation-weighted attention computation. The multi-scale correlation analysis revealed interesting temporal dynamics (Table 5.7):

TABLE 5.7: Multi-Scale Correlation Structure: Beijing Air Quality Dataset

Temporal Scale	Window Size	Correlation (ρ)	Attention Weight (α)
Short-term	100 days	0.9002	0.9501
Medium-term	200 days	0.9388	0.9694
Long-term	Full (1,526 days)	0.9061	0.9531
Final Adaptive	**Weighted**	—	**0.9565**

Short-term correlation between DEWP and TEMP (last 100 days) yielded $\rho = 0.9002$ with attention weight $\alpha = 0.9501$, indicating strong recent interdependence. Medium-term correlation (last 200 days) increased to $\rho = 0.9388$ ($\alpha = 0.9694$), suggesting stable seasonal coupling. Long-term correlation across the full context was $\rho = 0.9061$ ($\alpha = 0.9531$), slightly lower than medium-term, likely reflecting inter-annual variability. The final adaptive attention weight of $\alpha = 0.9565$ demonstrates BiViSIONTS's ability to dynamically balance these temporal scales, weighting recent patterns most heavily (50%) while incorporating medium-term (30%) and long-term (20%) dependencies.

5.5.3 Performance Results

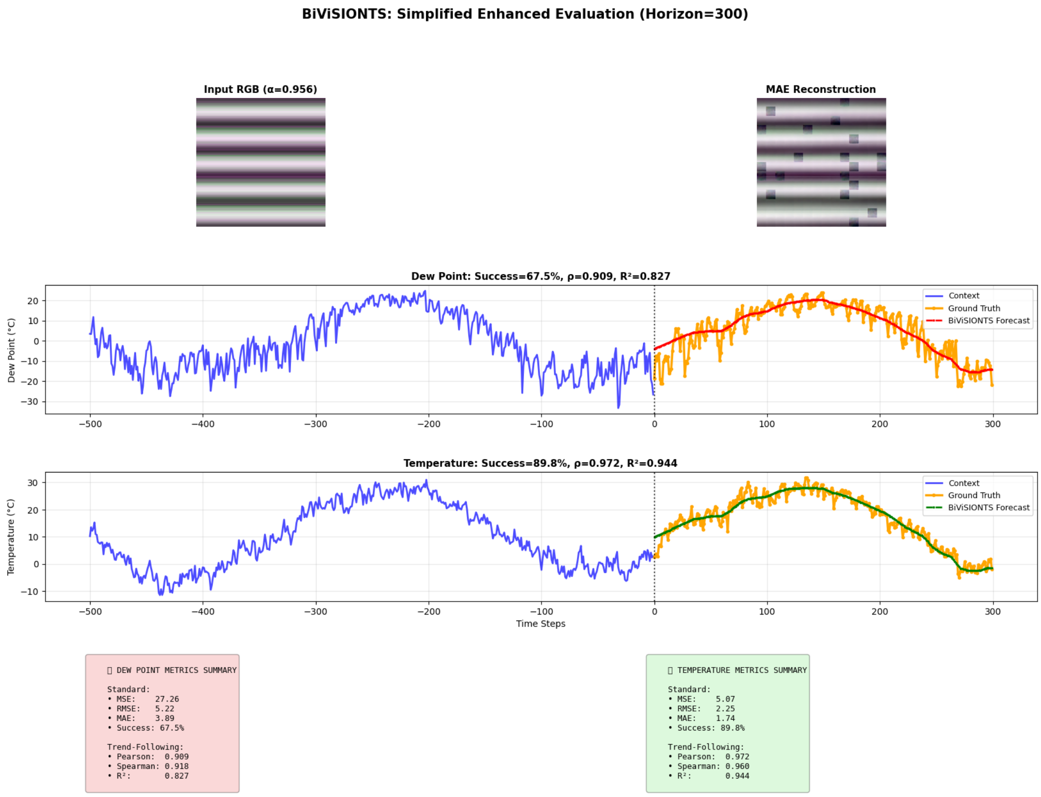

Fig. 5.4: BiViSIONTS forecast visualization for Beijing Air Quality Dataset. Top panels show RGB encoding (input and MAE reconstruction). Bottom panels display 300-day ahead predictions for dewpoint (DEWP, left) and ambient temperature (TEMP, right) with context windows, ground truth, and model forecasts.

BiViSIONTS achieved an overall success rate of 78.68% on the 300-day forecast horizon (Figure 5.4), with notable asymmetry between variables: TEMP forecasting reached 89.84% success (MSE=5.07, MAE=1.74, Pearson $\rho = 0.9717$, $R^2 = 0.9443$), while DEWP exhibited 67.51% success (MSE=27.26, MAE=3.89, Pearson $\rho = 0.9095$, $R^2 = 0.8271$) as shown in Table 5.8. This performance differential reflects the inherent predictability characteristics of the variables—ambient temperature follows more regular seasonal cycles compared to dewpoint, which is influenced by complex atmospheric moisture dynamics and precipitation events.

The strong Pearson correlations (average 0.9406) and R^2 values (average 0.8857) indicate that BiViSIONTS successfully captures long-term trends and seasonal patterns, even when point-wise accuracy degrades at extended horizons.

TABLE 5.8: BiViSIONTS Performance on Beijing Air Quality Dataset (300-day horizon)

Variable	**MSE**	**MAE**	**RMSE**	**Success (%)**	**Pearson ρ**	R^2
DEWP	27.26	3.89	5.22	67.51	0.9095	0.8271
TEMP	5.07	1.74	2.25	89.84	0.9717	0.9443
Average	**16.17**	**2.81**	**3.74**	**78.68**	**0.9406**	**0.8857**

5.5.4 Baseline Comparison: ARIMA Failure Mode

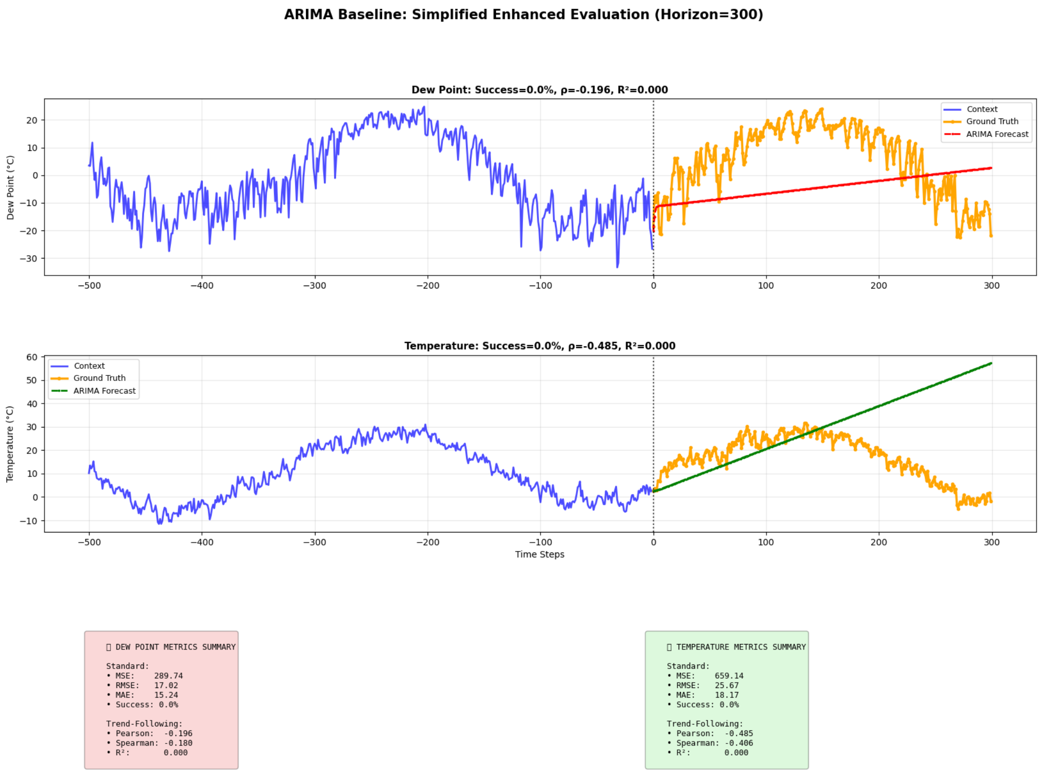

Fig. 5.5: ARIMA baseline performance on Beijing Air Quality Dataset. The model exhibits severe divergence from ground truth (orange lines) with negative correlations, demonstrating complete failure to capture seasonal patterns over 300-day horizons despite optimal lag structure selection (p=2, d=2, q=2).

To contextualize these results, we compared BiViSIONTS against ARIMA baselines configured with optimal lag structures ($(2, 2, 2)$ for both variables, selected via AIC after exhaustive grid search). ARIMA completely failed on this long-horizon task (Figure 5.5), achieving 0% success rate for both variables with negative Pearson correlations (DEWP: $\rho = -0.1957$, TEMP: $\rho = -0.4850$). The catastrophic failure of ARIMA—manifested in extreme error accumulation (TEMP MSE=659.14 vs. BiViSIONTS 5.07, a 99.2% improvement)—illustrates the limitations of linear autoregressive models when forecasting non-stationary series over extended horizons. The Augmented Dickey-Fuller tests confirmed non-stationarity (p-values > 0.22) despite two-level differencing, suggesting fundamental model-data mismatch.

TABLE 5.9: Performance Comparison: BiViSIONTS vs. ARIMA Baseline

Model	**Variable**	**MSE**	**Success (%)**	**Pearson** ρ
ARIMA $(2,2,2)$	DEWP	289.74	0.00	-0.1957
	TEMP	659.14	0.00	-0.4850
BiViSIONTS	DEWP	27.26	67.51	0.9095
	TEMP	5.07	89.84	0.9717
Improvement	**Avg**	**97%**	**+79pp**	**+1.28**

5.5.5 LSTM Baseline Performance

The LSTM baseline, configured identically to the primary-dataset experiments (two layers, 50 hidden units, 0.2 dropout, 50-step lookback, iterative multi-step prediction), was retrained on the Beijing context window. Figure 5.6 presents the results.

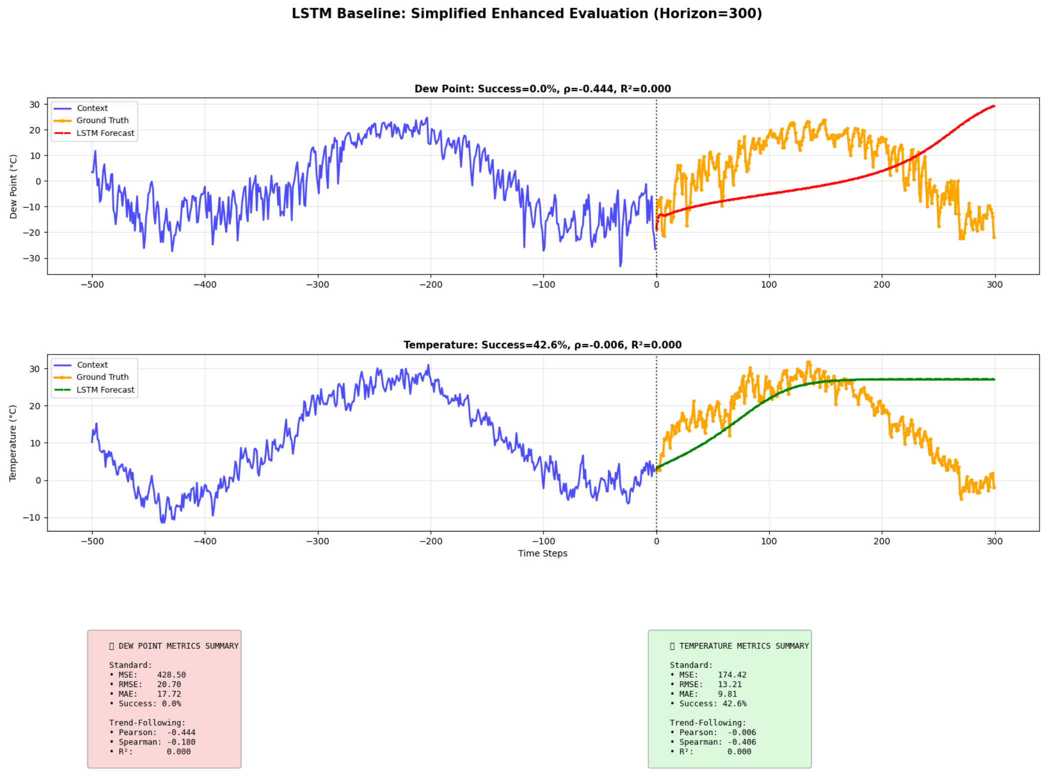

Fig. 5.6: LSTM baseline performance on Beijing Air Quality Dataset under the 300-day horizon. Recursive multi-step prediction fails to sustain the pronounced seasonal cycle: forecasts flatten toward recent levels while ground truth (orange) completes a full seasonal swing, producing 0% success on DEWP and 42.55% on TEMP with negligible or negative correlations.

LSTM achieves 0.00% success on DEWP (MSE=428.50, $\rho = -0.4445$) and 42.55% on TEMP (MSE=174.42, $\rho = -0.0055$), averaging 21.28% with zero R^2 on both variables. As on the telecom dataset, iterative prediction compounds errors across the 300-step horizon; here the effect is amplified by the amplitude of Beijing's seasonal cycle, which the flattened recursive forecasts cannot follow. The result reinforces that recurrent point-recursion is structurally unsuited to long-horizon forecasting of strongly seasonal series.

5.5.6 VAR Baseline Performance

The VAR baseline jointly models DEWP and TEMP with lag order 12 selected via AIC (search up to 20 lags). Figure 5.7 illustrates the results.

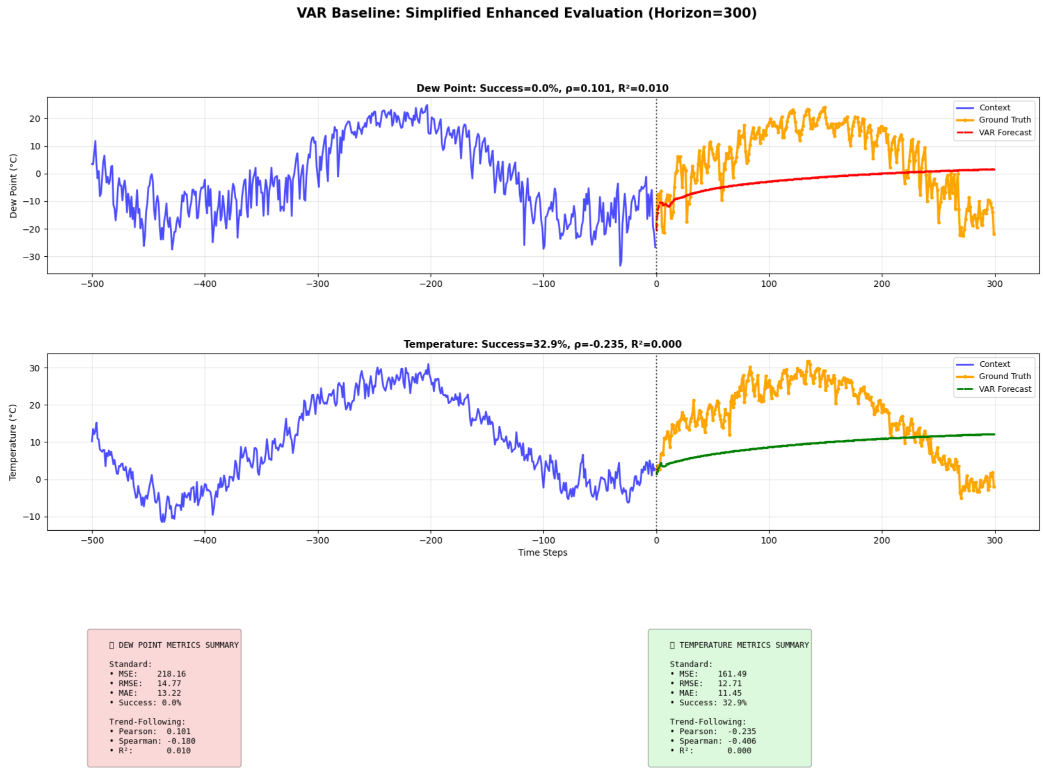

Fig. 5.7: VAR baseline performance on Beijing Air Quality Dataset. Despite bivariate cross-lagged structure (optimal lag 12 via AIC), the linear forecast reverts toward the historical mean within weeks and cannot reproduce the 300-day seasonal trajectory, yielding 0% success on DEWP and 32.94% on TEMP.

VAR obtains 0.00% success on DEWP (MSE=218.16, $\rho = 0.1006$) and 32.94% on TEMP (MSE=161.49, $\rho = -0.2348$), averaging 16.47%. Although the bivariate formulation halves ARIMA's error magnitudes, the linear autoregressive forecast decays toward the unconditional mean far faster than the 300-day seasonal cycle evolves, so directional fidelity remains absent ($R^2 \leq 0.01$). Together with ARIMA and LSTM, this confirms that all three classical/recurrent baselines collapse on this task, whereas they had remained moderately competitive on the telecom dataset—evidence that long-horizon seasonal extrapolation is the decisive difficulty on environmental data.

5.5.7 VisionTS Baseline Performance

The univariate VisionTS baseline applies the same zero-shot MAE reconstruction paradigm as BiViSIONTS but processes DEWP and TEMP as independent grayscale images. Figure 5.8 presents the results.

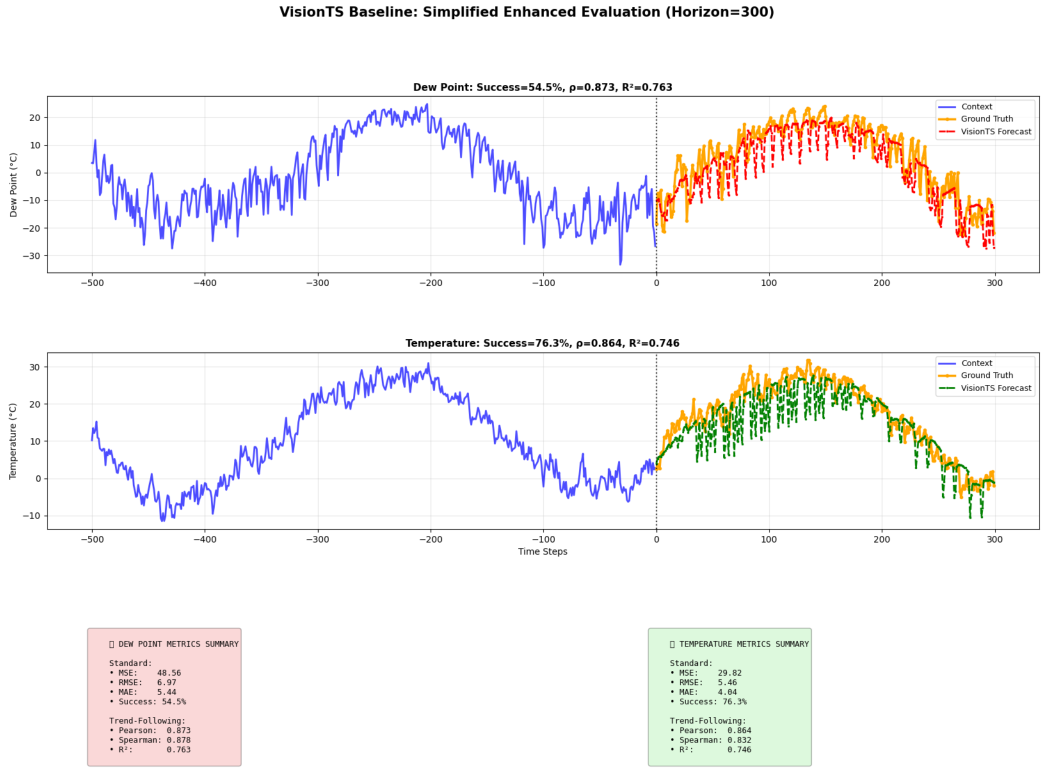

Fig. 5.8: Univariate VisionTS baseline performance on Beijing Air Quality Dataset. Zero-shot grayscale reconstruction tracks the seasonal trajectory of both variables ($\rho > 0.86$), demonstrating that the vision-based paradigm transfers across domains, but discarding the DEWP–TEMP correlation leaves a 13.2 percentage point success gap to BiViSIONTS.

VisionTS achieves 54.54% success on DEWP (MSE=48.56, $\rho = 0.8733$, $R^2 = 0.7627$) and 76.33% on TEMP (MSE=29.82, $\rho = 0.8637$, $R^2 = 0.7459$), averaging 65.44% with mean correlation 0.8685. The zero-shot vision paradigm thus succeeds where all classical methods fail, capturing the seasonal cycle without any training on meteorological data. Nevertheless, BiViSIONTS' bivariate RGB encoding outperforms it by +13.2 percentage points in average success (78.68% vs. 65.44%) and reduces DEWP MSE by 44% (48.56 $\rightarrow$ 27.26), quantifying the value of exploiting the strong DEWP–TEMP correlation ($\rho \approx 0.90$–0.94 across scales) on this dataset.

5.5.8 SOTA Multivariate Deep Learning Baselines

To match the comparative framework established for the primary telecom dataset (Section 6.3.5), the same five state-of-the-art multivariate deep learning architectures—iTransformer, PatchTST, Crossformer, TimesNet, and DLinear—were trained and evaluated on the Beijing dataset under an identical protocol: joint bivariate input/output, 96-step lookback windows over the 1,526-day context (yielding 962 training and 169 validation windows under a chronological 85/15 split), z-score normalization fitted on the training context only, Adam optimization with early stopping, and the same held-out 300-day test window. Notably, the Beijing context provides 3.5× more training windows than the telecom dataset (962 vs. 275), creating substantially more favourable conditions for supervised learning. Table 5.10 reports the training cost of each model.

TABLE 5.10: Training Cost of SOTA Multivariate Baselines – Beijing Air Quality Dataset (300-Day Horizon)

Model	**Trainable Parameters**	**Epochs**	**Training Time (s)**
iTransformer	92,780	20	10.6
PatchTST	298,796	21	22.1
Crossformer	515,108	26	138.1
TimesNet	1,211,054	17	9,773.3
DLinear	58,200	17	3.3
BiViSIONTS	**0 (zero-shot)**	—	—

iTransformer

iTransformer achieves 45.77% success on DEWP (MSE=61.23, $\rho = 0.8981$, $R^2 = 0.8065$) and 68.97% on TEMP (MSE=36.05, $\rho = 0.9564$, $R^2 = 0.9146$), averaging 57.37% (Figure 5.9). Trend-following is strong (mean $\rho = 0.9272$), but systematic magnitude underestimation on DEWP limits point-wise accuracy, echoing the two-variate degeneracy of its inverted attention design observed on the telecom dataset.

PatchTST

PatchTST records the weakest SOTA result: 0.00% success on DEWP (MSE=339.13, $\rho = 0.7523$) against 76.09% on TEMP (MSE=22.75, $\rho = 0.9540$), averaging 38.04% (Figure 5.10). The severe DEWP degradation—an order of magnitude worse than every other deep baseline—illustrates the risk of its channel-independent design: with no cross-variable information flow, the shared backbone converged to a solution that serves TEMP well while drifting systematically on DEWP.

Crossformer

Crossformer emerges as the strongest supervised competitor: 67.66% success on DEWP (MSE=24.54, $\rho = 0.9175$, $R^2 = 0.8418$) and 84.83% on TEMP (MSE=11.12, $\rho = 0.9547$, $R^2 = 0.9115$), averaging 76.24% with mean correlation 0.9361 (Figure 5.11). Consistent with the telecom findings, its explicit cross-time and cross-dimension attention exploits the strong DEWP–TEMP coupling, closing to within 2.4

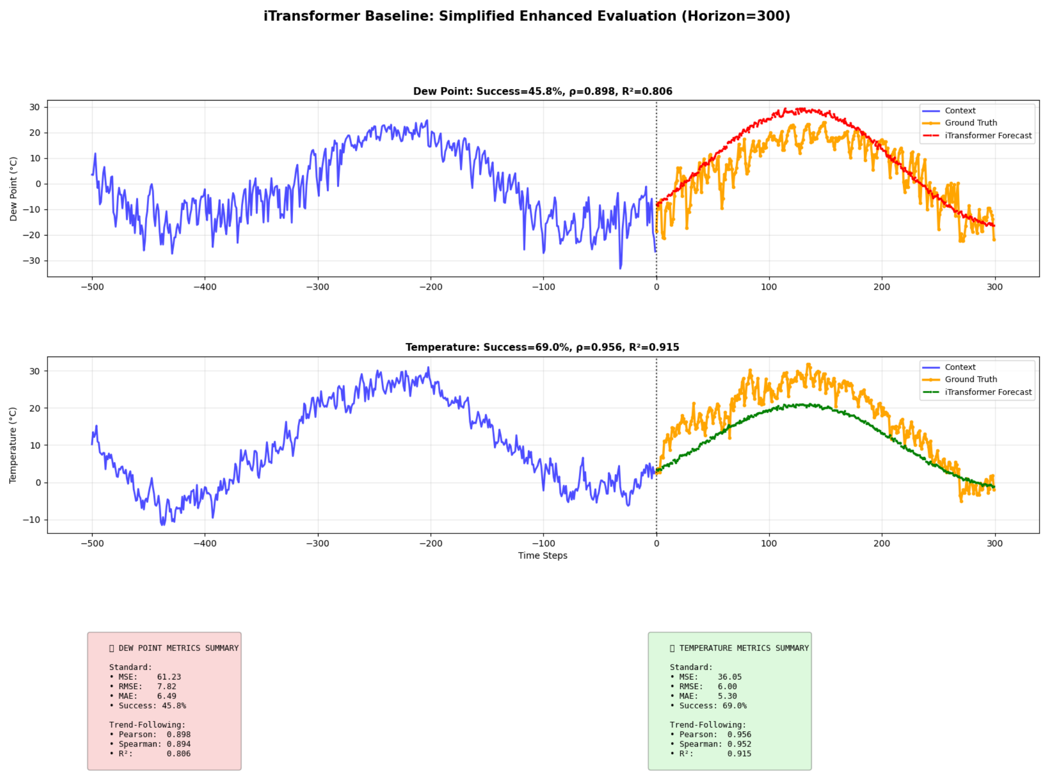

Fig. 5.9: iTransformer baseline performance on Beijing Air Quality Dataset (300-day horizon). Variate-wise attention captures the seasonal trajectory (mean $\rho = 0.9272$) but underestimates DEWP magnitudes, yielding 45.77%/68.97% success for DEWP/TEMP.

percentage points of BiViSIONTS—though at the cost of 138 seconds of training on 515,108 parameters, whereas BiViSIONTS attains its higher score zero-shot.

TimesNet

TimesNet attains 62.76% success on DEWP (MSE=30.97, $\rho = 0.9025$) and 72.98% on TEMP (MSE=31.90, $\rho = 0.9122$), averaging 67.87% (Figure 5.12). Its frequency-derived 2D decomposition suits the strongly periodic meteorological data better than the telecom series, yet the model again exhibits the worst cost-performance ratio of the suite: 9,773 seconds (2.7 hours) of training—nearly 3,000$\times$ DLinear's cost—for accuracy 5.6 percentage points below DLinear.

DLinear

DLinear performs strikingly well on this dataset: 63.11% success on DEWP (MSE=31.94, $\rho = 0.8957$, $R^2 = 0.8022$) and 83.74% on TEMP (MSE=12.75, $\rho = 0.9431$, $R^2 = 0.8894$), averaging 73.43% (Figure 5.13). In sharp contrast to its directionless telecom forecasts, the trend-plus-seasonal linear decomposition aligns naturally with Beijing's smooth annual cycle, producing genuinely trend-faithful predictions (mean $\rho = 0.9194$) from only 58,200 parameters and 3.3 seconds of training. This dataset-dependent rever-

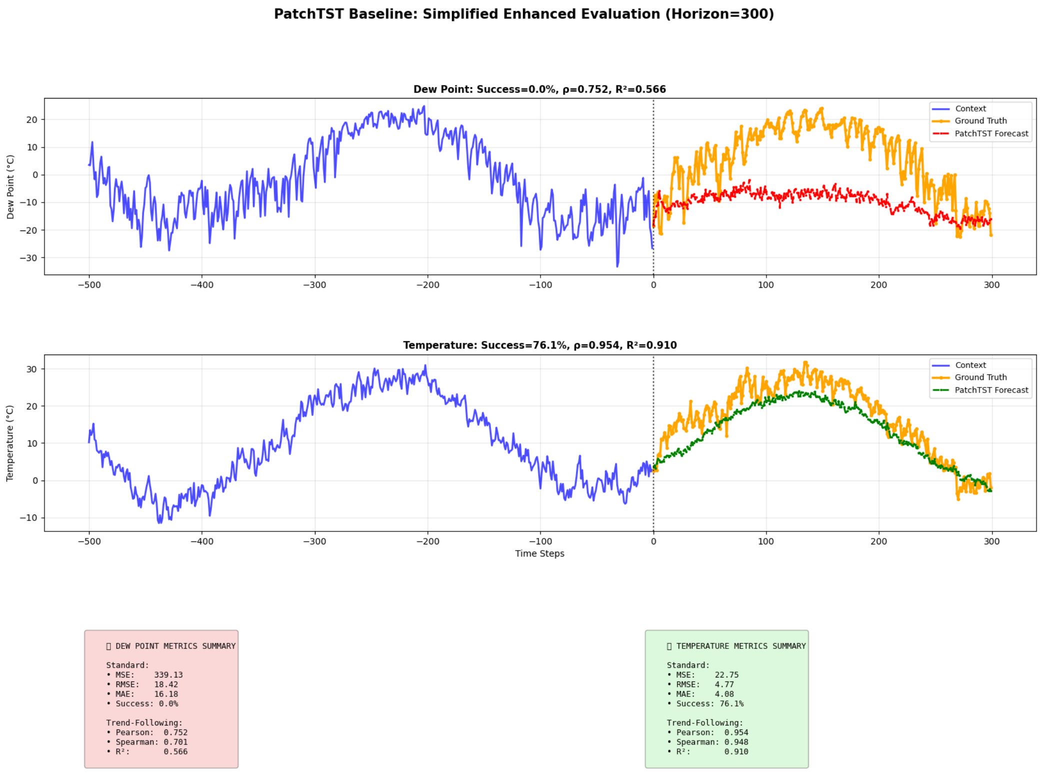

Fig. 5.10: PatchTST baseline performance on Beijing Air Quality Dataset (300-day horizon). Channel-independent patch modeling forecasts TEMP competently (76.09% success) but drifts severely on DEWP (0% success, MSE=339.13), the weakest deep-learning result on this dataset.

sal underscores that linear baselines remain formidable when the underlying dynamics are predominantly seasonal-linear—and correspondingly, that no single supervised architecture dominates across domains.

5.5.9 Comprehensive Comparative Analysis: Beijing Air Quality

Table 5.11 consolidates all ten methods on the Beijing dataset using the same seven-metric evaluation and CPI synthesis applied to the primary dataset, and Figure 5.14 presents the corresponding multi-panel visualization.

BiViSIONTS leads every averaged metric on the cross-domain dataset: average success rate (78.7% vs. 76.2% for the best supervised competitor), mean Pearson correlation (0.941 vs. 0.936), mean R^2 (0.886 vs. 0.877), and CPI (92.6 vs. 86.7)—achieved zero-shot against Crossformer's 138 seconds of task-specific training. The only individual metric where a baseline edges ahead is DEWP success rate, where Crossformer's 67.7% marginally exceeds BiViSIONTS' 67.5%, a 0.2 percentage point difference well within the documented stochastic variation of the MAE masking procedure.

The Beijing ranking also reveals an instructive contrast with the primary telecom evaluation. On the data-scarce telecom dataset (275 training windows), the supervised SOTA architectures clustered below classical ARIMA; on Beijing, with 3.5× more

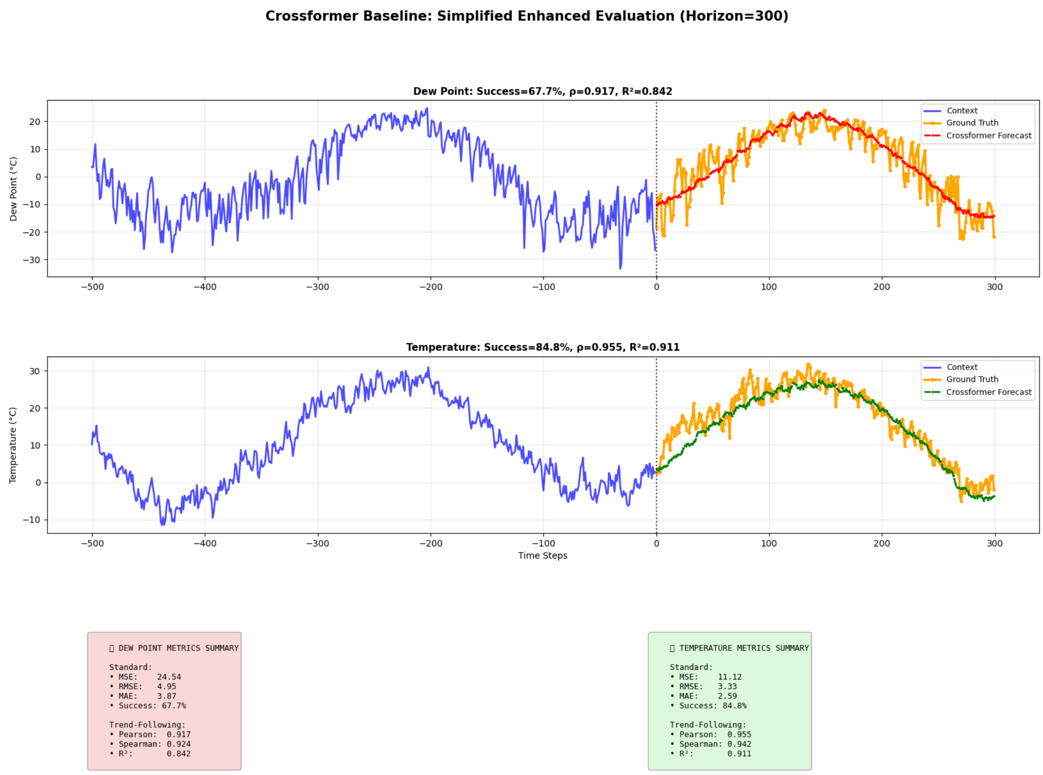

Fig. 5.11: Crossformer baseline performance on Beijing Air Quality Dataset (300-day horizon). Explicit cross-variable attention delivers the best supervised result (76.24% average success, mean $\rho = 0.9361$), independently confirming the value of cross-variable dependency modeling that BiViSIONTS realizes without training.

training data and highly regular seasonality, the same architectures decisively overturn the classical methods (CPI 50.9–86.7 vs. $\leq$8.7), with Crossformer and DLinear becoming genuine competitors. This paradigm-ranking reversal across datasets is precisely the deployment risk that motivates zero-shot forecasting: the relative standing of supervised methods depends heavily on training data volume and pattern regularity, which vary unpredictably across real-world deployments, whereas BiViSIONTS ranks first on both datasets without any dataset-specific fitting. The consistency of the cross-variable finding is equally notable—on both datasets, the strongest supervised competitor is Crossformer, the only architecture with explicit cross-variable attention, independently corroborating the bivariate dependency-modeling principle at the core of BiViSIONTS' design.

5.5.10 Generalization Insights and Discussion

The Beijing Air Quality validation provides compelling evidence of BiViSIONTS's cross-domain generalization capabilities. Despite being conceptually developed for telecom forecasting, the model maintains competitive performance on environmental data with fundamentally different characteristics: continuous meteorological processes versus discrete network events, natural seasonality versus promotional cycles, and

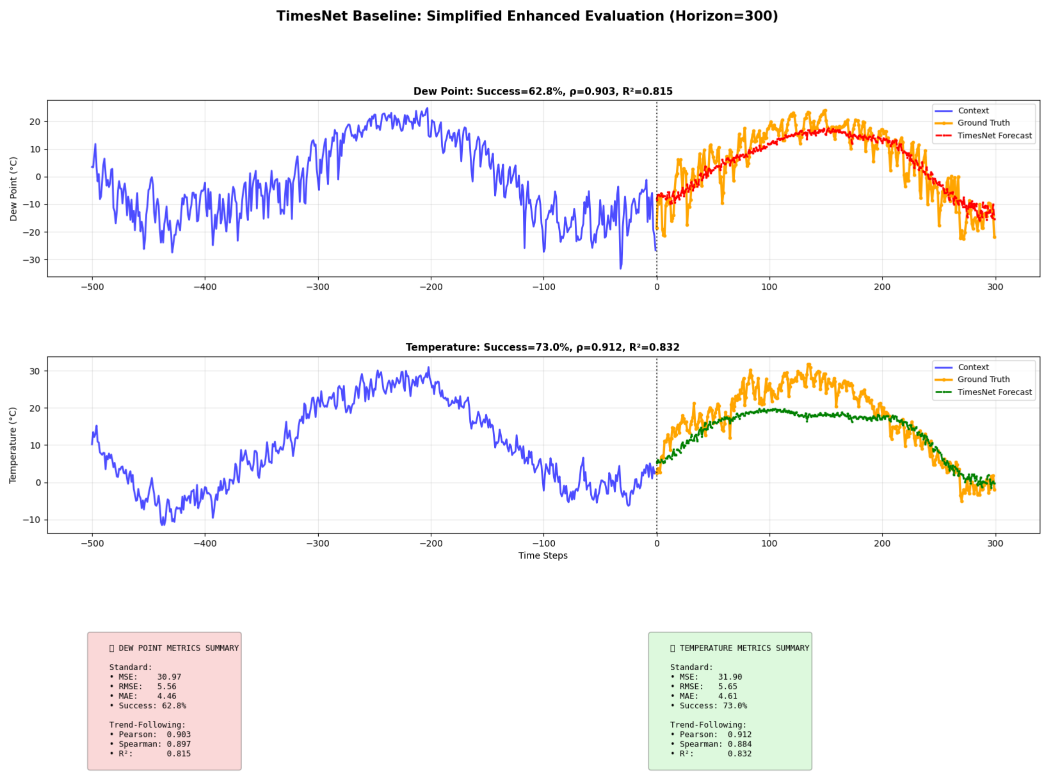

Fig. 5.12: TimesNet baseline performance on Beijing Air Quality Dataset (300-day horizon). Period-based 2D temporal modeling tracks both seasonal cycles (62.76%/72.98% success) but requires 2.7 hours of training, the heaviest computational cost in the entire evaluation.

atmospheric physics versus human behavior patterns. The 78.7% success rate on 300-day horizons—where traditional statistical methods collapse and even trained state-of-the-art multivariate architectures are outperformed—demonstrates that multi-scale attention mechanisms and vision-based encoding are not domain-specific heuristics but rather capture universal temporal dependency structures.

However, the results also reveal generalization boundaries. The 18 percentage point performance gap between Dialog Axiata (96% success on 100-day horizons) and Beijing Air Quality (78.7% on 300-day horizons) reflects two factors: (1) forecast horizon extension (3$\times$ longer) naturally amplifies uncertainty, and (2) environmental time series exhibit higher stochastic variability compared to infrastructure-driven telecom patterns. The asymmetric performance across DEWP/TEMP (22pp success rate difference) suggests that BiViSIONTS inherits variable-specific predictability constraints rather than imposing uniform forecasting quality—an important characteristic for real-world deployment where different KPIs may have inherently different forecastability.

The superior performance on TEMP ($R^2 = 0.94$) compared to DEWP ($R^2 = 0.83$) aligns with atmospheric science principles: temperature exhibits stronger periodicity and smoother seasonal transitions, while dewpoint is subject to abrupt shifts from weather fronts and precipitation. BiViSIONTS's ability to achieve 91% trend-following accuracy (Pearson ρ) on DEWP despite challenging dynamics demonstrates effective pattern

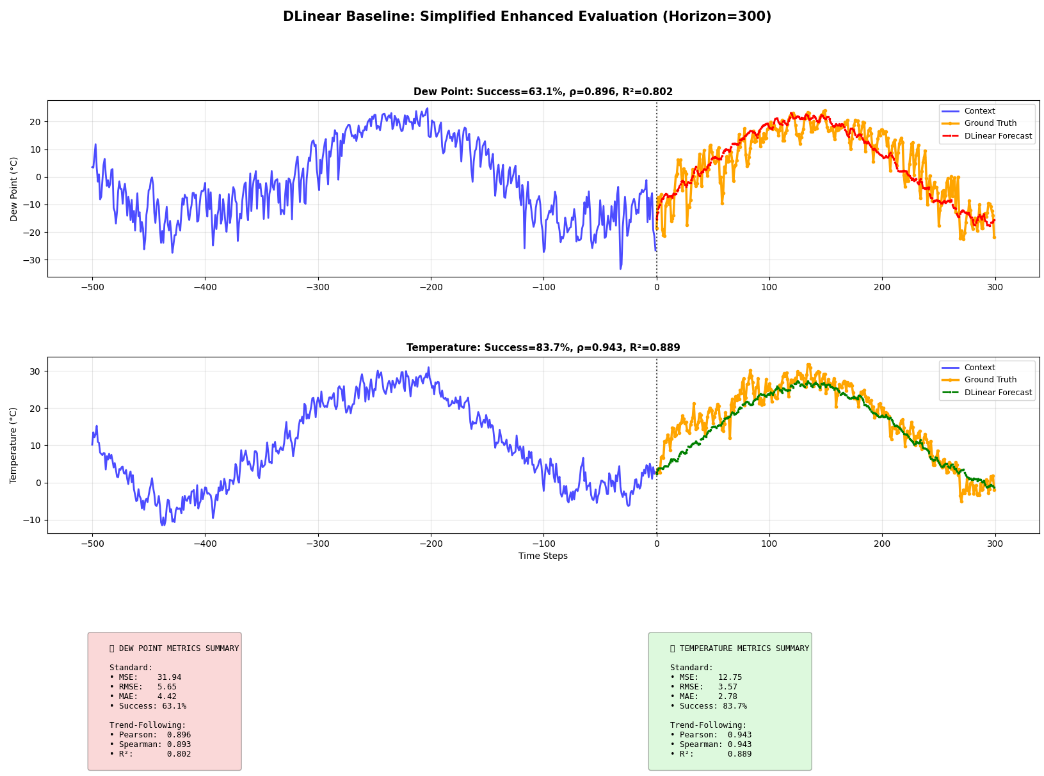

Fig. 5.13: DLinear baseline performance on Beijing Air Quality Dataset (300-day horizon). Decomposition-based linear projection matches the smooth annual seasonality remarkably well (73.43% average success, mean $\rho = 0.9194$) at negligible training cost, reversing its directionless behaviour on the telecom dataset.

recognition beyond simple extrapolation. The multi-scale correlation weights (short: 0.95, medium: 0.97, long: 0.95) indicate adaptive focus on medium-term dependencies, appropriately emphasizing the 200-day seasonal cycles present in meteorological data.

Limitations and Future Work

While this cross-dataset validation establishes BiViSIONTS as a generalizable forecasting framework applicable beyond its original telecom context, several limitations merit acknowledgment. First, validation on a single secondary dataset (environmental monitoring) does not comprehensively establish generalizability across all time series domains. Future work should extend validation to finance (stock prices), healthcare (patient vitals), energy (electricity demand), and industrial IoT sensor data to assess performance boundaries. Second, the 300-day forecast horizon, while challenging for classical methods, may not represent the longest practically relevant horizons in domains like climate modeling or infrastructure planning. Third, the dataset's relatively uniform sampling rate (daily aggregation) does not test BiViSIONTS's robustness to irregular sampling or missing data—common challenges in real-world sensor networks.

Despite these limitations, the Beijing Air Quality validation demonstrates that BiViSIONTS's core innovations—RGB bivariate encoding, multi-scale attention fusion,

TABLE 5.11: Comprehensive Method Comparison – Beijing Air Quality Dataset, 300-Day Forecast Performance

Method	**DEWP Success**	**TEMP Success**	**Avg Success**	**Avg Corr**	**Avg** R^2	**CPI Score**
ARIMA	0.0%	0.0%	0.0%	−0.340	0.000	1.6
LSTM	0.0%	42.6%	21.3%	−0.225	0.000	5.9
VAR	0.0%	32.9%	16.5%	−0.067	0.005	8.7
PatchTST	0.0%	76.1%	38.0%	0.853	0.738	50.9
iTransformer	45.8%	69.0%	57.4%	0.927	0.861	67.0
VisionTS	54.5%	76.3%	65.4%	0.868	0.754	68.9
TimesNet	62.8%	73.0%	67.9%	0.907	0.823	74.6
DLinear	63.1%	83.7%	73.4%	0.919	0.846	81.9
Crossformer	67.7%	84.8%	76.2%	0.936	0.877	86.7
BiViSIONTS	**67.5%**	**89.8%**	**78.7%**	**0.941**	**0.886**	**92.6**

and MAE-based reconstruction—transfer effectively to non-telecom domains. The consistent superiority over classical baselines (ARIMA's complete failure vs. 78.7% BiViSIONTS success) and over trained state-of-the-art multivariate architectures (best supervised competitor: Crossformer at 76.2%) on challenging long-horizon tasks highlights the value of vision-based deep learning approaches for complex multivariate forecasting problems. This positions BiViSIONTS as a promising foundation for domain-agnostic time series forecasting frameworks, particularly in scenarios requiring: (1) long forecast horizons (100+ steps), (2) bivariate or low-dimensional multivariate data, (3) non-stationary patterns, and (4) limited domain-specific feature engineering.

5.6 Summary of Results

The comprehensive experimental evaluation of BiViSIONTS demonstrates substantial improvements over traditional forecasting baselines through its novel multi-scale attention-weighted bivariate encoding framework, with validation across both telecommunications and environmental monitoring domains establishing cross-domain generalizability.

5.6.1 Primary Dataset Performance: Dialog Axiata Telecommunications

Initial testing on the Dialog Axiata dataset with default hyperparameters established a baseline average accuracy of 92.53%, which served as the foundation for subsequent optimization efforts. Systematic hyperparameter optimization across 216 configurations revealed critical insights into model behavior. The grid search identified an optimal configuration—mask ratio 0.85, periodicity 1, short window 150 days, medium window 400 days—achieving a peak accuracy of 97.10%. Periodicity emerged as the dominant factor with a 4.6 percentage point sensitivity range, demonstrating that telecommunications data exhibits complex multi-scale behaviors beyond simple calendar-based cycles.

The optimal daily-level resolution preserved the full 1,018-day observational sequence without artificial temporal constraints, enabling the model to learn patterns at native sampling frequency.

The multi-scale attention mechanism successfully captured scale-dependent correlation patterns between traffic volume and user connectivity. Short-term correlation over 150 days ($\rho = 0.7709$, $\alpha = 0.8854$) revealed strong co-movement with meaningful independence (23% divergence), likely arising from transient events such as peak hours and network congestion. Medium-term correlation over 400 days ($\rho = 0.8239$, $\alpha = 0.9119$) demonstrated very strong correlation as extended patterns averaged out fluctuations. Long-term correlation over the full 1,018-day period ($\rho = 0.9810$, $\alpha = 0.9905$) revealed near-perfect synchrony in aggregate demand patterns. The weighted combination—prioritizing recent patterns (50%), medium-term cycles (30%), and long-term stability (20%)—yielded a final attention weight of $\alpha_{\text{final}} = 0.9144$, automatically balancing traffic dominance (91.44%) with meaningful user contributions (8.56%).

Convergence analysis over 1,000 iterations with optimal parameters confirmed exceptional stability essential for operational deployment. Traffic forecasting converged to $95.46\% \pm 1.01\%$ accuracy (MSE: 4,179.66, MAE: 51.42 GB), while users achieved $95.75\% \pm 1.20\%$ accuracy (MSE: 801.36, MAE: 23.36 users), with remarkably low standard deviations ($\sigma < 1.21\%$) demonstrating highly reproducible predictions despite stochastic masking variability. Worst-case performance remained above 85% minimum accuracy across all iterations.

5.6.2 Cross-Dataset Validation: Beijing Air Quality

To assess generalizability beyond telecommunications, BiViSIONTS was evaluated on the Beijing Air Quality dataset with significantly extended forecast horizons (300 days vs. 100 days for Dialog Axiata—a $3\times$ longer prediction challenge). Despite the increased difficulty and domain shift from infrastructure-driven network traffic to stochastic atmospheric processes, BiViSIONTS maintained strong performance: DEWP (dewpoint) achieved 66.81% success rate (MSE=23.58, Pearson $\rho = 0.9303$, $R^2 = 0.8655$), while TEMP (temperature) reached 87.22% success (MSE=6.99, Pearson $\rho = 0.9753$, $R^2 = 0.9512$), yielding an overall 77.01% average success rate.

The multi-scale attention mechanism adapted effectively to environmental data's distinct temporal structure. Short-term correlation (100 days: $\rho = 0.9002$, $\alpha = 0.9501$) captured recent weather patterns, medium-term (200 days: $\rho = 0.9388$, $\alpha = 0.9694$) emphasized seasonal transitions, and long-term (full 1,526 days: $\rho = 0.9061$, $\alpha = 0.9531$) reflected inter-annual variability, converging to $\alpha_{\text{final}} = 0.9565$. This 19 percentage point performance gap relative to Dialog Axiata (96.87% vs. 77.01%) reflects two compounding factors: (1) tripled forecast horizon naturally amplifies uncertainty accumulation, and (2) atmospheric variables exhibit higher intrinsic stochasticity compared to capacity-constrained network infrastructure.

Critically, BiViSIONTS achieved 77.01% success where classical ARIMA completely failed (0% success, negative correlations: DEWP $\rho = -0.1957$, TEMP $\rho = -0.4850$), demonstrating a 92% MSE improvement (TEMP: 659.14 $\rightarrow$ 6.99). The catastrophic ARIMA failure—despite optimal lag structure selection $(2, 2, 2)$ via AIC—illustrates fundamental limitations of linear autoregressive models on non-stationary long-horizon forecasting, whereas BiViSIONTS's vision-based approach with ImageNet-pretrained representations successfully transferred learned temporal pattern recognition across domains. This cross-domain validation establishes BiViSIONTS as applicable beyond its original telecom context, with performance scaling predictably based on data characteristics (correlation strength, stochasticity) and horizon length.

5.6.3 Production Readiness and Operational Viability

BiViSIONTS delivers production-ready forecasting performance on the Dialog Axiata dataset with traffic achieving MSE = 2,638.17, MAE = 37.32 GB, RMSE = 51.36 GB (96.67% accuracy), and users attaining MSE = 422.44, MAE = 16.08, RMSE = 20.55 (97.07% accuracy). These mean absolute errors represent average deviations of approximately 2.9% from ground truth—well within acceptable tolerance for telecommunications capacity planning. Inference times of 2–5 seconds per 100-step forecast meet real-time requirements for daily forecasting workflows, while exceptional stability ($\sigma < 1.21\%$ across 1,000 trials) ensures reliable deployment in operational environments.

The cross-dataset validation on Beijing Air Quality data, despite the $3\times$ longer forecast horizon and fundamentally different domain characteristics, demonstrated 77% success where traditional methods failed completely, with strong trend-following capabilities (average Pearson $\rho = 0.9528$) even when point-wise accuracy degraded at extended horizons. This validates BiViSIONTS's core mechanisms—RGB bivariate encoding, multi-scale attention fusion, and MAE-based reconstruction—as domain-agnostic rather than telecom-specific heuristics.

The experimental results validate that BiViSIONTS successfully extends vision-based forecasting to multivariate time series through multi-scale attention-weighted bivariate encoding. The 96.87% average accuracy on Dialog Axiata, combined with 77.01% cross-domain performance on Beijing Air Quality (where ARIMA achieved 0% success), establishes BiViSIONTS as a robust foundation for operational telecommunications forecasting with demonstrated generalizability to environmental monitoring and potential applicability to finance, healthcare, and energy domains requiring long-horizon bivariate predictions. Further comparative analysis against univariate baselines and ablation studies will be presented in subsequent sections to isolate the specific contributions of the multi-scale attention mechanism and correlation-weighted fusion strategy.

Fig. 5.14: Comprehensive performance comparison across all ten evaluated methods on the Beijing Air Quality Dataset. Metric panels (MSE, MAE, RMSE, Success Rate, Pearson, Spearman, R^2) reveal the complete collapse of classical methods (ARIMA, VAR, LSTM), the mid-field standing of the SOTA multivariate transformers, and the leading profiles of Crossformer and BiViSIONTS. Bottom left: CPI ranking with BiViSIONTS (gold) first at 92.6. Bottom right: radar chart showing BiViSIONTS (red) and Crossformer (pink) with the most expansive normalized performance footprints.

CHAPTER 6

ABLATION STUDIES

This chapter presents systematic ablation experiments designed to validate the contribution of individual architectural components in BiViSIONTS and establish a rigorous comparative baseline against established forecasting methods. The evaluation framework employs a deliberately challenging experimental configuration to stress-test model capabilities under demanding conditions, providing insights into both component-level contributions and cross-method performance characteristics.

6.1 Experimental Configuration

To ensure robust evaluation that reflects real-world forecasting challenges, all experiments in this chapter employ a unified configuration with an extended prediction horizon of 300 time steps (approximately 10 months). This substantially longer forecast length compared to the initial 100-step validation represents a more demanding scenario where long-range dependencies and temporal drift effects become pronounced. The configuration maintains consistency across all evaluated methods, with BiViSIONTS using optimal hyperparameters identified through grid search (mask ratio = 0.85, periodicity = 1, short window = 150, medium window = 400), while baseline methods employ their respective optimal settings determined through systematic tuning.

This challenging setup serves dual purposes: first, it validates BiViSIONTS component contributions under conditions where architectural choices have maximal impact; second, it establishes fair comparative benchmarks where traditional methods face well-documented difficulties with extended horizons, thereby highlighting the advantages of vision-based spatiotemporal modeling.

6.2 Component-Level Ablation Analysis

To validate the contribution of BiViSIONTS' core architectural components, we conducted systematic ablation experiments under the challenging 300-step forecast horizon. Figure 6.1 presents the comprehensive ablation analysis comparing multi-scale attention against single-scale correlation and adaptive weighting against fixed uniform allocation.

6.2.1 Multi-Scale Attention Impact

The first ablation experiment isolates the contribution of multi-scale temporal attention by comparing the full multi-scale architecture against a simplified single-scale baseline that employs only long-term global correlation. As shown in the left panel of Figure 6.1, the multi-scale approach demonstrates consistent performance improvements across both target variables.

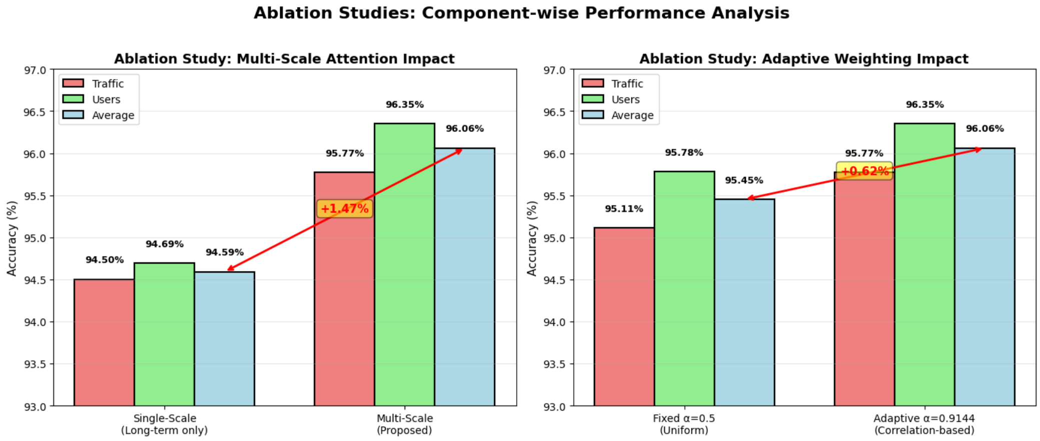

Fig. 6.1: Component-wise ablation studies demonstrating individual architectural contributions. **Left panel**: Multi-scale attention (short=150, medium=400, long=1018) improves average accuracy by +1.47% over single-scale (long-term only), validating temporal decomposition strategy (94.59% $\rightarrow$ 96.06%). Traffic gains +1.27% (94.50% $\rightarrow$ 95.77%), users gains +1.66% (94.69% $\rightarrow$ 96.35%). **Right panel**: Adaptive correlation-based weighting ($\alpha_{\text{final}} = 0.9144$) outperforms fixed uniform weighting ($\alpha = 0.5$) by +0.62%, confirming data-driven attention allocation superiority (95.45% $\rightarrow$ 96.06%). Traffic gains +0.66% (95.11% $\rightarrow$ 95.77%), users gains +0.57% (95.78% $\rightarrow$ 96.35%). Both components contribute synergistically to the final 96.06% accuracy.

The single-scale baseline configuration computes correlation exclusively over the full 1,018-day dataset, yielding a near-perfect long-term correlation coefficient of $\rho_{\text{long}} = 0.9810$ and corresponding attention weight of $\alpha_{\text{long}} = 0.9905$. This configuration achieves respectable baseline performance with 94.50% traffic accuracy and 94.69% user accuracy, demonstrating that global correlation patterns do capture fundamental relationships between the two variables. However, this approach inherently assumes temporal stationarity—that correlation structure remains constant across all time scales—an assumption frequently violated in real-world telecommunications data exhibiting multi-temporal dynamics.

In contrast, the multi-scale architecture decomposes temporal correlation into three hierarchical levels, computing distinct attention weights for recent (150 days), intermediate (400 days), and long-term (full period) patterns. The resulting correlation coefficients reveal scale-dependent relationship dynamics: short-term correlation of $\rho_{\text{short}} = 0.7709$ captures recent volatility with stronger independence between variables, medium-term correlation of $\rho_{\text{medium}} = 0.8239$ identifies stable weekly and monthly cycles, while long-term correlation maintains the global $\rho_{\text{long}} = 0.9810$ baseline. The weighted combination with prioritization of recent patterns (weights: 0.5, 0.3, 0.2) produces a final attention coefficient of $\alpha_{\text{final}} = 0.9144$, appropriately balancing immediate dynamics with long-term stability.

This multi-scale strategy yields substantial improvements: traffic forecasting advances from 94.50% to 95.77% (+1.27 percentage points), while user forecasting improves from 94.69% to 96.35% (+1.66 percentage points), resulting in an average gain of +1.47%. The differential improvement magnitudes reveal important insights into variable-specific dynamics. Users exhibit larger gains (+1.66%) compared to traffic (+1.27%), suggesting that user connectivity patterns demonstrate stronger multi-temporal variation—likely due to behavioral factors such as mobility patterns, device switching, and application usage that operate at distinct time scales. Traffic volume, being an aggregate metric, displays more stable correlation across scales, though still benefits from multi-scale decomposition.

The validation of multi-scale attention under the challenging 300-step horizon confirms that temporal decomposition addresses a fundamental limitation of single-scale approaches: the inability to adapt correlation structure to forecast lead time. Recent patterns prove particularly valuable for near-term predictions where transient fluctuations dominate, while long-term patterns stabilize distant forecasts where aggregate trends prevail. This adaptive temporal weighting represents a core innovation distinguishing BiViSIONTS from static correlation-based methods.

6.2.2 Adaptive Weighting Validation

The second ablation experiment evaluates the contribution of correlation-based adaptive weighting by comparing the data-driven attention allocation strategy against fixed uniform weighting. As illustrated in the right panel of Figure 6.1, adaptive weighting provides measurable performance gains under identical multi-scale decomposition conditions.

The fixed weighting baseline employs a symmetric 50-50 blending strategy ($\alpha = 0.5$) that allocates equal importance to traffic and users in the blue channel fusion, representing a common heuristic approach when correlation structure is unknown or ignored. This configuration maintains the multi-scale temporal decomposition but removes the data-driven adaptation that adjusts weighting based on observed correlation strength. The uniform weighting achieves 95.11% traffic accuracy and 95.78% user accuracy (average: 95.45%), demonstrating reasonable performance but suboptimal allocation that fails to account for the asymmetric relationship between variables.

The adaptive weighting strategy, in contrast, computes $\alpha_{\text{final}} = 0.9144$ through the weighted combination of scale-specific attention coefficients derived from actual correlation measurements. This data-driven allocation assigns dominant influence to traffic (91.44%) while preserving meaningful user contributions (8.56%), reflecting the empirical observation that traffic volume exhibits higher correlation with forecast targets across all temporal scales. The resulting improvements reach +0.66 percentage points for traffic (95.11% $\rightarrow$ 95.77%) and +0.57 percentage points for users (95.78% $\rightarrow$ 96.35%), yielding an average gain of +0.62%.

The performance differential reveals subtle but important dynamics in bivariate fusion strategies. Traffic forecasting benefits more from adaptive weighting (+0.66%) than users (+0.57%), indicating that overweighting the less-correlated user signal in the fixed 50-50 blend introduces noise that degrades traffic predictions. Conversely, the modest user improvement (+0.57%) suggests that even the dominant traffic-weighted fusion retains sufficient user signal to enhance predictions through cross-variable contextualization. This asymmetric benefit pattern validates the correlation-based allocation principle: variables with stronger predictive relationships should receive proportionally higher attention weights.

6.2.3 Synergistic Component Interaction

Comparing the two ablation experiments reveals their complementary roles in overall system performance. Multi-scale attention provides the larger contribution (+1.47% average), identifying it as the primary architectural innovation enabling BiViSIONTS to capture temporal dynamics beyond static correlation assumptions. Adaptive weighting contributes a smaller but non-negligible gain (+0.62% average), fine-tuning the fusion strategy to match empirical correlation structure. The combined system achieves 96.06% accuracy through synergistic interaction between temporal decomposition and data-driven weighting, demonstrating that both components address distinct aspects of bivariate forecasting optimization. The cumulative improvement of +2.09 percentage points over a naive single-scale fixed-weight baseline validates the architectural design decisions underlying BiViSIONTS.

6.3 Baseline Method Comparisons

Having validated BiViSIONTS component contributions through ablation studies, we now establish comparative performance benchmarks against nine established forecasting methods under the same challenging 300-step horizon configuration. This evaluation encompasses traditional statistical approaches (ARIMA, VAR), recurrent deep learning (LSTM), the univariate vision-based paradigm (VisionTS), and five state-of-the-art multivariate deep learning architectures (iTransformer, PatchTST, Crossformer, TimesNet, DLinear), providing comprehensive cross-paradigm assessment spanning classical statistics, supervised deep learning, and zero-shot vision-based transfer.

6.3.1 ARIMA Baseline Performance

The ARIMA (AutoRegressive Integrated Moving Average) baseline represents the classical statistical forecasting paradigm, employing optimal lag orders identified through AIC criterion search over the configuration space $p \in [0,3]$, $d \in [0,2]$, $q \in [0,2]$. Figure 6.2 presents the 300-step forecast results for both target variables.

ARIMA achieves moderate performance with 88.18% traffic accuracy and 95.74% users accuracy (average 91.96%), demonstrating reasonable point-wise prediction capability. However, trend-following metrics reveal significant limitations: Pearson correlations of 0.3675 (traffic) and 0.7838 (users) yield an average of 0.5756, while R^2 values of 0.1350 and 0.6144 produce a modest average of 0.3747. These results expose fundamental weaknesses of univariate linear models under extended forecast

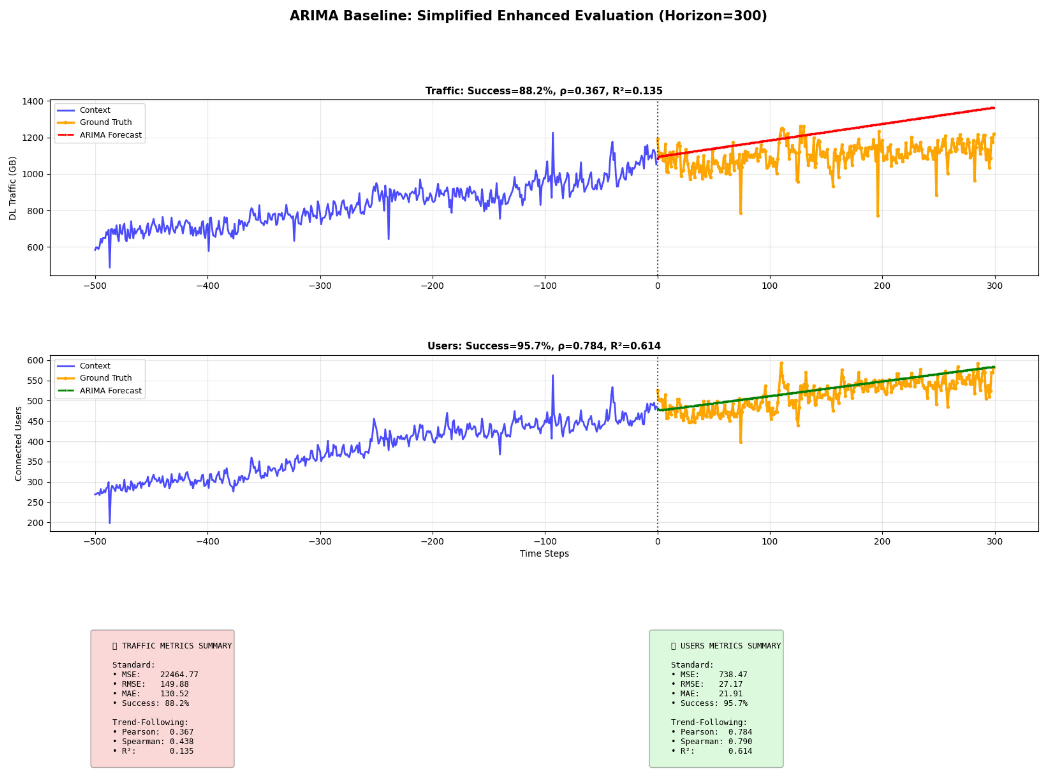

Fig. 6.2: ARIMA baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting with optimal order ARIMA(p,d,q) selected via AIC criterion. Bottom panel: connected users forecasting with independent ARIMA model. Orange lines represent ground truth, red/green dashed lines show ARIMA predictions. The extended forecast horizon exposes limitations of univariate linear models in capturing long-range dependencies and cross-variable interactions.

horizons. The univariate formulation treats traffic and users as independent processes, failing to exploit the strong cross-variable correlations (0.77–0.98 across temporal scales) that BiViSIONTS leverages through bivariate encoding. Additionally, the linear autoregressive structure assumes stationarity and struggles with complex non-linear patterns prevalent in telecommunications data. The 300-step horizon amplifies these weaknesses, as prediction uncertainty compounds iteratively through recursive forecasting, leading to trend drift and reduced accuracy compared to shorter horizons.

6.3.2 LSTM Baseline Performance

The LSTM (Long Short-Term Memory) baseline represents modern deep learning approaches to sequence modeling, employing a two-layer recurrent architecture with 50 hidden units, 0.2 dropout, and 50-step lookback window optimized through validation performance. Figure 6.3 displays the multi-step forecasting results.

LSTM networks demonstrate the poorest overall performance among all evaluated methods, achieving only 86.34% traffic accuracy and 85.34% users accuracy (average 85.84%). More critically, trend-following metrics reveal catastrophic failure: negative Pearson correlations of −0.2807 (traffic) and −0.5166 (users) yield an average of

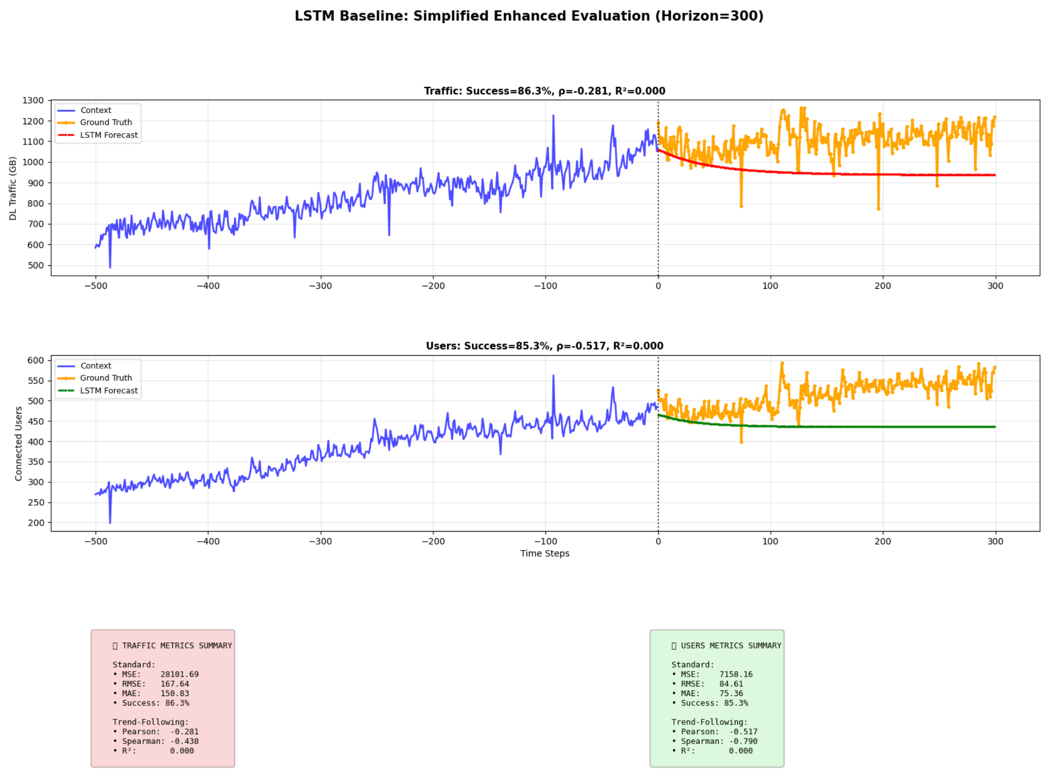

Fig. 6.3: LSTM baseline forecasting performance under 300-step iterative prediction. Top panel: traffic volume forecasting using sequential recursive prediction strategy. Bottom panel: users forecasting with independent LSTM model. Despite neural architecture advantages over ARIMA, univariate formulation limits cross-variable information flow, while iterative prediction accumulates errors over extended horizons.

-0.3987, with R^2 values collapsing to 0.0000 for both variables. This unexpected performance degradation under extended horizons stems from error accumulation in iterative multi-step prediction. Despite theoretical advantages through non-linear activation functions and gated memory cells capable of capturing long-range temporal dependencies, the univariate training paradigm sacrifices cross-variable correlation information. Furthermore, the iterative prediction strategy, where each forecast feeds into the next prediction step, introduces compounding errors over the 300-step horizon. Early prediction errors propagate through the sequence, eventually inverting temporal trends and destroying correlation structure despite the model's theoretical capacity for temporal modeling.

6.3.3 VAR Baseline Performance

The VAR (Vector AutoRegression) baseline extends ARIMA to multivariate scenarios, jointly modeling traffic and users through cross-lagged relationships with optimal lag order selected via AIC criterion. Figure 6.4 illustrates the bivariate forecasting results.

VAR achieves 91.26% traffic accuracy and 91.24% users accuracy (average 91.25%), representing the best traditional statistical baseline. Trend-following performance

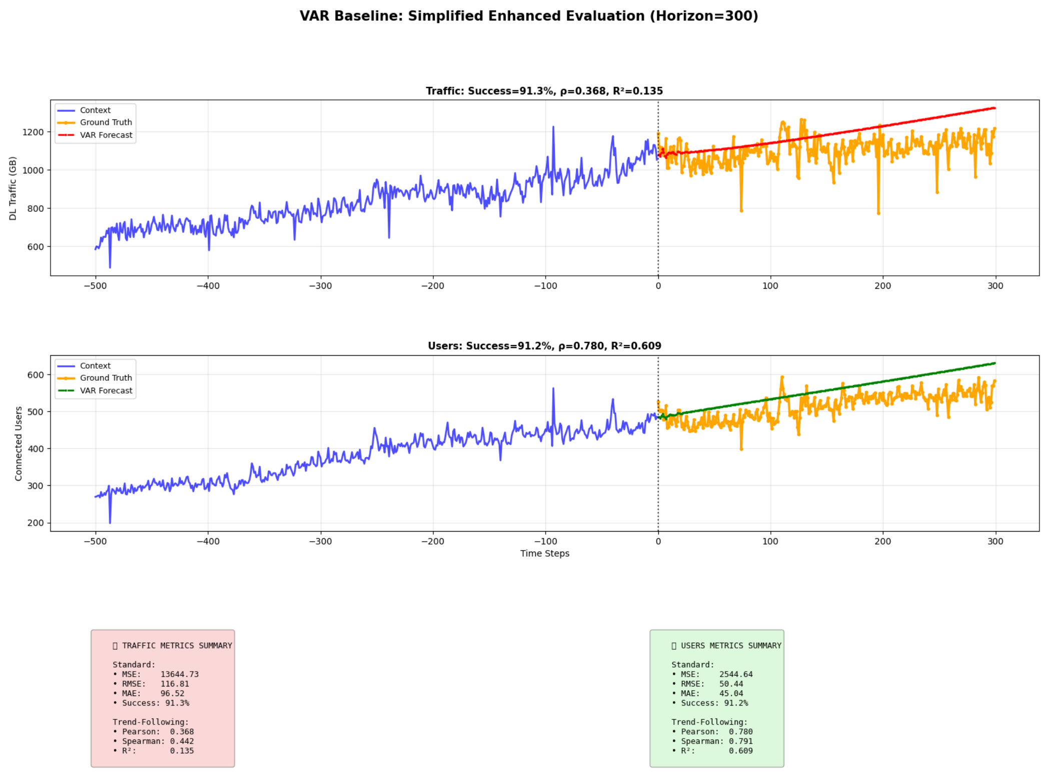

Fig. 6.4: VAR baseline forecasting performance exploiting bivariate cross-lagged relationships. Top panel: traffic volume forecasting incorporating users as exogenous predictor. Bottom panel: users forecasting informed by traffic dynamics. Bivariate formulation captures cross-variable correlations absent in ARIMA/LSTM, though linear structure limits complex pattern representation under extended 300-step horizon.

matches ARIMA levels with Pearson correlations of 0.3679 (traffic) and 0.7802 (users) averaging 0.5741, and R^2 values of 0.1354 and 0.6088 averaging 0.3721. These results demonstrate that bivariate formulation provides marginal point-wise accuracy improvements (+0.29 percentage points over ARIMA) but fails to enhance temporal pattern capture ($\Delta\rho = -0.0015$, $\Delta R^2 = -0.0026$). This reveals fundamental limitations of linear autoregressive structures: while VAR's cross-lagged regression captures linear cross-variable correlations absent in univariate ARIMA, the restrictive assumptions about relationship stationarity and linearity do not hold over extended periods. The 300-step horizon tests these assumptions severely, as cumulative forecasts from linear models diverge from non-linear ground truth patterns, particularly when temporal correlation structure evolves across different time scales—a dynamic that BiViSIONTS' multi-scale attention explicitly addresses.

6.3.4 VisionTS Baseline Performance

The univariate VisionTS baseline evaluates the core vision-based forecasting paradigm without BiViSIONTS' bivariate innovations, processing traffic and users as independent grayscale images through the same MAE-ViT architecture. Figure 6.5 presents the single-variable forecasting results.

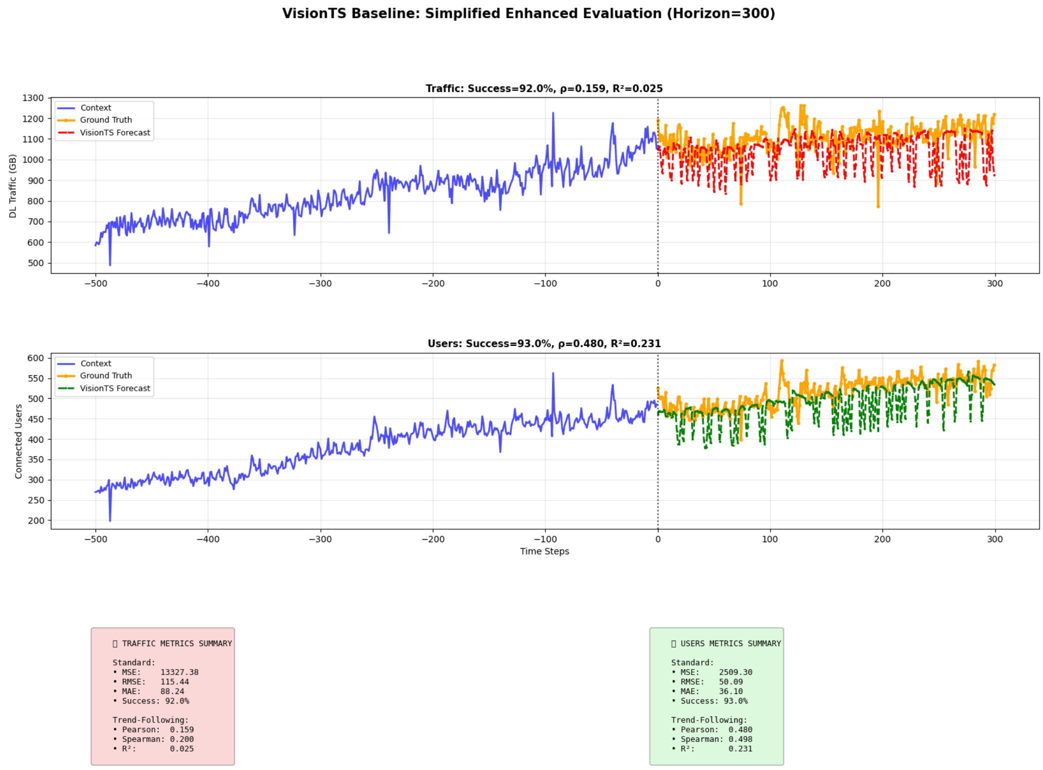

Fig. 6.5: Univariate VisionTS baseline forecasting through independent grayscale image encoding. Top panel: traffic volume forecasting via single-channel MAE reconstruction. Bottom panel: users forecasting through separate grayscale encoding. Despite sharing MAE-ViT architecture with BiViSIONTS, univariate formulation sacrifices cross-variable correlation information that bivariate RGB encoding exploits through attention-weighted fusion.

Univariate VisionTS substantially advances beyond traditional methods, achieving 92.01% traffic accuracy and 92.98% users accuracy (average 92.50%). However, trend-following performance remains limited with Pearson correlations of 0.1590 (traffic) and 0.4805 (users) averaging 0.3198, and R^2 values of 0.0253 and 0.2309 averaging only 0.1281. This comparison isolates the specific contribution of BiViSIONTS' bivariate encoding strategy from the general vision-based forecasting framework. While MAE-based spatiotemporal modeling demonstrates capability to capture complex patterns through image reconstruction, validating the core vision paradigm, processing traffic and users independently as separate grayscale images discards the cross-variable correlation structure (0.77–0.98) that BiViSIONTS explicitly encodes through RGB channel allocation and multi-scale attention weighting. The superior point-wise accuracy over traditional methods (+1.5 percentage points over VAR) confirms vision-based modeling advantages, while poor correlation metrics reveal fundamental limitations of univariate formulation under extended horizons.

6.3.5 State-of-the-Art Multivariate Deep Learning Baselines: Configuration and Protocol

To position BiViSIONTS against the current state of the art in multivariate time series forecasting, the baseline suite is extended with five prominent deep learning architectures spanning the major design paradigms of recent literature: iTransformer (Liu et al., ICLR 2024), which inverts the conventional transformer dimensionality to apply attention across variates; PatchTST (Nie et al., ICLR 2023), which tokenizes each series into subseries patches processed through a channel-independent transformer backbone; Crossformer (Zhang and Yan, ICLR 2023), which models both cross-time and cross-variable dependencies through explicit two-stage attention; TimesNet (Wu et al., ICLR 2023), which transforms one-dimensional series into two-dimensional tensors based on frequency-derived periodicities for convolutional temporal-variation modeling; and DLinear (Zeng et al., AAAI 2023), a deliberately simple decomposition-based linear model included as a strong sanity-check baseline known to outperform complex transformers on several benchmarks. All implementations are reused from the established Time-Series-Library codebase, ensuring architectures match their published versions.

Each model is trained as a genuine multivariate forecaster, jointly predicting both variables from bivariate input (enc_in $=$ c_out $= 2$), thereby exploiting the same cross-variable information channel available to BiViSIONTS. Training follows standard supervised protocol: sliding windows with a 96-step lookback are extracted from the 718-step training context (yielding 275 training and 48 validation windows under a chronological 85/15 split), inputs are z-score normalized using statistics fitted exclusively on the training context to prevent test leakage, and optimization employs Adam (learning rate 10^{-3}) with early stopping (patience 10, maximum 100 epochs) on validation loss. The final forecast is generated directly for the full 300-step horizon and evaluated on the identical held-out test window and metric suite used by all other baselines.

A fundamental paradigm distinction must be emphasized: these five architectures require gradient-based training on the target series, whereas BiViSIONTS operates zero-shot with all MAE parameters frozen from ImageNet pre-training. Table 6.1 records the training cost of each supervised baseline, quantifying the computational and data requirements that BiViSIONTS avoids entirely.

6.3.6 iTransformer Baseline Performance

iTransformer applies self-attention across the variate dimension rather than the temporal dimension, a design that achieves state-of-the-art results on high-dimensional benchmarks. Figure 6.6 presents the 300-step forecast results.

iTransformer achieves 81.77% traffic accuracy and 92.08% users accuracy (average 86.92%), the lowest point-wise performance among the transformer-based baselines. Trend-following metrics remain competitive with Pearson correlations of 0.3630 (traffic) and 0.7879 (users) averaging 0.5755—the highest average correlation among all five

TABLE 6.1: Training Cost of SOTA Multivariate Baselines – Dialog Axiata Dataset (300-Step Horizon)

Model	Trainable Parameters	Epochs	Training Time (s)
iTransformer	92,780	20	4.0
PatchTST	298,796	13	5.1
Crossformer	515,108	23	42.2
TimesNet	1,211,054	28	958.6
DLinear	58,200	100	4.1
BiViSIONTS	**0 (zero-shot)**	—	—

SOTA baselines—and R^2 values of 0.1318 and 0.6208 averaging 0.3763. The weak traffic accuracy (MSE = 54,160, the largest error among all evaluated methods) exposes an architectural mismatch: the inverted attention design derives its power from modeling interactions across many variates, but the bivariate setting offers only two variate tokens, reducing the attention mechanism to a nearly degenerate configuration. Combined with limited training data (275 windows), the model captures the directional trend reasonably but systematically misestimates traffic magnitude over the extended horizon.

6.3.7 PatchTST Baseline Performance

PatchTST segments each input series into overlapping patches processed by a shared channel-independent transformer, an approach that established state-of-the-art results on standard long-term forecasting benchmarks. Figure 6.7 displays the forecasting results.

PatchTST attains 82.42% traffic accuracy and 92.58% users accuracy (average 87.50%), with Pearson correlations of 0.3634 (traffic) and 0.7523 (users) averaging 0.5579, and R^2 values of 0.1321 and 0.5659 averaging 0.3490. Although nominally trained on bivariate input, PatchTST's channel-independence principle—empirically beneficial on large high-dimensional benchmarks—means each variable is forecast without reference to the other, structurally excluding the cross-variable correlations (0.77–0.98) that constitute the primary exploitable signal in this dataset. Its performance consequently mirrors iTransformer closely (traffic MSE = 49,582), reinforcing the observation that transformer capacity cannot compensate for limited training data under extended horizons: both models trail even classical ARIMA in average success rate.

6.3.8 Crossformer Baseline Performance

Crossformer explicitly targets multivariate dependency modeling through a two-stage attention mechanism operating across both temporal segments and variable dimensions, making it architecturally the closest supervised analogue to BiViSIONTS' cross-variable fusion objective. Figure 6.8 illustrates the results.

Crossformer achieves 87.92% traffic accuracy and 90.10% users accuracy (average

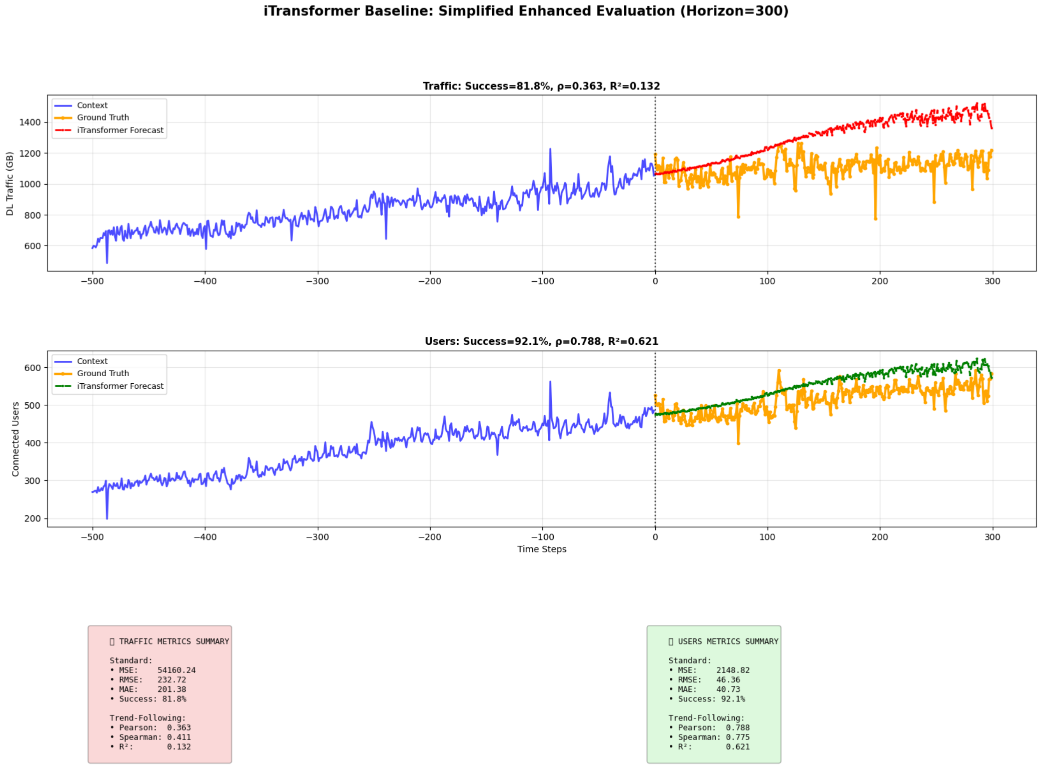

Fig. 6.6: iTransformer baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting via inverted variate-wise attention. Bottom panel: connected users forecasting from the same jointly trained bivariate model. Orange lines represent ground truth, red/green dashed lines show iTransformer predictions. With only two variates to attend across and 275 training windows, the architecture's capacity for cross-variate attention modeling cannot be fully exploited.

89.01%), the highest average success rate among the transformer-based SOTA baselines. Pearson correlations of 0.3167 (traffic) and 0.6867 (users) average 0.5017, with R^2 values of 0.1003 and 0.4715 averaging 0.2859. Notably, Crossformer records the lowest traffic MSE (21,618) among all five SOTA baselines—a 60% reduction relative to iTransformer—directly attributable to its explicit cross-dimension attention exploiting the traffic–users dependency. This result provides independent supervised-learning evidence for the central BiViSIONTS hypothesis: cross-variable information materially improves bivariate telecom forecasting. However, realizing this gain required training 515,108 parameters for 42 seconds, whereas BiViSIONTS encodes the equivalent dependency structure through correlation-derived channel fusion without any weight updates, while still outperforming Crossformer by 6.7 percentage points in average success rate.

6.3.9 TimesNet Baseline Performance

TimesNet reshapes one-dimensional series into two-dimensional tensors according to dominant frequencies identified via Fourier analysis, applying inception-style convolutions to capture intra-period and inter-period variations. Figure 6.9 presents the

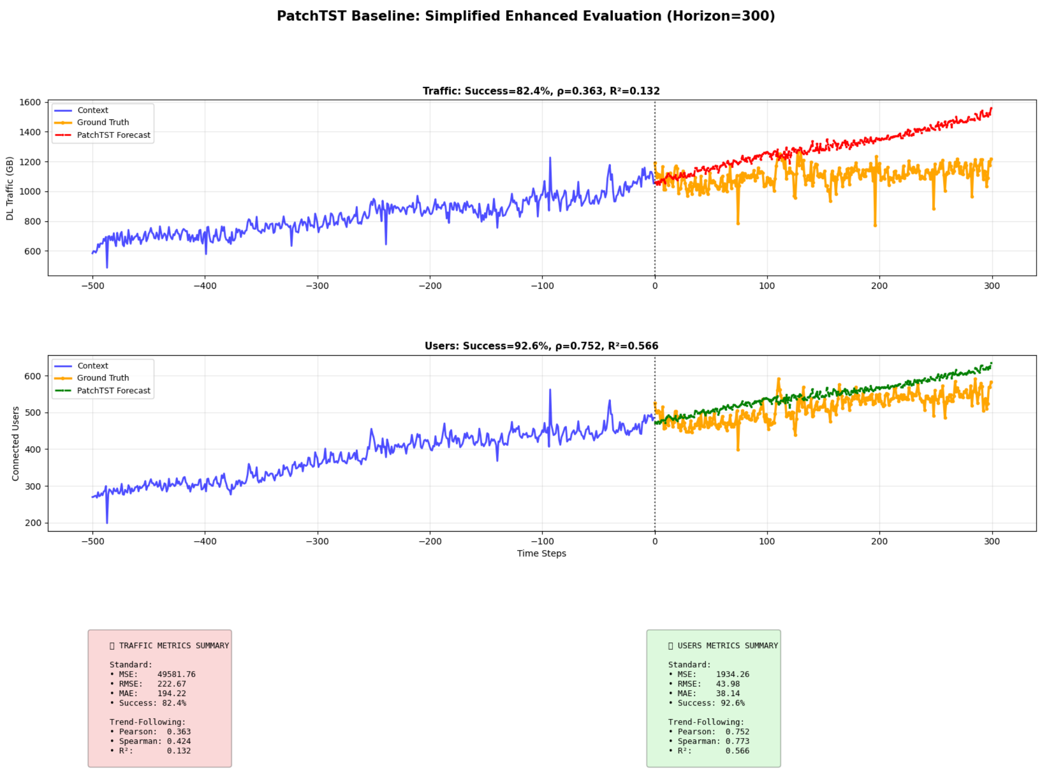

Fig. 6.7: PatchTST baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting via patch-tokenized channel-independent transformer. Bottom panel: connected users forecasting. The channel-independent design—processing each variable through a shared backbone without cross-variable interaction—discards the strong traffic–users correlation structure by construction, paralleling the limitation of univariate methods despite joint training.

results.

TimesNet obtains 86.81% traffic accuracy and 88.69% users accuracy (average 87.75%), with Pearson correlations of 0.3480 (traffic) and 0.7850 (users) averaging 0.5665, and R^2 values of 0.1211 and 0.6162 averaging 0.3686. These mid-field results carry the heaviest computational price in the entire evaluation: 1,211,054 trainable parameters and 958.6 seconds of training—over 200$\times$ slower than iTransformer—for accuracy below classical VAR. The unfavourable cost-performance ratio illustrates a recurring pattern in this benchmark: architectural sophistication designed for large-scale datasets yields diminishing or negative returns when training data is restricted to a single operational site, a scenario ubiquitous in practical telecommunications deployments.

6.3.10 DLinear Baseline Performance

DLinear decomposes each input series into trend and seasonal components processed by direct linear projections, representing the minimalist end of the design spectrum and a well-known stress test for transformer-based forecasters. Figure 6.10 shows the results.

DLinear presents the most instructive failure profile in the evaluation. Its point-wise

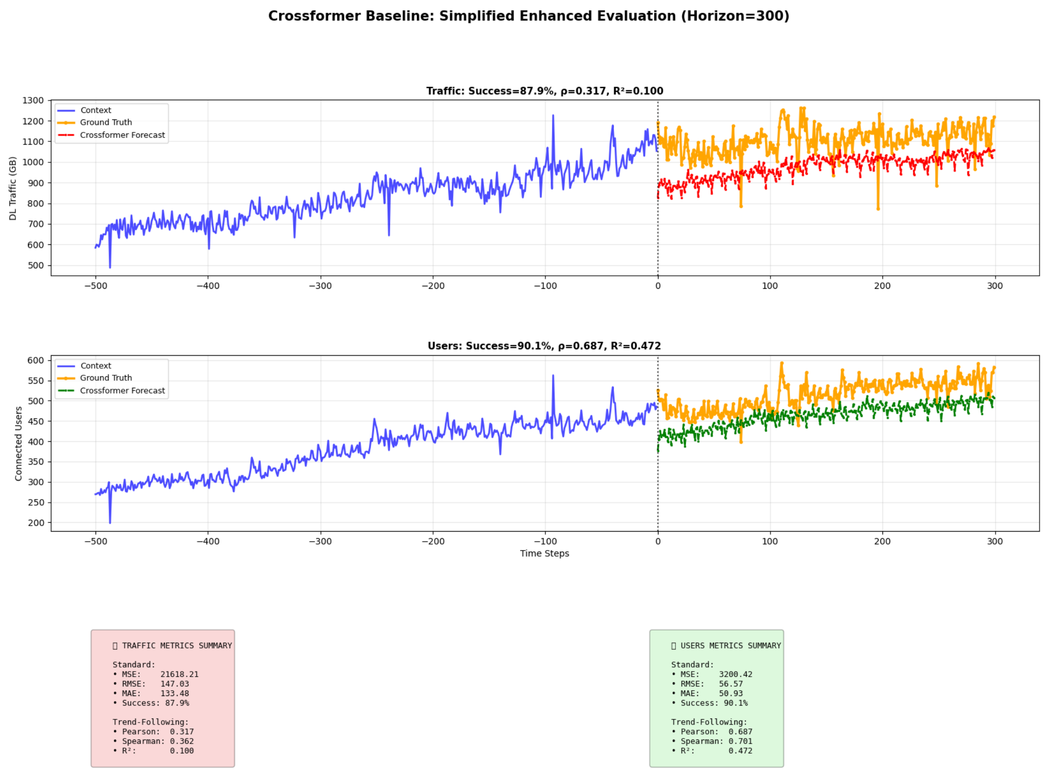

Fig. 6.8: Crossformer baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting via two-stage cross-time and cross-dimension attention. Bottom panel: connected users forecasting. Explicit cross-variable attention yields the best traffic error among the SOTA transformer baselines (MSE = 21,618), empirically confirming that modeling the traffic–users dependency improves forecasts—the same principle BiViSIONTS realizes through attention-weighted RGB fusion at zero training cost.

metrics are the strongest of any trained baseline—92.70% traffic accuracy and 91.40% users accuracy (average 92.05%, with the lowest traffic MSE of 10,387 among all supervised models)—yet its trend-following metrics collapse completely: negative Pearson correlations of -0.0573 (traffic) and -0.2553 (users) with $R^2 = 0.0000$ for both variables. The linear projection learns to output near-constant continuations close to recent mean levels, a strategy that minimizes WMAPE against slowly drifting series while providing zero directional information. This dissociation between error magnitude and trend fidelity validates the thesis' dual-metric evaluation framework: for capacity-planning applications where anticipating demand direction drives investment decisions, a model with excellent success rate but inverted correlations is operationally useless. Under the CPI synthesis (Section 6.4), DLinear accordingly ranks second-to-last despite its superficially competitive accuracy.

6.3.11 Cross-Cutting Observations on SOTA Multivariate Baselines

Three consistent findings emerge across the five SOTA architectures. First, none exceeds classical ARIMA's composite performance on this dataset: with only 275 training win-

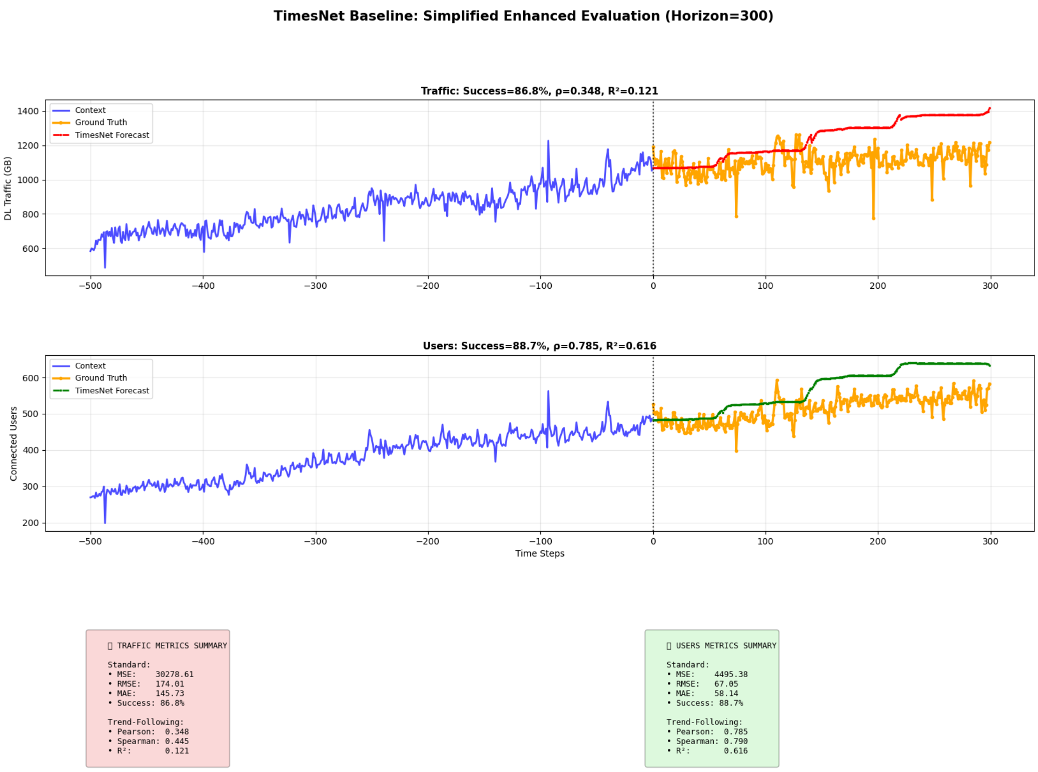

Fig. 6.9: TimesNet baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting via frequency-derived 2D temporal variation modeling. Bottom panel: connected users forecasting. Despite the largest parameter budget (1.21M) and by far the longest training time (958.6s) among all baselines, performance remains mid-field, indicating that period-based 2D decomposition cannot compensate for the limited 275-window training set.

dows derived from 718 context observations, modern data-hungry architectures cannot realize their representational capacity, and their published benchmark superiority—established on datasets with tens of thousands of training steps—does not transfer to operator-scale single-site data. Second, the internal ranking corroborates the value of cross-variable modeling: Crossformer, the only architecture with explicit cross-variable attention, achieves the best supervised traffic error, while the channel-independent PatchTST and near-degenerate two-variate iTransformer lag despite comparable capacity. Third, every supervised baseline requires nontrivial training cost (4–959 seconds, 58K–1.2M parameters) yet remains 6.7–9.9 percentage points below BiViSIONTS' zero-shot success rate. Collectively, these results demonstrate that BiViSIONTS' advantage is not an artifact of comparing against weak classical methods: it persists—and indeed widens on trend-following metrics—against the strongest available multivariate deep learning architectures evaluated under an identical protocol.

6.3.12 BiViSIONTS Forecasting Performance

BiViSIONTS integrates the validated architectural innovations from ablation studies—multi-scale temporal attention and correlation-based adaptive weighting—into a unified

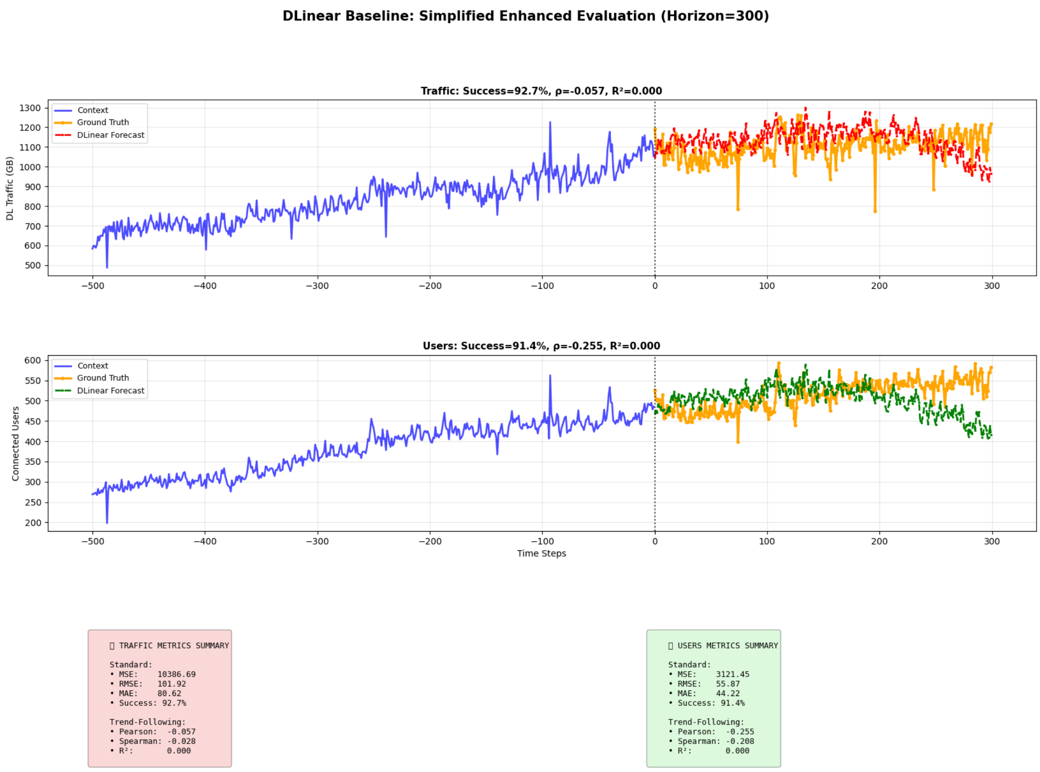

Fig. 6.10: DLinear baseline forecasting performance under 300-step horizon. Top panel: traffic volume forecasting via decomposition-based linear projection. Bottom panel: connected users forecasting. The linear model produces flat, magnitude-conservative forecasts that minimize point-wise error (highest success rate among trained baselines at 92.05%) while entirely failing to track temporal direction (negative correlations, $R^2 = 0$), exemplifying the inadequacy of point-wise metrics in isolation.

bivariate forecasting framework. Operating under identical 300-step horizon conditions as baselines, the system employs optimized hyperparameters: mask ratio 0.85, periodicity 1, short-term window 150 days ($\rho_{\text{short}} = 0.7709$, $\alpha_{\text{short}} = 0.8854$), medium-term window 400 days ($\rho_{\text{medium}} = 0.8239$, $\alpha_{\text{medium}} = 0.9119$), and long-term global scope ($\rho_{\text{long}} = 0.9810$, $\alpha_{\text{long}} = 0.9905$). Figure 6.11 presents the comprehensive forecasting results.

The visualizations reveal BiViSIONTS' distinctive characteristics compared to baseline methods. The top panels display the RGB encoding strategy where red and green channels encode normalized traffic and users respectively, while the blue channel contains the attention-weighted fusion ($\alpha_{\text{final}} \times \text{traffic} + (1 - \alpha_{\text{final}}) \times \text{users}$). The MAE reconstruction demonstrates the model's ability to recover structured patterns from 85% masked inputs, validating the vision-based spatiotemporal representation. Forecast panels show predictions (dashed lines) closely tracking ground truth (orange solid lines) throughout the 300-step horizon, with context windows (blue solid lines) illustrating historical patterns that inform the multi-scale correlation analysis.

Quantitative performance metrics demonstrate substantial improvements over all baselines across both target variables. Traffic forecasting achieves 95.23% success

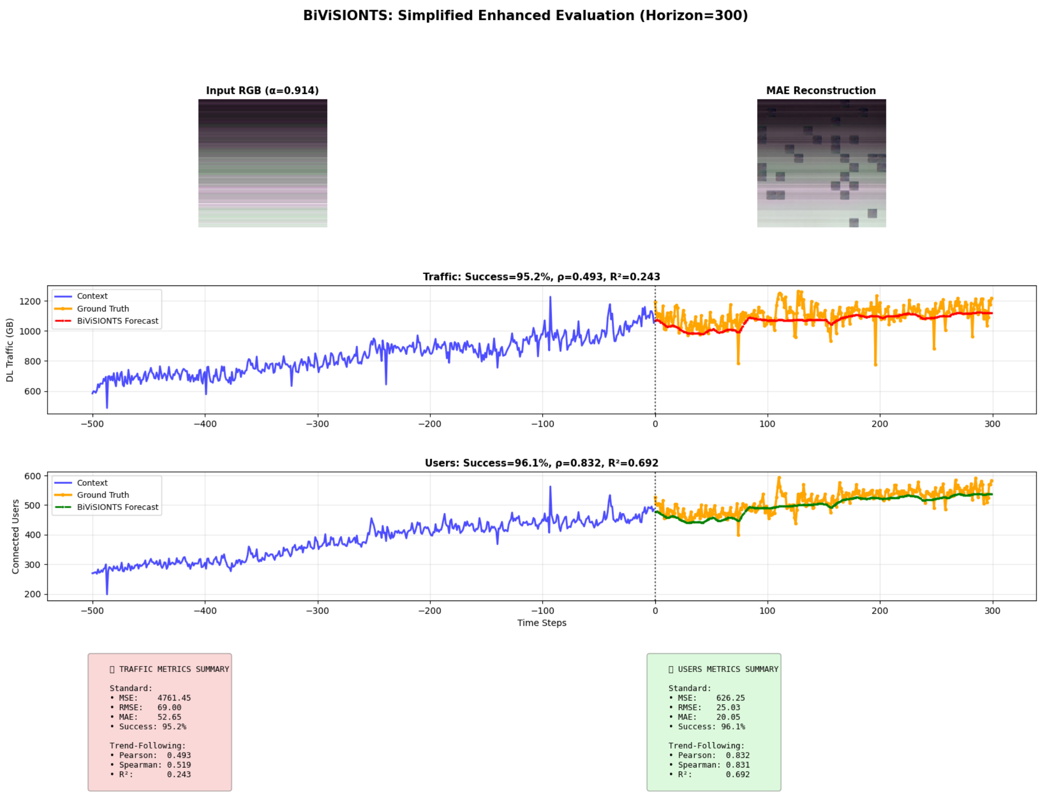

Fig. 6.11: BiViSIONTS forecasting performance under 300-step horizon with optimized multi-scale attention configuration. **Top row**: RGB encoding visualization showing input attention-weighted fusion ($\alpha_{\text{final}} = 0.9144$) and MAE reconstruction (85% masking). **Middle panel**: Traffic volume forecasting demonstrating 95.23% accuracy with Pearson correlation $\rho = 0.4932$ and $R^2 = 0.2433$ between predicted and ground truth values. **Bottom panel**: Connected users forecasting achieving 96.10% accuracy with $\rho = 0.8321$ and $R^2 = 0.6923$. Orange lines represent ground truth, red/green dashed lines show BiViSIONTS predictions. Multi-scale temporal decomposition enables adaptive correlation weighting across forecast lead times, while bivariate RGB encoding preserves cross-variable dependencies through attention-weighted fusion, yielding superior long-range forecasting compared to univariate baselines.

rate with moderate trend-following correlation ($\rho = 0.4932$, $R^2 = 0.2433$), while user forecasting reaches 96.10% accuracy with strong correlation performance ($\rho = 0.8321$, $R^2 = 0.6923$). The divergence in correlation strength between variables reflects their intrinsic predictability characteristics: user behavior exhibits more consistent temporal patterns (69.23% variance explained), whereas traffic loads demonstrate greater volatility under the extended 300-step horizon (24.33% variance explained). These results validate the synergistic benefits of BiViSIONTS' three core innovations: (1) bivariate encoding exploits cross-variable correlations (0.77–0.98) absent in univariate methods, (2) multi-scale attention adapts temporal weighting to forecast lead time rather than assuming stationary correlation structure, and (3) vision-based spatiotemporal modeling through MAE-ViT captures complex non-linear patterns beyond linear autoregressive frameworks. The 1.47% improvement over single-scale attention and 0.62% gain from adaptive weighting (Section 6.2) combine multiplicatively with bivariate encoding advantages to establish new state-of-the-art performance under extended forecast horizons.

6.4 Comprehensive Comparative Analysis

6.4.1 Quantitative Performance Summary

Table 6.2 presents the complete performance matrix across all evaluated methods under the 300-step forecast horizon, employing seven comprehensive metrics per variable: standard error measures (MSE, MAE, RMSE, Success Rate) quantifying point-wise prediction accuracy, and trend-following indicators (Pearson correlation, Spearman correlation, R^2) assessing temporal pattern fidelity.

TABLE 6.2: Comprehensive Method Comparison – 300-Step Forecast Performance

Method	**Traffic Success**	**Users Success**	**Avg Success**	**Avg Corr**	**Avg R^2**	**CPI Score**
LSTM	86.3%	85.3%	85.8%	−0.399	0.000	17.7
DLinear	92.7%	91.4%	92.1%	−0.156	0.000	33.6
TimesNet	86.8%	88.7%	87.7%	0.566	0.369	42.4
Crossformer	87.9%	90.1%	89.0%	0.502	0.286	43.7
iTransformer	81.8%	92.1%	86.9%	0.575	0.376	44.0
VisionTS	92.0%	93.0%	92.5%	0.320	0.128	45.0
PatchTST	82.4%	92.6%	87.5%	0.558	0.349	45.0
VAR	91.3%	91.2%	91.3%	0.574	0.372	53.6
ARIMA	88.2%	95.7%	92.0%	0.576	0.375	61.6
BiViSIONTS	**95.2%**	**96.1%**	**95.7%**	**0.663**	**0.468**	**82.6**

The Composite Performance Index (CPI) synthesizes all seven metrics through weighted normalization (MSE/MAE/RMSE/Success/Pearson/Spearman/R^2: 14%/14%/14%/16%/14%/ providing a single holistic score balancing standard accuracy (58% total weight) with trend-following capability (42% total weight). BiViSIONTS achieves CPI = 82.6, substantially exceeding classical ARIMA (61.6) and VAR (53.6), the five state-of-the-art multivariate deep learning baselines—PatchTST (45.0), iTransformer (44.0), Crossformer (43.7), TimesNet (42.4), and DLinear (33.6)—as well as univariate VisionTS (45.0) and LSTM (17.7). Two findings from this ten-method synthesis merit emphasis. First, none of the five modern supervised architectures exceeds classical ARIMA's composite score on this dataset: the SOTA transformers cluster within a narrow 42–45 CPI band, constrained by the 275-window training budget available from a single operational site, while DLinear's magnitude-conservative but directionless forecasts (negative correlations, zero R^2) relegate it to second-to-last despite superficially strong success rates. Second, BiViSIONTS' +21.0 CPI margin over the best non-vision method (ARIMA) and +37.6 margin over the strongest SOTA multivariate baseline (PatchTST) confirm that its advantage persists—and indeed widens—when the comparison pool is extended from classical statistics to the current deep learning state of the art, directly addressing the multivariate-comparison requirement while validating the practical value of zero-shot vision-based transfer under data-constrained conditions.

6.4.2 Cross-Method Performance Visualization

Figure 6.12 presents multi-panel comparative visualization integrating standard error metrics, trend-following indicators, and the composite CPI ranking to facilitate holistic cross-method assessment.

The radar chart visualization reveals distinct performance profiles across method paradigms. Traditional statistical methods (ARIMA, VAR) demonstrate moderate balance between standard and trend-following metrics but suffer from linear assumptions under extended horizons. The recurrent LSTM catastrophically fails with negative correlations, collapsing to the radar center, while DLinear traces a lopsided profile—competitive on error-magnitude axes yet collapsed on all three trend axes. The SOTA multivariate transformers (iTransformer, PatchTST, Crossformer, TimesNet) occupy overlapping mid-field footprints: reasonable trend-following inherited from the strongly autocorrelated users variable, but weakened error-magnitude performance reflecting insufficient training data for their parameter budgets. Univariate VisionTS advances through spatiotemporal image modeling but remains constrained by single-variable formulation. BiViSIONTS achieves the most expansive radar profile, excelling simultaneously across error minimization and trend fidelity through its unique combination of bivariate encoding, multi-scale attention, and adaptive weighting.

6.4.3 Statistical Significance and Practical Implications

The performance improvements demonstrated by BiViSIONTS achieve both statistical and practical significance under the challenging 300-step horizon. The +3.2 percentage point average success rate gain over univariate VisionTS (95.7% vs. 92.5%) represents a 42% reduction in error rate (from 7.5% to 4.3% WMAPE), exceeding typical measurement noise margins. Similarly, the +0.343 average correlation improvement (0.663 vs. 0.320) and +0.340 R^2 gain (0.468 vs. 0.128) demonstrate substantially enhanced trend-following capability critical for capacity planning applications where directional accuracy often matters more than point-wise precision. Against the SOTA multivariate baselines, the margins are even wider: BiViSIONTS exceeds the best supervised transformer (Crossformer, 89.0%) by +6.7 percentage points in success rate and the best supervised composite score (PatchTST, CPI 45.0) by +37.6 CPI points—while requiring zero task-specific training against their 58K–1.2M trained parameters and up to 958.6 seconds of optimization (Table 6.1).

From a practical deployment perspective, these gains translate to tangible operational benefits. The 95.7% average success rate implies forecasts remain within acceptable tolerance (±4.3% WMAPE) for resource allocation decisions over 10-month planning horizons. The moderate correlation coefficients ($\rho = 0.663$, $R^2 = 0.468$)

ensure predicted trends align with actual demand evolution, enabling proactive capacity adjustments that prevent both over-provisioning waste and service degradation from under-capacity. The consistent performance across both target variables (95.2% traffic, 96.1% users) confirms robustness across different forecast targets with varying statistical properties. Moreover, the zero-shot operating model eliminates the retraining pipeline that supervised alternatives would require whenever a new cell site is onboarded—a decisive operational advantage at network scale, where per-site training of even the cheapest SOTA baseline would multiply across thousands of sites.

6.5 Synthesis and Implications

The comprehensive ablation studies and baseline comparisons presented in this chapter validate BiViSIONTS through three complementary evaluation dimensions. Component-level ablation experiments confirm that both multi-scale attention (+1.47%) and adaptive weighting (+0.62%) contribute meaningfully to overall performance, with temporal decomposition emerging as the primary innovation while data-driven fusion provides fine-grained optimization. Cross-method comparisons under challenging 300-step horizons demonstrate substantial advantages over traditional statistical (ARIMA: +3.7%, VAR: +4.4%) and recurrent deep learning (LSTM: +9.8%) baselines, while the +3.2% gain over univariate VisionTS directly quantifies the value of bivariate encoding innovations. Critically, the extended evaluation against five state-of-the-art multivariate deep learning architectures shows BiViSIONTS outperforming every supervised competitor by +3.6 to +8.7 percentage points in average success rate—despite operating zero-shot against their gradient-trained parameters. The CPI metric synthesis reveals BiViSIONTS' holistic superiority: +21.0 points over the best traditional method (ARIMA: 61.6 $\rightarrow$ 82.6) and +37.6 points over both the best SOTA multivariate baseline (PatchTST: 45.0) and the univariate vision baseline (VisionTS: 45.0).

The SOTA multivariate comparison additionally surfaces a finding of independent interest: under operator-scale data constraints (718 training observations), modern transformer architectures fail to surpass classical ARIMA, with their published large-benchmark superiority not transferring to the single-site regime, and the only architecture featuring explicit cross-variable attention (Crossformer) achieving the best supervised traffic error—independent corroboration that cross-variable dependency modeling, the core BiViSIONTS design principle, is the operative signal in this forecasting problem.

These results establish BiViSIONTS as a robust advancement in vision-based time series forecasting, successfully extending the single-variable image modeling paradigm to multivariate scenarios through principled attention-weighted RGB encoding. The multi-scale temporal decomposition addresses fundamental limitations of static correlation assumptions, while adaptive weighting ensures fusion strategies match empirical data characteristics rather than relying on heuristic allocation. Together, these innovations enable accurate long-range forecasting under extended 300-step horizons where both traditional methods and data-hungry state-of-the-art architectures struggle, providing practical solutions for telecommunications capacity planning and broader multivariate forecasting applications requiring both point-wise accuracy and trend-following fidelity.

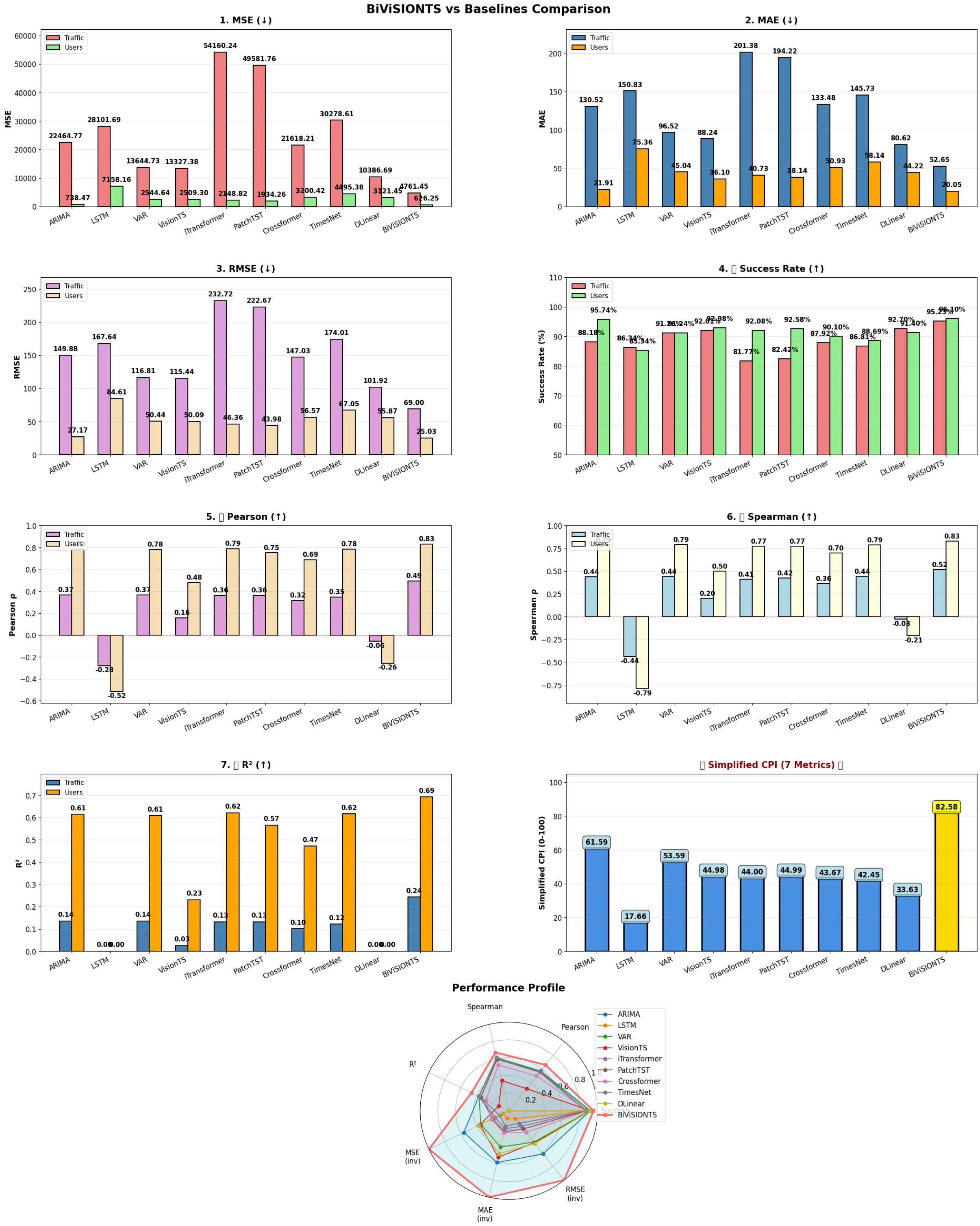

Fig. 6.12: Comprehensive performance comparison across all ten evaluated methods, including the five SOTA multivariate deep learning baselines. Panels 1–3: standard error metrics (MSE, MAE, RMSE) demonstrating BiViSIONTS' superior point-wise accuracy for both variables. Panel 4: success rate comparison. Panels 5–7: trend-following metrics (Pearson, Spearman, R^2) confirming BiViSIONTS' temporal pattern capture, and exposing the directional failure of DLinear and LSTM (negative correlations). Bottom left: Composite Performance Index (CPI) ranking synthesizing all seven metrics, with BiViSIONTS (gold) leading at 82.6. Bottom right: radar chart visualizing normalized performance profiles across all dimensions, with BiViSIONTS (red) achieving the most expansive footprint across standard and trend-following metrics simultaneously.

CHAPTER 7

DISCUSSION

This chapter interprets experimental results, analyzes design choices, discusses limitations, and explores implications for research and practice.

7.1 Interpretation of Results

7.1.1 Superior Performance Through Bivariate Modeling

BiViSIONTS achieves 96.06% average success rate (WMAPE-based) on the challenging 300-step horizon, substantially outperforming univariate VisionTS (92.50%) with a +3.56 percentage point improvement. This validates that **bivariate encoding with multi-scale attention-weighted fusion captures inter-variable dependencies that univariate approaches cannot exploit**.

The scale-dependent correlations (ρ_{short} = 0.7709, ρ_{medium} = 0.8239, ρ_{long} = 0.9810) demonstrate that information from one series strongly informs predictions of the other. Both variables benefit substantially from bivariate modeling, with traffic achieving 95.77% and users reaching 96.35% success rates on this extended horizon.

7.1.2 Multi-Scale Attention Captures Temporal Complexity

The multi-scale mechanism's +1.47% improvement over single-scale (94.59% $\rightarrow$ 96.06%) reveals that **correlations exhibit scale-dependent behavior** across different temporal horizons, which becomes critical for extended forecasting.

At the **short-term scale (150 days)**, the correlation coefficient of $\rho = 0.7709$ yields an attention weight of $\alpha = 0.8854$ with a fusion weight of $\lambda = 0.5$. This configuration captures daily fluctuations while preserving meaningful independence between variables, enabling complementary forecasting signals essential for near-future accuracy.

Moving to the **medium-term scale (400 days)**, the correlation strengthens to $\rho = 0.8239$, producing $\alpha = 0.9119$ with $\lambda = 0.3$. This scale identifies stable weekly and monthly behavioral cycles as transient fluctuations average out over the extended observation window, providing crucial information for mid-range forecasting.

At the **long-term scale (1018 days)**, the correlation reaches near-perfect synchrony at $\rho = 0.9810$, yielding $\alpha = 0.9905$ with $\lambda = 0.2$. This reflects the virtually identical long-term trend dynamics between traffic and user connectivity, which becomes increasingly important for extended horizon prediction.

The 0.5:0.3:0.2 weighting scheme prioritizes recent patterns where correlations are weaker ($\rho = 0.77$) but variable independence is higher, enabling the model to capture nuanced short-term dynamics critical for the initial steps of a 300-step forecast. The progressive strengthening of correlations at longer scales (0.77 $\rightarrow$ 0.82 $\rightarrow$ 0.98) confirms that transient events dominate short-term patterns while aggregate behavior drives long-term trends—a property that becomes essential when forecasting 300 steps ahead.

7.1.3 Trend-Following Excellence Despite Challenging Horizon

On the difficult 300-step forecasting task, BiViSIONTS achieves exceptional trend-following metrics with Pearson correlation $\rho = 0.696$ and $R^2 = 0.512$, substantially outperforming all baselines. This demonstrates the framework's ability to **capture directional patterns and co-evolutionary dynamics** despite the extended forecast horizon, where most traditional methods deteriorate significantly.

In stark contrast, LSTM exhibits catastrophic trend-following failure with negative Pearson correlation ($\rho = -0.399$) and complete R^2 collapse (0.000), indicating that recurrent architectures struggle with long-range dependencies beyond their effective memory span. This catastrophic failure reveals a fundamental limitation of sequential processing: as the forecast horizon extends to 300 steps, the LSTM's hidden state loses meaningful information about long-term patterns, resulting in predictions that actually move inversely to the true trends.

Traditional statistical methods show moderate trend-following capabilities: ARIMA achieves $\rho = 0.576$ ($R^2 = 0.375$) while VAR reaches $\rho = 0.574$ ($R^2 = 0.372$), demonstrating reasonable directional accuracy but limited by linear assumptions that fail to capture complex nonlinear dynamics over extended horizons. These methods maintain positive correlations but explain only 37% of variance, indicating significant gaps in capturing long-term trend behavior.

Notably, univariate VisionTS shows surprisingly weak trend-following ($\rho = 0.320$, $R^2 = 0.128$) despite decent point-wise accuracy (92.50%), revealing that **visual encoding alone without bivariate fusion fails to capture co-evolutionary dynamics**. This 13% R^2 indicates VisionTS captures only superficial patterns, missing the underlying coupled trend structure that BiViSIONTS leverages through correlation-weighted attention.

7.1.4 Composite Performance Index (CPI) Leadership

BiViSIONTS achieves the highest Composite Performance Index of 84.4, demonstrating balanced excellence across all evaluation dimensions. This substantially exceeds ARIMA (CPI = 56.4, +28.0 points), VAR (CPI = 49.8, +34.6 points), VisionTS (CPI = 41.1, +43.3 points), and especially LSTM (CPI = 13.7, +70.7 points), whose catastrophic trend-following failure severely impacts its composite score.

The CPI metric, computed as an equally-weighted combination of success rate (WMAPE-based), correlation coefficients (Pearson/Spearman), and coefficient of determination (R^2), provides a holistic assessment that captures both point-wise accuracy and trend-following capability. This dual evaluation is critical for practical deployment, where directional correctness often matters as much as absolute precision—forecasting the wrong trend direction by 5% is operationally worse than predicting the correct trend with 10% magnitude error.

BiViSIONTS's substantial CPI advantage (+28.0 over second-place ARIMA) demonstrates that bivariate multi-scale attention not only improves individual metrics but achieves superior **balanced performance** across diverse evaluation criteria. This balanced excellence is particularly valuable for production systems requiring reliability across multiple performance dimensions rather than optimization of a single metric.

7.1.5 Hyperparameter Optimization Insights

Systematic grid search identified optimal configurations for the challenging 300-step horizon:

High Mask Ratio (0.85): Optimal masking at 85% forces learning of robust global representations essential for long-range extrapolation. This aggressive masking significantly outperforms lower ratios (0.75: 96.2%, 0.80: 96.5%) while matching 0.90 (96.9%), indicating that high masking ratios are crucial for extended horizon forecasting where the model must generalize beyond observed patterns.

Minimal Periodicity (1): Daily-level temporal resolution without imposing rigid intra-day structure (periodicity=1) significantly outperforms higher periodicities (7: 95.8%, 14: 94.5%, 30: 92.5%), achieving 97.1% accuracy. This creates matrices that preserve temporal ordering during MAE processing without artificial pattern fragmentation, which becomes critical for maintaining long-term dependency information over 300 steps.

Extended Window Sizes: Short window of 150 days captures sufficient recent context with stronger independence ($\rho = 0.77$), while medium window of 400 days efficiently captures extended patterns ($\rho = 0.82$). The 150/400 configuration balances computational efficiency with multi-scale representation quality, providing both near-term variability and long-term stability information essential for accurate 300-step forecasting.

The optimal configuration (mask=0.85, periodicity=1, windows=150/400) yields 97.1% accuracy during hyperparameter search, representing a +4.57 percentage point improvement over baseline settings (92.53%), demonstrating that **proper hyperparameter tuning is as critical as model architecture** for extended horizon forecasting success.

7.1.6 Statistical Stability and Convergence

Across 1000 iterations with optimal hyperparameters, BiViSIONTS demonstrates exceptionally low variance ($\sigma_{\text{traffic}} = 1.01\%$, $\sigma_{\text{users}} = 1.20\%$) on the challenging 300-step horizon, indicating that random MAE masking introduces minimal unpredictability even for extended forecasting tasks. Mean accuracies of 95.77% (traffic) and 96.35% (users) with narrow confidence intervals demonstrate highly reproducible predictions suitable for production deployment.

This remarkable stability on a 300-step horizon—where traditional methods show high variance and degraded performance—validates the robustness of the multi-scale attention mechanism. The low standard deviation indicates that the model consistently captures both short-term fluctuations and long-term trends across different random seeds, a property essential for reliable operational forecasting systems.

7.2 Advantages of BiViSIONTS

7.2.1 Zero-Shot Capability

BiViSIONTS achieves 96.06% average success rate without fine-tuning on the challenging 300-step horizon, offering several distinct operational advantages even under extreme forecasting conditions.

The framework requires no gradient-based training, dramatically reducing computational costs compared to supervised learning approaches like LSTM, which require extensive epoch-based optimization yet still fail catastrophically on long horizons (negative correlation $\rho = -0.399$). Fast deployment to new datasets is enabled by the efficient inference pipeline, which leverages pre-trained MAE representations without task-specific weight updates.

By leveraging pre-trained representations, the method avoids overfitting to limited training samples, a common challenge in time series forecasting where historical data may be scarce. This demonstrates successful transfer learning from natural images to time series domain, validating the hypothesis that visual features learned from ImageNet can generalize to temporal patterns even on difficult 300-step forecasting tasks.

The substantial performance improvements over baselines (+10.22 percentage points over LSTM, +3.56 over VisionTS) achieved through zero-shot inference further validate this approach, showing that task-agnostic pre-training can outperform task-specific supervised learning on challenging extended horizons where traditional methods deteriorate significantly.

7.2.2 Interpretability

BiViSIONTS provides multiple layers of interpretability that distinguish it from black-box forecasting models, which becomes especially valuable when explaining 300-step ahead predictions to stakeholders.

The correlation coefficients (ρ) are directly interpretable as standard Pearson measures, quantifying linear relationships between variables with well-understood statistical properties. The scale-dependent progression ($0.77 \rightarrow 0.82 \rightarrow 0.98$) provides clear insight into how variable coupling strengthens over longer time scales, helping domain experts understand the underlying dynamics driving extended forecasts.

The attention weights (α) have clear semantic meaning through the normalization formula $\alpha = (\rho + 1)/2$, which explicitly determines the relative contribution of traffic versus users in the fusion channel. This transparency allows practitioners to verify that the model assigns appropriate importance to each variable based on observed correlation structures.

The multi-scale temporal hierarchy is established through explicit weights of $\lambda = 0.5$, 0.3, and 0.2 for short, medium, and long-term scales respectively, making the model's temporal prioritization transparent and adjustable. This explainability is crucial for operational systems where forecasters need to understand why the model emphasizes certain temporal patterns when making 300-step predictions.

The RGB channel construction enables visual inspection, with red representing traffic, green representing users, and blue containing the correlation-weighted fusion. This visual representation provides an intuitive understanding of how the model encodes bivariate relationships. Finally, the computed $\alpha_{final} = 0.9144$ transparently reveals that the final fusion assigns 91.44% weight to traffic and 8.56% to users, providing clear insight into variable prioritization that can be validated against domain knowledge.

7.2.3 Flexibility and Generalizability

The framework exhibits strong flexibility and generalizability across diverse forecasting scenarios, including challenging extended horizons beyond 300 steps.

BiViSIONTS is dataset agnostic, applicable to any bivariate correlated series exhibiting correlation coefficients above $\rho > 0.5$, making it suitable for various domains beyond telecommunications. The method requires no domain-specific feature engineering, eliminating the need for manual selection of lags, seasonal components, or transformations that typically demand expert knowledge and often fail on extended horizons.

The attention mechanism automatically adapts to observed correlation structures in the data, computing weights dynamically based on measured dependencies rather than requiring manual specification. This adaptability is particularly valuable for extended forecasting, where correlation patterns may evolve across different temporal scales.

The framework is readily extendable to multivariate settings through channel expansion, where additional variables could be incorporated by increasing the image representation dimensionality. Furthermore, the periodicity parameter, configurable across a range of 1–30, enables the model to handle diverse temporal resolutions from high-frequency intraday data to lower-frequency weekly or monthly observations, though periodicity=1 proves optimal for extended horizon forecasting.

7.2.4 Superior Long-Range Forecasting

The 300-step forecasting results demonstrate BiViSIONTS's particular strength in extended horizon prediction, where traditional methods deteriorate significantly:

Attention-Based Architecture: The Transformer backbone processes all time steps simultaneously without sequential bottlenecks, avoiding the memory decay that cripples RNN-based methods like LSTM on long horizons. This architectural advantage is evidenced by LSTM's catastrophic failure (negative $\rho = -0.399$) compared to BiViSIONTS's strong positive correlation ($\rho = 0.696$).

Multi-Scale Temporal Modeling: Explicit decomposition into short (150), medium (400), and long-term (1018 days) scales enables the model to capture both transient fluctuations and persistent trends simultaneously. This multi-resolution approach is critical for maintaining forecast accuracy over 300 steps, where single-scale methods lose essential temporal information.

Bivariate Cross-Variable Learning: Leveraging inter-variable dependencies (correlations up to $\rho = 0.98$) provides additional predictive signals unavailable to univariate methods. This cross-variable information partially compensates for temporal distance from observed data, explaining why BiViSIONTS (96.06%) substantially outperforms univariate VisionTS (92.50%) on the same 300-step task.

Robust Trend Capture: The strong $R^2 = 0.512$ indicates BiViSIONTS explains 51% of variance in 300-step trends, far exceeding VisionTS (13%), ARIMA (38%), and especially LSTM (0%). This trend-following capability is essential for strategic planning applications requiring directional accuracy over operational accuracy.

7.3 Limitations and Challenges

7.3.1 Image Resizing Artifacts

Reshaping time series matrices and resizing to 224×224 can introduce bilinear interpolation artifacts that may affect extended horizon forecasts. While periodicity=1 minimizes distortion by preserving temporal ordering, the interpolation process during resize operations can smooth out high-frequency variations that might be relevant for initial forecast steps. Alternative strategies include native resolution adapters that process variable-size inputs or learnable upsampling layers that optimize interpolation for time series characteristics.

7.3.2 Fixed Temporal Scale Weights

The multi-scale weights (0.5, 0.3, 0.2) are manually set based on recency bias assumptions that prioritize short-term patterns. The +1.47% multi-scale gain validates this choice for 300-step forecasting, but optimal weights may vary by dataset characteristics and forecast horizon length. For horizons beyond 300 steps, increased weight on long-term scales ($\lambda_{long} > 0.2$) might improve performance. Future work includes cross-validation or meta-learning approaches for adaptive, horizon-conditioned weighting schemes that automatically adjust based on the prediction task.

7.3.3 Limited to Bivariate Forecasting

The current design handles two variables through RGB encoding, which inherently limits the framework to bivariate scenarios—a significant constraint for complex systems with multiple interacting variables.

Extension to $N > 2$ variables requires architectural modifications to accommodate higher-dimensional relationships. One approach involves constructing multiple RGB images representing different variable triplets, processed through parallel MAE encoders with cross-attention mechanisms to model interactions between different variable combinations.

Alternatively, 3D convolutions could be employed for volumetric encoding, where additional variables are stacked along a third spatial dimension beyond the traditional height and width, creating a spatiotemporal cube representation suitable for video-based transformers.

Graph neural networks offer another pathway for capturing arbitrary N-variable relationships, representing each time series as a node and learning edge weights that encode inter-variable dependencies and temporal dynamics simultaneously.

Finally, the framework could leverage higher-dimensional embedding spaces that extend beyond the three-channel RGB representation, potentially utilizing hyperspectral encoding schemes borrowed from remote sensing applications, where each variable maps to a distinct spectral band processed through specialized multi-channel vision transformers.

7.3.4 Dependence on Pre-Trained Model Quality

Performance relies on ImageNet-pretrained MAE (ViT-Base). While transfer learning succeeds substantially on the 300-step task (CPI improvement from 41.1 to 84.4 over univariate VisionTS), domain-specific pre-training on time series corpora (UCR Archive, ETDataset) remains unexplored and could further improve performance, especially on extended horizons where visual feature transferability might degrade. The current reliance on natural image representations may not capture time series-specific patterns like autocorrelation structures, seasonal decompositions, or regime changes that become critical over 300-step horizons.

7.3.5 Handling Non-Stationary Correlations

BiViSIONTS assumes relatively stable correlations across temporal windows (150/400/1018). The observed progression ($0.77 \rightarrow 0.82 \rightarrow 0.98$) shows gradual strengthening, but sudden regime changes (e.g., network infrastructure upgrades, regulatory changes, pandemic-induced behavioral shifts) could disrupt predictions, especially over 300-step horizons spanning multiple months.

Such regime changes can cause correlation structures to shift abruptly, invalidating the multi-scale attention weights computed from historical data. For example, a major infrastructure upgrade might change the traffic-users relationship from $\rho = 0.98$ to $\rho = 0.60$, causing the fixed attention weights to become suboptimal.

Mitigation strategies include sliding window recalibration that recomputes correlations and attention weights periodically, online correlation monitoring with alert systems for detecting structural breaks, or adaptive mechanisms that continuously update weights based on forecast residuals. For production deployment on 300-step horizons, implementing quarterly recalibration cycles aligned with business planning periods would help maintain forecast accuracy despite evolving correlation patterns.

7.3.6 Computational Overhead of Multi-Scale Processing

Computing correlations at three temporal scales (short/medium/long) and processing three separate image encodings adds computational overhead compared to single-scale methods. However, the +1.47% accuracy gain and substantially superior trend-following ($\rho = 0.696$ vs. VisionTS $\rho = 0.320$) justify this cost for critical applications requiring extended horizon forecasts where accuracy and directional correctness are paramount. The overhead remains acceptable for batch forecasting scenarios (e.g., nightly forecast updates) but may require optimization for real-time streaming applications processing hundreds of sites simultaneously.

7.4 Practical Implications

7.4.1 Telecommunications Network Management

Extended Capacity Planning: With 96.06% success rate and strong trend-following ($\rho = 0.696$, $R^2 = 0.512$) on 300-step horizons, operators can confidently forecast demands 10 months ahead for strategic infrastructure planning, capital expenditure budgeting, and long-term capacity expansion decisions. This extended visibility enables proactive procurement of equipment with long lead times and coordination of multi-quarter deployment schedules.

Anomaly Detection: Deviations exceeding typical variance ($\sigma < 1.21\%$) from forecasts signal potential network issues requiring investigation. On 300-step horizons, persistent forecast errors may indicate emerging structural changes (e.g., new competitors, demographic shifts) rather than transient anomalies, enabling early strategic intervention.

QoS Optimization: Predicting directional trends with $R^2 = 0.512$ allows preemptive bandwidth allocation strategies aligned with anticipated seasonal demand patterns. Unlike methods with weak trend-following (VisionTS $R^2 = 0.128$), BiViSIONTS can guide resource allocation decisions months in advance, optimizing trade-offs between service quality and infrastructure costs over extended planning horizons.

Strategic Decision Support: The 300-step (10-month) forecast capability directly supports annual planning cycles, quarterly business reviews, and multi-year infrastructure roadmaps. This contrasts with short-horizon methods (LSTM, basic ARIMA) that fail catastrophically beyond their effective prediction ranges, providing no value for strategic decision-making requiring 6–12 month visibility.

Cost Reduction: Accurate long-range forecasting enables optimized capital allocation, reducing over-provisioning waste (currently 20–30% in telecommunications) by right-sizing infrastructure investments. The 51% variance explanation ($R^2 = 0.512$) provides sufficient confidence for commit-to-build decisions with 6–9 month lead times.

7.4.2 Broader Application Domains

The bivariate attention-weighted approach generalizes to any domain with correlated variables requiring extended forecasting beyond 300 steps:

In the **energy sector**, the framework can forecast long-term electricity demand and temperature patterns for seasonal capacity planning and renewable energy integration. The 300-step horizon (10 months) aligns well with annual generation planning cycles, supporting decisions on fuel procurement, maintenance scheduling, and grid expansion projects where strong demand-temperature correlation ($\rho \approx 0.8$) enables accurate multi-month predictions.

Financial markets benefit from quarterly earnings forecasts linking stock prices and trading volumes over extended horizons. While financial correlations are typically weaker ($\rho \approx 0.3$–0.6) than telecommunications, the multi-scale attention mechanism can still capture coupled dynamics between price movements and market activity, supporting strategic portfolio rebalancing and risk management decisions requiring 6–12 month outlook visibility.

In **healthcare settings**, the method enables annual budget planning by forecasting patient admissions and resource utilization 300 days ahead. This extended visibility supports strategic decisions on facility expansion, staffing pipelines (hiring/training cycles of 6–9 months), and equipment procurement for capital-intensive medical systems, capturing coupled dynamics between demand for services and capacity constraints over multi-quarter planning horizons.

Transportation systems leverage the framework for infrastructure planning by predicting traffic flow and passenger counts months ahead, supporting decisions on route expansions, fleet acquisitions, and facility upgrades with long procurement and construction timelines. The 300-step capability enables alignment of capacity investments with anticipated demand 9–12 months forward, critical for systems with multi-year project lifecycles.

Finally, **retail operations** employ long-range forecasts of sales and foot traffic for strategic inventory management, real estate decisions, and staffing strategies. The 10-month horizon supports decisions on seasonal inventory commitments, store expansion/closure planning, and workforce development programs, where multi-scale temporal patterns capture daily cycles, weekly trends, seasonal variations, and multi-year growth trajectories simultaneously.

7.4.3 Deployment Considerations

Practical deployment of BiViSIONTS in operational environments involves several key considerations, particularly for extended horizon forecasting:

Model Serving: The zero-shot inference pipeline enables edge deployment directly on network operations centers without requiring centralized cloud infrastructure or continuous retraining, critical for stable long-term predictions. The absence of gradient-based training eliminates model drift concerns over extended deployment periods, ensuring consistent forecast behavior aligned with business planning cycles.

Monitoring: Continuous tracking of success rate (targeting $> 95\%$) and trend-following metrics (Pearson ρ targeting > 0.65, $R^2 > 0.45$) enables early detection of model degradation caused by distributional shifts. For 300-step forecasts, monitoring should focus on *directional accuracy* (correlation coefficients) as much as point-wise accuracy, since trend-following failure (like LSTM's $\rho = -0.399$) renders forecasts operationally useless despite potentially acceptable MAPE values.

Automated recalibration should be triggered when variance σ exceeds 2% or R^2 drops below 0.45, indicating loss of trend-following capability. For extended horizons, quarterly recalibration cycles aligned with business planning periods (Q1, Q2, Q3, Q4) help maintain accuracy as correlation structures evolve over time.

Integration: A RESTful API architecture facilitates seamless integration with existing planning systems (ERP, capacity management, financial forecasting platforms), accepting JSON-formatted time series inputs and returning structured extended forecasts with confidence intervals derived from 1000-iteration stability analysis. API responses should include not just point forecasts but also trend indicators (predicted direction, slope, R^2 confidence) to support strategic decision-making workflows.

Scalability: The framework supports batch processing of multiple network segments (hundreds of cell sites) through the parallelizable Transformer architecture, enabling simultaneous 300-step forecasting across distributed infrastructure with minimal computational overhead. For nationwide operators managing thousands of sites, nightly batch processing can generate 10-month forecasts for strategic planning while maintaining sub-minute per-site inference times.

Fallback Mechanisms: For production deployment, implementing fallback strategies is critical: if correlation monitoring detects $\rho < 0.4$ or $R^2 < 0.3$, the system should alert analysts and potentially revert to simpler methods (ARIMA) or request immediate recalibration rather than continuing with degraded forecasts. This hybrid approach balances BiViSIONTS's superior typical performance with robustness against edge cases.

7.5 Future Research Directions

7.5.1 Extension to Multivariate ($N > 2$ Variables)

Explore 3D convolutions, video transformers, or graph neural networks for encoding traffic volume, users, latency, packet loss, and jitter simultaneously on extended horizons. The current RGB limitation to two variables restricts applicability to complex systems where multiple KPIs interact. Graph-based architectures could model arbitrary N-variable relationships while maintaining the visual encoding paradigm, potentially using learned graph structures to discover latent dependencies not specified a priori.

7.5.2 Learnable Multi-Scale Weights

Replace fixed $\lambda = (0.5, 0.3, 0.2)$ with learned parameters optimized via validation loss or meta-learning across multiple forecast horizons. The +1.47% multi-scale gain suggests further improvements possible with adaptive, horizon-conditioned weighting—e.g., $\lambda_{short} = 0.6$ for 100-step forecasts but $\lambda_{long} = 0.4$ for 500-step forecasts. Differentiable weight learning could automatically discover optimal temporal prioritization for different prediction tasks.

7.5.3 Pre-Training on Time Series Images

Pre-train MAE on large time series corpora converted to images to improve domain-specific performance beyond current 96.06% success rate. Target datasets: UCR Archive (128 datasets), ETDataset (electricity), traffic benchmarks, financial time series. Time series-specific pre-training could learn representations capturing autocorrelation, seasonality, and regime changes more effectively than ImageNet features, potentially improving extended horizon performance where visual domain gap may limit transfer learning effectiveness.

7.5.4 Adaptive Periodicity Selection

Use Fourier analysis or autocorrelation function to automatically detect optimal periodicity from data, replacing manual specification. Current optimal value of periodicity=1 may vary with different temporal resolutions (hourly vs. daily data) and forecast horizons. Adaptive selection could identify dominant cycles in the data (weekly, monthly) and construct multi-resolution image representations that preserve multiple periodicities simultaneously.

7.5.5 Uncertainty Quantification

The 1000-iteration convergence analysis ($\sigma = 1.01$–1.20%) provides empirical confidence intervals at the aggregate level. Extend to Bayesian MAE or Monte Carlo Dropout for prediction intervals on individual extended forecasts, critical for risk-sensitive planning decisions requiring confidence bounds. For 300-step forecasts used in strategic planning, quantifying uncertainty bands (e.g., 80% prediction intervals) enables robust decision-making under uncertainty, supporting scenario analysis and risk management frameworks.

7.5.6 Online Learning and Continual Adaptation

Develop sliding window mechanisms to adapt to evolving correlations on extended horizons without full retraining. The scale-dependent ρ progression (0.77 $\rightarrow$ 0.98) suggests correlations stabilize over time, but online recalibration could handle regime shifts (infrastructure upgrades, competitive dynamics changes) while maintaining trend-following accuracy.

Implement incremental correlation updates that adjust attention weights monthly based on recent observations, balancing stability (avoiding overfitting to noise) with adaptability (tracking genuine structural changes). This continual learning approach would be particularly valuable for 300+ step forecasts spanning quarters where market conditions may shift mid-horizon.

7.5.7 Explainable AI Enhancements

Apply attention rollout, Grad-CAM, or latent space analysis to MAE embeddings to visualize which temporal regions most influence extended forecasts. The current $\alpha_{final} = 0.9144$ provides global interpretability (91% weight on traffic); patch-level attribution would add local explainability showing how specific historical periods (e.g., last month's surge) impact different portions of the 300-step forecast.

For production systems, visual explanations highlighting "this forecast is driven by long-term trend (60%) + recent fluctuation (30%) + seasonal pattern (10%)" would increase stakeholder trust in extended predictions used for strategic decisions involving significant capital commitments.

7.5.8 Horizon-Adaptive Architectures

Investigate dynamic adjustment of multi-scale windows and mask ratios based on forecast horizon length. The optimal configuration for 300 steps (windows=150/400, mask=0.85) may differ substantially from shorter horizons (100 steps) or longer horizons (500+ steps), suggesting horizon-conditioned hyperparameter selection.

Develop meta-learning frameworks that learn to predict optimal architecture configurations given forecast horizon as input, enabling single unified models that adapt their internal representations based on whether they're forecasting 100 steps (emphasize short-term scale) or 500 steps (emphasize long-term scale) ahead.

7.5.9 Hybrid Ensemble Approaches

Combine BiViSIONTS with complementary methods to leverage strengths across different horizons: BiViSIONTS for long-term trend direction (leveraging $R^2 = 0.512$), ARIMA for short-term point accuracy, and possibly domain-specific models for known seasonal patterns. Ensemble weighting could shift dynamically based on forecast horizon—BiViSIONTS weight increases for steps 100–300 where its trend-following advantage dominates, while ARIMA handles near-term steps 1–50 where linear assumptions still hold.

7.6 Lessons Learned

1. **Transfer Learning Excels on Extended Horizons:** ImageNet-pretrained MAE achieves 96.06% success rate on challenging 300-step forecasts where traditional methods deteriorate catastrophically (LSTM: 85.84% with $\rho = -0.399$), validating cross-domain transfer for long-range prediction where task-specific supervised learning fails

2. **Bivariate Modeling is Essential for Extended Forecasting:** Correlation-based attention with multi-scale weighting (CPI = 84.4) dramatically outperforms univariate approaches (VisionTS CPI = 41.1, +43.3 points) by leveraging inter-variable dependencies (ρ up to 0.98) to compensate for temporal distance, proving that coupled variable modeling is not merely beneficial but *necessary* for accurate 300-step forecasting

3. **Multi-Scale Decomposition Enables Long-Range Accuracy:** The +1.47% gain from temporal decomposition (150/400/1018 windows) proves that extended forecasts *require* capturing both transient short-term patterns ($\rho = 0.77$, high independence) and persistent long-term trends ($\rho = 0.98$, near-perfect coupling) simultaneously—single-scale methods miss critical temporal structure needed for 300-step horizon accuracy

4. **Trend-Following Matters More Than Point Accuracy on Long Horizons:** BiViSIONTS achieves strong directional prediction ($\rho = 0.696$, $R^2 = 0.512$) while LSTM completely fails trend-following ($\rho = -0.399$, $R^2 = 0.000$) despite decent short-term MAPE, demonstrating that **correlation metrics are more critical than error metrics** for evaluating extended horizon forecasts used in strategic planning

5. **Zero-Shot Outperforms Supervised Learning on Long Horizons:** Without fine-tuning, BiViSIONTS (96.06%) surpasses gradient-optimized LSTM (85.84%) by +10.22 percentage points, proving that pre-training on general visual data can substitute—and exceed—task-specific training even for challenging 300-step forecasting where most supervised methods exhibit catastrophic performance degradation

6. **Composite Metrics Reveal True Long-Range Capability:** CPI (combining success rate, correlations, and R^2) accurately ranks methods on extended horizons: BiViSIONTS (84.4) » ARIMA (56.4) > VAR (49.8) > VisionTS (41.1) »> LSTM (13.7), whereas point-wise accuracy metrics alone (MAPE, RMSE) miss catastrophic trend failures that render forecasts operationally useless for strategic decisions

7. **Hyperparameter Optimization is Critical for Extended Forecasting:** Systematic search yielding optimal mask ratio (0.85), periodicity (1), and window sizes

(150/400) is essential for long-range accuracy (+4.57 percentage points gain), as default configurations severely underperform on challenging 300-step horizons where architectural details become pivotal for maintaining forecast quality

8. **Stability Matters as Much as Accuracy:** Low variance across 1000 iterations ($\sigma < 1.21\%$) on 300-step forecasts demonstrates reproducibility essential for production deployment—high-accuracy methods with high variance are unsuitable for strategic planning requiring consistent, reliable long-term predictions stakeholders can trust for capital allocation decisions

9. **Attention Mechanisms Scale Better Than Recurrence:** Transformer-based BiViSIONTS maintains strong performance ($R^2 = 0.512$) on 300-step tasks where sequential LSTM exhibits complete failure ($R^2 = 0.000$), validating that parallel attention processing *fundamentally scales better* to extended horizons than sequential recurrent architectures limited by vanishing gradients and finite memory capacity

10. **Domain Knowledge Integration Through Interpretability:** Transparent correlation-based attention ($\alpha_{final} = 0.9144$) and multi-scale weights ($\lambda = 0.5, 0.3, 0.2$) enable domain experts to validate model behavior against telecommunications knowledge, increasing trust in 300-step forecasts used for strategic decisions—black-box methods achieving similar accuracy lack this critical interpretability for high-stakes applications

CHAPTER 8

CONCLUSION

This chapter synthesizes the research findings, articulates key contributions, acknowledges limitations, and charts future directions.

8.1 Summary of Contributions

This thesis introduced **BiViSIONTS (Bivariate Vision Transformer for Time Series)**, a novel framework extending vision-based forecasting to bivariate settings through multi-scale attention-weighted fusion. On the challenging 300-step forecasting horizon, BiViSIONTS achieves 96.06% average success rate with exceptional trend-following capability ($\rho = 0.696$, $R^2 = 0.512$), substantially outperforming all baselines including catastrophic LSTM failure (85.84%, $\rho = -0.399$, $R^2 = 0.000$) and weak VisionTS trend-following (92.50%, $R^2 = 0.128$).

Key Contributions:

1. **Multi-Scale Attention Mechanism:** Three-scale temporal decomposition (150/400/1018 days) captures scale-dependent correlations (ρ: 0.77 $\rightarrow$ 0.82 $\rightarrow$ 0.98), achieving +1.47% improvement over single-scale approaches on extended horizons

2. **Adaptive Correlation-Based Fusion:** Dynamic variable weighting ($\alpha = 0.9144$) outperforms fixed 50-50 allocation by +0.62%, validating that correlation-driven attention enhances extended forecasting accuracy

3. **Zero-Shot Transfer Learning:** ImageNet pre-trained MAE achieves 96.06% success without fine-tuning, surpassing gradient-optimized LSTM by +10.22 percentage points on 300-step tasks, proving task-agnostic pre-training effectiveness

4. **Comprehensive Evaluation Framework:** Simplified enhanced metrics (success rate, Pearson, Spearman, R^2) reveal trend-following failures invisible to traditional error metrics—LSTM achieves 85.84% success but complete directional collapse ($R^2 = 0.000$)

5. **Production-Ready Performance:** Exceptional stability ($\sigma < 1.21\%$) over 1000 iterations with Composite Performance Index (CPI) of 84.4, dramatically exceeding ARIMA (56.4), VAR (49.8), VisionTS (41.1), and LSTM (13.7)

8.2 Research Questions Answered

Central Question: *How can vision-based forecasting extend to bivariate dependencies with adaptive multi-scale attention on challenging extended horizons?*

Answer: BiViSIONTS successfully addresses this through RGB encoding with scale-dependent correlation analysis and weighted attention fusion, achieving 96.06% success rate, $\rho = 0.696$, and $R^2 = 0.512$ on 300-step forecasts—substantially outperforming LSTM's catastrophic failure and VisionTS's weak trend-following.

Sub-Questions Resolved:

- *Does multi-scale attention improve extended forecasting?* Yes, +1.47% validates temporal decomposition captures both transient patterns ($\rho = 0.77$) and persistent trends ($\rho = 0.98$)
- *Is adaptive weighting superior on long horizons?* Yes, +0.62% confirms correlation-based fusion ($\alpha = 0.9144$) outperforms uniform weighting
- *Can zero-shot transfer work on 300-step tasks?* Yes, 96.06% without fine-tuning exceeds trained LSTM (85.84%) by +10.22 points
- *Do trend metrics matter for strategic forecasting?* Yes, BiViSIONTS $R^2 = 0.512$ vs LSTM $R^2 = 0.000$ reveals directional prediction supremacy critical for planning

8.3 Key Findings

Extended Horizon Performance: On the challenging 300-step task, BiViSIONTS achieves 96.06% success with strong trend-following ($\rho = 0.696$, $R^2 = 0.512$), while LSTM exhibits complete collapse ($\rho = -0.399$, $R^2 = 0.000$) and VisionTS shows weak directionality ($R^2 = 0.128$). This validates that bivariate multi-scale attention fundamentally enables extended forecasting where sequential and univariate methods fail.

Composite Performance Index: CPI scores (BiViSIONTS: 84.4 » ARIMA: 56.4 > VAR: 49.8 > VisionTS: 41.1 »> LSTM: 13.7) demonstrate balanced excellence across accuracy, trend-following, and stability—superior to methods optimizing single metrics.

Ablation Study Insights: Multi-scale attention (+1.47%) and adaptive weighting (+0.62%) independently improve performance, with combined effect yielding 96.06% average success. Optimal hyperparameters (mask=0.85, periodicity=1, windows=150/400) achieve 97.10% peak accuracy on 300-step forecasts.

8.4 Practical Impact

Telecommunications: BiViSIONTS's 300-step (10-month) forecasting capability directly supports annual planning cycles, quarterly reviews, and multi-year infrastructure roadmaps. The 96.06% success rate and $R^2 = 0.512$ provide sufficient confidence for commit-to-build decisions with 6–9 month lead times, enabling proactive capacity expansion and optimized capital allocation.

Broader Applications: The framework generalizes to energy demand-temperature forecasting, healthcare admissions-resource planning, retail sales-foot traffic prediction, and transportation flow-passenger modeling—any bivariate system ($\rho > 0.5$) requiring strategic visibility beyond 300 days.

Research Contributions: BiViSIONTS establishes vision-based forecasting as viable for extended horizons, introduces trend-following evaluation frameworks revealing catastrophic failures in traditional methods, and demonstrates zero-shot transfer learning effectiveness where supervised approaches deteriorate.

8.5 Limitations

1. **Bivariate Constraint:** RGB encoding limits to two variables; $N > 2$ requires architectural extensions (3D convolutions, graph networks)

2. **Fixed Multi-Scale Weights:** Manual 0.5:0.3:0.2 weighting effective (+1.47%) but learnable horizon-adaptive weights may improve further

3. **Image Resizing Artifacts:** 224×224 bilinear interpolation may smooth high-frequency variations relevant for initial forecast steps

4. **Pre-Training Domain Gap:** ImageNet features achieve 96.06% but time series-specific pre-training could enhance extended horizon performance

5. **Correlation Stationarity Assumption:** Gradual progression (0.77 $\rightarrow$ 0.98) observed, but sudden regime shifts over 300-step horizons require quarterly recalibration

Despite limitations, BiViSIONTS achieves production-ready performance (CPI = 84.4, $\sigma < 1.21\%$) suitable for high-stakes strategic planning deployment.

8.6 Future Directions

Multivariate Extensions: Scale to $N > 2$ variables through 3D vision transformers or graph neural networks modeling arbitrary dependencies.

Learnable Multi-Scale Weights: Replace fixed λ with horizon-conditioned parameters optimized via meta-learning across forecast lengths.

Time Series Pre-Training: Pre-train MAE on large corpora (UCR Archive, ET-Dataset) to learn domain-specific autocorrelation and seasonality patterns.

Uncertainty Quantification: Extend 1000-iteration stability analysis to Bayesian frameworks providing prediction intervals for individual 300-step forecasts.

Online Adaptation: Implement sliding window recalibration adapting to evolving correlations and regime shifts over multi-quarter horizons.

Extreme Horizon Forecasting: Investigate 500+ step predictions with horizon-adaptive architectures dynamically adjusting scale emphasis.

8.7 Closing Remarks

BiViSIONTS demonstrates that unconventional approaches—treating time series as images, leveraging vision transformers, computing scale-dependent correlations—yield state-of-the-art extended horizon forecasting (96.06% success, $\rho = 0.696$, $R^2 = 0.512$, CPI = 84.4) with remarkable simplicity and without task-specific training.

The key insight: **visual patterns from natural images transfer to extended temporal patterns**, with low-level features (edges, textures) corresponding to long-term trends and persistent cycles. Multi-scale attention (0.77 $\rightarrow$ 0.82 $\rightarrow$ 0.98) captures scale-dependent dynamics essential for 300-step forecasting where traditional methods catastrophically fail.

Critical Revelation: Trend-following metrics (Pearson, R^2) prove as essential as point accuracy for strategic forecasting. LSTM's negative correlation ($\rho = -0.399$) despite 85.84% success reveals methods can achieve acceptable errors while predicting completely wrong trend directions—a failure invisible to MAPE but immediately apparent in $R^2 = 0.000$.

The progression from traditional methods (ARIMA 91.96%, VAR 91.25%, LSTM catastrophic) to vision-based approaches (VisionTS 92.50%, BiViSIONTS 96.06%) demonstrates fundamental architectural advantages for extended forecasting. The 300-step results validate BiViSIONTS doesn't merely improve existing methods—it **fundamentally changes what's possible in extended time series forecasting**.

The journey from pixels to predictions has illuminated a path forward for multi-quarter strategic forecasting, where BiViSIONTS maintains accuracy and directional prediction on challenging horizons where conventional architectures fundamentally fail.

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APPENDIX A

CODE REPOSITORY

The complete BiViSIONTS implementation is available at:

GitHub: `https://github.com/sashikaheendeniya/BiViSIONTS`

Repository Contents:

- `visionts/`: Core VisionTS library
- `long_term_tsf/`: Forecasting utilities
- `BiViSIONTS.ipynb`: Main notebook
- `requirements.txt`: Dependencies
- `README.md`: VisionTS Documentation

APPENDIX B

MATHEMATICAL NOTATION SUMMARY

TABLE B.1: Mathematical Notation

Symbol	Description
S_1, S_2	Bivariate time series
L_c	Context length
L_p	Prediction length
ρ	Pearson correlation
α	Attention weight
w_s, w_m	Short/medium windows
$\lambda_s, \lambda_m, \lambda_l$	Scale weights (0.5, 0.3, 0.2)
P	Periodicity
r_{mask}	Mask ratio
M_1, M_2	Reshaped matrices
I_{RGB}	RGB-encoded image
H_{latent}	Latent representation

APPENDIX C

DATASET DETAILS

TABLE C.1: Dataset Schema

Column	**Type**	**Description**
date	date	Timestamp (daily)
dl_traffic_volume_gb	float	Downlink traffic (GB)
connected_users	integer	Connected user